

CONTRACTIBILITY OF THE COMPLEX OF INCOMPRESSIBLE SEIFERT SURFACES: THE KNOT CASE OF KAKIMIZU’S PROBLEM

Apex Intelligence

ABSTRACT. Let $K \subset S^3$ be a non-trivial knot and let $\text{IS}(K)$ be the simplicial complex whose vertices are the ambient isotopy classes of incompressible Seifert surfaces in the exterior $E(K)$, a finite set of distinct vertices spanning a simplex when its classes admit simultaneously pairwise disjoint representatives. Kakimizu proved that $\text{IS}(K)$ is connected; we prove that it is contractible, together with every truncation on the vertices of genus at most ℓ , and with $\text{IS}(L)$ for links satisfying a linking condition. The argument factors through a combinatorial theorem: every exchange complex is contractible, where an exchange complex carries a complexity subject to two axioms requiring only that an exchanging vertex *exist*, never that one be *selected*. That is what circumvents the obstruction recorded by Przytycki and Schultens. Theorem A relies on the boundary regularity of relative area minimisers implicit in [17, Cor. A.4]; a self-contained proof is given in Appendix A.

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1. INTRODUCTION

Let $K \subset S^3$ be a knot with exterior $E(K) = S^3 \setminus \text{Int } N(K)$. A *Seifert surface* for K is a compact connected orientable surface properly embedded in $E(K)$ whose boundary is a longitude of K ; it is *incompressible* if the inclusion is injective on fundamental groups. Kakimizu [10] attached to K two simplicial complexes: the complex $\text{IS}(K)$ whose vertices are the ambient isotopy classes of incompressible Seifert surfaces, and the full subcomplex $\text{MS}(K)$ on the classes of minimal genus. In both, a finite set of distinct vertices spans a simplex exactly when its classes admit simultaneously pairwise disjoint representatives. Kakimizu proved that $\text{IS}(L)$ and $\text{MS}(L)$ are connected for every non-split oriented link L [10, Thm. A], and that the edge-path metric of $\text{IS}(L)$ is the distance defined through the infinite cyclic cover [10, Prop. 3.1(1)]. Przytycki and Schultens proved that MS is contractible [16, Thm. 1.1].

Their proof runs through a projection π , which assigns to an ordered pair of vertices at distance $d > 0$ a vertex adjacent to the second and at distance $d - 1$ from the first, and which they verify to be well defined on isotopy classes [16, Prop. 4.4]. On IS the projection has to be composed with a further operation, and of that composite they write: “*we do not know how to make [it] well-defined on the set of isotopy classes of surfaces. This is why we do not know if we can extend Theorem 1.5 or even Theorem 1.1 to the complex $\text{IS}(L)$.*” The question whether $\text{IS}(K)$ is contractible is raised there [16].

The device of this paper is to require existence and not selection. We isolate two axioms on a complexity c on the vertices of a flag complex, with values in a well-ordered set: every pair u, w at distance 2 admits an *apex* of the two-interval $\mathcal{I}(u, w) = N(u) \cap N(w)$, that is a member equal or adjacent to every member of $\mathcal{I}(u, w)$; and *some* apex x satisfies $c(x) < \max\{c(u), c(w)\}$. Neither axiom names a map $(u, w) \mapsto x$, and no argument below transports one from one stage to

the next. Contractibility then follows by transfinite induction along the well-order: at a successor the complex is the union of a sublevel set and the cone on the descending link of the top vertex, and the descending link satisfies the same two axioms. For $\text{IS}(K)$ the complexity is the pair $(g(v), A(v))$ of the genus and the least relative area, ordered lexicographically, and the second axiom is supplied by an exchange lemma for incompressible surfaces proved by minimal-surface methods.

Our main result is the following.

Theorem A. *Let $K \subset S^3$ be a non-trivial knot. Then $\text{IS}(K)$ is contractible.*

The hypothesis excludes only the unknot, for which IS is a single point by §4.1. The proof factors through a combinatorial statement that involves no three-dimensional topology, stated as Theorem B at the head of §2: every exchange complex is contractible. Two further results are proved.

Theorem C. *For every integer $\ell \geq g(K)$ the truncation $\text{IS}_\ell(K)$, the full subcomplex of $\text{IS}(K)$ on the vertices of genus at most ℓ , is isometrically embedded in $\text{IS}(K)$ and is contractible. The same holds for the full subcomplex on the vertices of genus at most ℓ and relative area at most a_0 , for every $a_0 > 0$.*

Taking $\ell = g(K)$ recovers the case $E = E(K)$ of [16, Thm. 1.1]; the general case, for an arbitrary (E, γ, α) , is not claimed here. The connectedness of $\text{IS}_\ell(K)$ is part of Theorem C and is obtained below from Corollary 2.3; it also has a direct source in [2, Prop. 5.15], which we do not use.

The second extension concerns links, and requires a condition. Call a link $L = L_1 \cup \cdots \cup L_n$ *linking-indecomposable* if for every partition $\{1, \dots, n\} = I \sqcup I^c$ into two non-empty parts there is an index m with $\text{lk}(L_m, L_J) \neq 0$, where L_J is the sublink indexed by the part *not* containing m ; the condition is vacuous for $n = 1$. Its role is to force every spanning surface to be connected, so that Kakimizu's vertices — which are allowed to be disconnected, and are required only to be incompressible componentwise — coincide with ours.

Theorem D. *Let L be a non-split, linking-indecomposable link in S^3 which is not the unknot. Then $\text{IS}(L)$ is contractible.*

Remark 1.1. Write (F) for the condition that every descending link $\{u \in \text{lk}(v) : c(u) < c(v)\}$ of an exchange complex is finite. Under (F), Theorem B is a restatement of [21, Lem. 2.3] together with the heredity of the axioms, Lemma 2.4, applied at the bottom level. We do not know whether $\text{IS}(K)$ satisfies (F).

Remark 1.2. The geometric input of §3 is used through the literal statement of [6, Thm. 6.12], that the area infimum over the piecewise smooth surfaces properly isotopic to F rel ∂F is attained. The boundary regularity of the resulting minimiser is not written out in the printed proof of that theorem, and it is needed here and in the derivation of [17, Cor. A.4]. Appendix A derives it from that existence statement alone, by the first variation of area and Hopf's boundary point lemma.

One question is left open: whether $\text{IS}(K)$ satisfies (F), equivalently whether a vertex of $\text{IS}(K)$ can have infinitely many neighbours of strictly smaller genus. Three extensions are not treated: general links, where the step that fails is the hypothesis $S \cap (S_+ \cup S_-) = \emptyset$ of Lemma 4.9, two distinct vertices of $\text{IS}(L)$ being able to share the isotopy class of a component; the equivariant statement, where the least-complexity apices of a two-interval span a G -invariant simplex whose vertices G may permute without fixing one; and Haken manifolds, where we know of no source for the connectedness of the corresponding complex.

Organisation. Section 2 is combinatorial and self-contained: it defines exchange complexes and proves Theorem B. Section 3 fixes the relative metric (2) and collects the three facts about relative area minimisers which are taken from the literature, (U), (E) and (Fin). Section 4 puts a complexity on $IS(K)$ and verifies the first axiom; §5 proves the exchange lemma, which is the second; and §6 assembles Theorems A, C and D. Appendix A proves the two boundary regularity statements (R1) and (R2) used in §5, and Appendix B carries the corner rounding estimates which the exchange lemma consumes.

Appendix C is used nowhere else in the paper. It treats the fixed boundary least area map along $\partial E(K)$; it supplies the local input at the boundary which [3, §7] lists as needed and leaves unargued; and it records precisely which statements would still have to be added before the alternative route of Remark A.8 could replace the citation used in §3. Its own analytic input is one clause asserted by the theorem cited for it rather than proved there, and §C.3 reduces that clause to a statement about a limit in moduli space without proving it. None of §§2–6 depends on that appendix.

Acknowledgements. The mathematical content of this paper was developed with substantial assistance from large language models, which were used for literature search, for drafting, and for adversarial checking of the arguments; the authors are responsible for all statements. Four combinatorial statements from the development of §2 have been formalised in Lean 4 against `mathlib`: the ordering statement of Corollary 2.3 for an arbitrary well-founded linear order, an injective complexity and a possibly infinite vertex set; the reduction to an injective complexity of Lemma 2.2, without any finiteness assumption; a combinatorial reformulation of dismantlability; and a seven-vertex exchange complex whose descending link at the top vertex is contractible but is not a cone. The development is 913 lines, contains no `sorry`, no `axiom` declaration and no appeal to `native_decide`. Theorem B itself has not been formalised and no such claim is made.

2. EXCHANGE COMPLEXES AND THEIR CONTRACTIBILITY

This section and its results use no three-dimensional topology. Nothing is quoted from it below except Definition 2.1.

Let X be a simplicial complex with vertex set $X^{(0)}$, and write dist for the edge-path metric on $X^{(0)}$. For u, w with $\text{dist}(u, w) = 2$ the *two-interval* is

$$\mathcal{I}(u, w) = N(u) \cap N(w),$$

the set of common neighbours; it is non-empty by the definition of the distance. A vertex $x \in \mathcal{I}(u, w)$ is an *apex* of the interval if x is equal or adjacent to every member of $\mathcal{I}(u, w)$; equivalently, if X is flag, the full subcomplex on $\mathcal{I}(u, w)$ is a cone with apex x .¹

Definition 2.1 (Exchange structure). Let W be a well-ordered set. An *exchange structure* on X is a map $c: X^{(0)} \rightarrow W$ such that

- (N1) every two-interval has an apex; and
- (N2) whenever $\text{dist}(u, w) = 2$, some apex x of $\mathcal{I}(u, w)$ satisfies $c(x) < \max\{c(u), c(w)\}$.

A non-empty connected flag complex carrying such a c is an *exchange complex*.²

¹The word “apex” is used in a different sense in [1], for a vertex of a graph adjacent to all others. We keep it here because it is the natural word for a cone point.

²Non-emptiness cannot be dropped: the induction below is based at order type 1, and the empty complex is not contractible.

Axiom (N2) asserts in particular that an apex exists, so it implies (N1); we keep both because they are used for different purposes — (N1) in the manufacture of the cone structure, (N2) in the descent — and because in §6 they are verified by different arguments. The axioms assert that an apex exists and never that one can be chosen: no map $(u, w) \mapsto x$ is part of the data, and none is constructed below.³

Theorem B. *Every exchange complex is contractible.*

Lemma 2.2 (Ties may be broken). *If X carries an exchange structure, then it carries one whose complexity is injective and well-orders $X^{(0)}$.*

Proof. Well-order each fibre $c^{-1}(a)$ and order $X^{(0)}$ lexicographically, first by the old value and then within the fibre; a lexicographic sum of well-orders indexed by a well-order is a well-order. Axiom (N1) does not mention c , and a witness x for the old (N2) lies in a value block strictly preceding the block of the larger endpoint, so the strict inequality survives. $\square$

From here on the complexity is assumed injective.

Corollary 2.3. *Let (X, c) be an exchange complex with c injective and let $D \subseteq X^{(0)}$ be downward closed for c . Then the full subcomplex X_D on D is isometrically embedded in X , and is connected. In particular, any two vertices p, q are joined by a geodesic all of whose vertices z satisfy $c(z) \leq \max\{c(p), c(q)\}$.*

This is a special case of [1, Lem. 9.13], whose hypotheses are weaker — no apex, no flagness, no local finiteness — and whose argument is the one we give; for self-containedness we give it.

Proof. Let $u, w \in D$ and $n := \text{dist}_X(u, w)$, finite because X is connected; assume $n \geq 2$. Among the paths $v_0 = u, v_1, \dots, v_n = w$ of length n in X — a non-empty set — consider $M(v_\bullet) := \max\{c(v_i) : 0 < i < n\} \in W$. The order on W is a well-order, so the set of attained values has a least element; fix a path attaining it.

Suppose some interior vertex has $c(v_i) > \max\{c(u), c(w)\}$, and let v_i be the interior vertex attaining M , unique because c is injective. Then $c(v_{i\pm 1}) < c(v_i)$: each of $v_{i\pm 1}$ is either an endpoint, and then its complexity is below $c(v_i)$ by assumption, or an interior vertex, and then its complexity is at most M , with equality excluded by injectivity. A path of length n between vertices at distance n admits no shortcut, so v_{i-1} and v_{i+1} are neither equal nor adjacent, whence $\text{dist}(v_{i-1}, v_{i+1}) = 2$, and (N2) supplies $x \in \mathcal{I}(v_{i-1}, v_{i+1})$ with $c(x) < \max\{c(v_{i-1}), c(v_{i+1})\} < c(v_i)$. Replace v_i by x : the result is a walk of length n from u to w , and it is a path, since a repeated vertex would allow a loop to be excised and give a walk of length $< n$. Every interior vertex of it other than x had complexity strictly below $c(v_i)$, by injectivity, and $c(x) < c(v_i)$; so M has strictly dropped, contradicting minimality.

So every vertex of the chosen path satisfies $c(v_i) \leq \max\{c(u), c(w)\}$, and D being downward closed with $u, w \in D$, the whole path lies in X_D . Hence $\text{dist}_{X_D}(u, w) \leq n$, and the reverse inequality is automatic; X_D is isometrically embedded, and connected because these distances are finite. The last sentence is the displayed property with $D = X^{(0)}$. $\square$

Lemma 2.4 (Hereditiy). *Let (X, c) be an exchange complex with c injective. For $D \subseteq X^{(0)}$ downward closed for c , let X_D be the full subcomplex on D ; for a vertex v let $X_{<v}$ be X_D for*

³The axiom of choice enters this paper once, in Lemma 2.2, to well-order the fibres of the complexity. For a total order, “every non-empty subset has a least element” implies “no infinite strictly descending sequence” in ZF, and the converse needs dependent choice; for the complexity of Definition 4.7 the passage to least elements is available in ZF, since the fibres are finite by Lemma 4.8(i). The proof of Theorem B treats $|X|$ as a CW complex with one open cell per simplex and never needs a total ordering of the vertices.

$D = \{u : c(u) < c(v)\}$, and let $L(v)$ be the full subcomplex on the descending link $L_c(v) = \{u \in N(v) : c(u) < c(v)\}$. Then

- (1) if non-empty, X_D is an exchange complex; in particular so is $X_{<v}$;
- (2) if non-empty, $L(v)$ is an exchange complex, and it is non-empty unless v is the least vertex.

Proof. A full subcomplex of a flag complex is flag, and a subset of a well-ordered set is well-ordered, so the restricted complexity still takes values in a well-ordered set. Let u, w be non-adjacent vertices, either at distance 2 in X_D or both in $L_c(v)$; in either case they are non-adjacent in X , the subcomplexes being full. In the first case a common neighbour inside D witnesses $\text{dist}_X(u, w) = 2$; in the second v itself is a common neighbour, so again $\text{dist}_X(u, w) = 2$. By (N2) there is an apex x of $\mathcal{I}(u, w)$ with

$$c(x) < \max\{c(u), c(w)\}, \quad (1)$$

and in the second case the right-hand side is in turn $< c(v)$. For X_D the inequality (1) places x in D , which is downward closed. For $L(v)$: the apex is equal or adjacent to every member of $\mathcal{I}(u, w)$, and $v \in \mathcal{I}(u, w)$; the strict inequality $c(x) < c(v)$ excludes $x = v$, so x is adjacent to v and lies in $L_c(v)$. In both cases the smaller two-interval is contained in $\mathcal{I}(u, w)$, so x lies in it and is an apex of it, which is (N1) there, and (1) is (N2).

Two non-adjacent members of $L(v)$ have the common neighbour x just produced, so $L(v)$ has graph diameter at most 2 and is connected; X_D is connected by Corollary 2.3. Finally, if v is not the least vertex v_0 , apply Corollary 2.3 to v and v_0 : the vertex adjacent to v on the resulting geodesic is a neighbour $z \neq v$ with $c(z) \leq c(v)$, and injectivity gives $c(z) < c(v)$, so $z \in L_c(v)$. $\square$

Proof of Theorem B. By Lemma 2.2 we may assume c injective, so that it well-orders the vertex set. For an exchange complex (Z, c) with c injective, the *order type* of (Z, c) is that of $(Z^{(0)}, c)$; if it is α we write $Z^{(0)} = \{z_\beta\}_{\beta < \alpha}$ in increasing order and let Z_γ be the full subcomplex on $\{z_\beta : \beta < \gamma\}$, so that $Z_0 = \emptyset$ and $Z_\alpha = Z$. By Lemma 2.4(1), Z_γ with $1 \leq \gamma \leq \alpha$ is again an exchange complex with injective complexity, of order type γ . We prove by transfinite induction on α :

- $(*_\alpha)$ Every exchange complex with injective complexity and order type at most α is contractible.

The statement is universal because the successor step applies the inductive hypothesis to a descending link, which is not one of the subcomplexes Z_γ of the complex being treated; Lemma 2.4(2) is what makes that application legitimate.

Base. Order type 1: a single vertex.

Successor. Let $\alpha \geq 1$, assume $(*_\alpha)$, and let Z be an exchange complex with injective complexity of order type $\alpha + 1$. Then Z has a c -largest vertex v ; put $Y = Z$, $A = Z_\alpha$ and $L = L(v)$. Since Y is the full subcomplex on an initial segment containing v , the neighbours of v in Y are exactly the vertices of L . A simplex of Y either omits v , and then lies in A ; or it is $\{v\} \cup \tau$ with τ a simplex of Y contained in $L_c(v)$, hence a simplex of L because L is full. Conversely a simplex of $v * L$ is $\{v\} \cup \tau$ with $\tau \subseteq L_c(v) \subseteq N(v)$, so $\{v\} \cup \tau$ is a clique of the 1-skeleton and hence, by flagness, a simplex of Y . Therefore

$$Y = A \cup (v * L), \quad A \cap (v * L) = L,$$

the second equality because a simplex of $v * L$ lies in A exactly when it omits v . Both A and L are exchange complexes with injective complexity of order type at most α (Lemma 2.4, using $\alpha \geq 1$ for the non-emptiness of L), so both are contractible by $(*_\alpha)$.

Since $|Z|$ is a CW complex with one open cell per simplex, the pairs $(v * L, L)$ and (Y, A) are CW pairs, and for a CW pair with contractible subcomplex the quotient map is a homotopy equivalence [7, Props. 0.16 and 0.17]. Apply this twice. First to $(v * L, L)$: the cone $v * L$ is contractible — the straight-line homotopy towards the cone point is continuous for the weak topology because $|v * L| \times I$ is again a CW complex, I being locally compact — and L is contractible, so $(v * L)/L$ is contractible. Second to (Y, A) : A is contractible, so $Y \rightarrow Y/A$ is a homotopy equivalence. Collapsing A collapses $L \subseteq A$, and no cell of $v * L$ outside L lies in A , so $Y/A = (v * L)/L$, which is contractible. Therefore Y is contractible.

Limit. Let λ be a limit ordinal, assume $(*_\alpha)$ for every $\alpha < \lambda$, and let Z be an exchange complex with injective complexity of order type λ . Then $|Z| = \bigcup_{1 \leq \alpha < \lambda} |Z_\alpha|$, each Z_α contractible by $(*_\alpha)$. Let $f: S^n \rightarrow |Z|$ be continuous, $n \geq 0$. Its image is compact, hence lies in a finite subcomplex [7, Prop. A.1]; the finitely many vertices involved have indices $< \lambda$, so there is $\alpha < \lambda$ with $f(S^n) \subseteq |Z_\alpha|$, λ being a limit. Since Z_α is contractible, f is null-homotopic in $|Z_\alpha|$, hence in $|Z|$. Thus $\pi_n(|Z|, z_0) = 0$ for all $n \geq 0$ and any basepoint, where z_0 is a vertex of Z_1 , which is non-empty because $\lambda \geq \omega$; for $n = 0$ this says $|Z|$ is path-connected. The inclusion $\{z_0\} \hookrightarrow |Z|$ is then a map of connected CW complexes inducing an isomorphism on π_n for every n , hence a homotopy equivalence by Whitehead's theorem [7, Thm. 4.5]; that is, $|Z|$ is contractible.

Applying $(*_\kappa)$, where κ is the order type of (X, c) , to X itself gives the theorem. $\square$

The topology above is the weak topology on $|X|$, under which a set is closed exactly when its intersection with every closed simplex is closed; for a complex that is not locally finite, and $\text{IS}(K)$ is not [10], this is strictly finer than the metric topology.

3. GEOMETRIC INPUT

Fix on $E(K)$ a *relative Riemannian metric*, that is, one making $\partial E(K)$ strictly convex. Such a metric exists: on a collar $[0, \varepsilon)_r \times \partial E(K)$, with r the inward coordinate, $\partial E(K) = \{r = 0\}$ and $\varepsilon < 1$, take

$$dr^2 + \varphi(r)^2 g_{\partial E(K)}, \quad \varphi(0) = 1, \quad \varphi'(0) < 0, \quad (2)$$

say $\varphi(r) = 1 - r$, and extend smoothly; strict convexity is the only source of a barrier in this paper. Along the collar ∂_r is the unit gradient of r , so $\text{Hess } r(\partial_r, \cdot) = \frac{1}{2}d(|\partial_r|^2) = 0$, while $\nabla_X \partial_r = (\varphi'/\varphi)X$ for X tangent to the level sets; hence

$$\text{Hess } r = \varphi\varphi' g_{\partial E(K)} < 0 \quad \text{on the level sets,} \quad (3)$$

the right-hand side read as the pullback of $g_{\partial E(K)}$ to the collar, which annihilates ∂_r . Consequently, for a unit vector $v = a\partial_r + X$ one has $\text{Hess } r(v, v) = \varphi\varphi'|X|_{g_{\partial E(K)}}^2 \leq 0$, with equality exactly when $X = 0$. So along any geodesic γ of $\tilde{E}$ contained in the collar the function $r \circ \gamma$ is concave, and strictly concave unless $\dot{\gamma}$ is radial, in which case γ is an integral curve of ∂_r and $r \circ \gamma$ is affine with non-zero slope. In particular $r \circ \gamma$ admits no interior local minimum at which it vanishes: at such a point $\dot{\gamma}$ would be tangent to $\{r = 0\}$, so $(r \circ \gamma)'' < 0$ there, and a strict local maximum is not a local minimum. A geodesic of $E(K)$ joining two points of $\partial E(K)$ therefore has $r > 0$ at its interior points, since an interior point with $r = 0$ would be such a minimum, the geodesic lying in the collar near it. Dually, with the sign convention $\text{II}(X, Y) = \langle \nabla_X Y, N \rangle$ and $N = \partial_r$ the *inward* normal, the shape operator of $\{r = 0\}$ is $-\text{Id}$ and its second fundamental form is $g_{\partial E(K)} > 0$; so $\partial E(K)$ has mean curvature $\text{tr II} = 2 > 0$ with respect to that normal — the trace, not its average over the two principal directions — which is the sign convention of [6, §6]. We write $\tilde{E}$ for the manifold obtained by extending the collar to $r \in (-\varepsilon, \varepsilon)$ by the same formula, so that $E(K) \subset \tilde{E}$ isometrically and $\partial E(K)$ is a two-sided smooth surface in $\text{Int } \tilde{E}$.

Fix also a real-analytic foliation J of the boundary torus by preferred longitudes; smoothness is all that the citations below require [17, Def. 3], and real-analyticity, free here, is used in §5. A properly embedded $F \subset E(K)$ with $\partial F \subset J$ is a *relative area minimiser* if $\text{Area}(F)$ is least in its proper isotopy class subject to the boundary lying in J : the boundary slides from leaf to leaf and is not pinned pointwise.

Three facts about relative area minimisers in $E(K)$ are taken from the literature. The hypotheses under which they are stated are that M be a compact P^2 -irreducible Riemannian 3-manifold with smooth strictly convex boundary, that the family of boundary curves be compact, and that the surface be connected, incompressible, and meet each boundary component of M in a single component of its own boundary [17, App.]. All of these hold for $M = E(K)$ and $\mathcal{F} = J$: the exterior is compact, orientable and irreducible [10], hence P^2 -irreducible, since an orientable 3-manifold contains no two-sided projective plane; $\partial E(K)$ is a torus, made strictly convex by (2); J is a real-analytic foliation of it by closed curves; and a Seifert surface is connected, incompressible and meets $\partial E(K)$ in its single boundary circle. No minimal genus condition occurs among them.

Theorem 3.1 ((U): universal disjointness). (i) *Relative area minimisers $F_1, F_2 \subset E(K)$ which are incompressible and whose proper isotopy classes contain disjoint representatives are themselves disjoint, or coincide.*
(ii) *If $f_i: S_i \rightarrow E(K)$, $i = 1, \dots, n$, are incompressible, pairwise non-isotopic and pairwise disjoint, then relative area minimisers g_i in the classes of the f_i have pairwise disjoint images.*

Clause (i) is [17, Thm. 4] and clause (ii) is [17, Thm. A.5]. The force of (U) is the universal quantifier, which lets minimisers be chosen one class at a time, before any pair is examined.

Theorem 3.2 ((E): the area infimum is attained). *Let $F \subset E(K)$ be properly embedded, incompressible, with $\partial F \subset J$.*

- (i) (Fixed boundary.) *Let $\mathcal{F}$ be the set of piecewise smooth surfaces isotopic to F rel ∂F and $I = \inf\{\text{Area}(G) : G \in \mathcal{F}\}$. Then there is $F' \in \mathcal{F}$ with $\text{Area}(F') = I$.*
- (ii) (Boundary sliding in J .) *The infimum of area over the surfaces properly isotopic to F with boundary in J is attained, and the minimiser is a member of $\mathcal{M}(S)$: a stable minimal surface lying in the proper isotopy class of a smooth proper embedding $f: (S, \partial S) \rightarrow (E(K), \partial E(K))$ with $f^{-1}(\partial E(K)) = \partial S$, whose boundary is a parametrised curve of J .*

Clause (i) is [6, Thm. 6.12], whose hypotheses hold for $M = E(K)$: compact, hence homogeneously regular; orientable and irreducible, hence P^2 -irreducible; and $\partial E(K)$ sufficiently convex in the sense of [6, §6], the three conditions there being met by (2), by the mean curvature computation above, and by the extension $\tilde{E} \supset E(K)$. Clause (ii) is [17, Thm. 2 = Cor. A.4]. Below, (E) without qualification means clause (ii), a vertex of $\text{IS}(K)$ being a class in which the boundary longitude may move.

Theorem 3.3 ((Fin): compactness and finiteness). *Fix a compact connected surface S and $a > 0$, and let $\mathcal{M}(S)$ be as in (E)(ii), topologised by C^1 convergence, with $\mathcal{M}_a \subset \mathcal{M}(S)$ the subset of surfaces of area at most a . Then $\mathcal{M}_a$ is compact [17, Prop. A.1] and is the disjoint union of finitely many open and closed subsets, each consisting of mutually isotopic surfaces [17, Cor. A.3].*

Remark 3.4. Two further statements are used: (R1), that the minimiser of (E)(ii) is smooth up to $\partial E(K)$ and meets it exactly in its own boundary; and (R2), that it meets $\partial E(K)$ transversally. Read with the definition of $\mathcal{M}(S)$, (R1) is part of the statement of (E)(ii); but the non-emptiness

of $\mathcal{M}(S)$ is obtained in [17, App.] by treating the fixed-boundary minimiser of (E)(i) as a member of $\mathcal{M}(S)$, and (E)(i) as stated produces a minimiser only in the piecewise smooth category. Appendix A therefore proves (R1) and (R2) from (E)(i) together with (Fin), using no other input from [6] and none from [17] beyond (Fin).

4. THE KAKIMIZU COMPLEX

4.1. Vertices and simplices. A *knot* is a smooth oriented embedding $K \hookrightarrow S^3$; we fix an open tubular neighbourhood $N(K)$ and write $E(K) = S^3 \setminus \text{Int } N(K)$ for the compact exterior, whose boundary is a torus. The *preferred longitude* is the oriented slope on $\partial E(K)$ of linking number zero with K . A *Seifert surface in the exterior* is a compact connected orientable neatly embedded surface $S \subset E(K)$ whose oriented boundary is a preferred longitude; having one boundary component,

$$\chi(S) = 1 - 2g(S), \quad (4)$$

and $g(K) = \min\{g(S)\}$. A *compressing disc* for S is an embedded disc $D \subset E(K)$ with $D \cap S = \partial D$ and ∂D essential on S ; if none exists, S is *incompressible*. Boundary incompressibility is not required, and Lemma 4.2 shows it is automatic. The exterior is irreducible [10].

An *ambient isotopy* is a smooth family of diffeomorphisms $H_t: E(K) \rightarrow E(K)$ with $H_0 = \text{id}$, preserving $\partial E(K)$ setwise; a *normal isotopy* is a smooth family of neat embeddings $f_t: S \rightarrow E(K)$. The boundary longitude is allowed to move on the torus. That an ambient isotopy restricts to a normal isotopy is immediate, and the converse is isotopy extension [20, Thm. 2.4.6], whose hypotheses — compactness of the surface and neatness of the embeddings — hold here; so the two equivalence relations agree on Seifert surfaces, and the relative area minimisation of §3, performed in a normal isotopy class, computes an invariant of a vertex.

Definition 4.1. The vertices of $\text{IS}(K)$ are the ambient isotopy classes of incompressible Seifert surfaces in $E(K)$. A finite set of pairwise distinct vertices spans a simplex if and only if its classes admit *simultaneously* pairwise disjoint representatives. For an integer $\ell \geq g(K)$,

$$\text{IS}_\ell(K) = \text{IS}(K)[\{[S] : g(S) \leq \ell\}]$$

is the full subcomplex on the vertices of genus at most ℓ ; in particular $\text{IS}_{g(K)}(K) = \text{MS}(K)$. We write $\text{dist}(u, w)$ for the graph distance in the 1-skeleton, and call a complex *flag* if every finite clique in its 1-skeleton spans a simplex.

These are Kakimizu's vertices and simplices [10, 1.3]: his spanning surfaces are allowed to be disconnected and are required only to be incompressible componentwise, which for a knot selects the same surfaces, since a spanning surface for a knot has no closed component and one boundary circle; and he spans an edge on two classes with disjoint representatives, which for a knot agrees with the Kakimizu-distance-one convention of [16]. What those authors add is ∂ -incompressibility, and that condition is automatic.

Lemma 4.2. *Let L be a link in S^3 and let $\Sigma \subset E(L)$ be an incompressible spanning surface, neatly embedded, with $\partial\Sigma$ the union of the longitudes. Then Σ is ∂ -incompressible: there is no embedded disc $D \subset E(L)$ with $D \cap \Sigma = a$ and $D \cap \partial E(L) = b$, where $\partial D = a \cup b$ and a is an arc that does not cut a disc off Σ together with an arc of $\partial\Sigma$.*

Proof. Suppose such a D exists. As $D \cap \Sigma = a$ and $\lambda := \Sigma \cap \partial E(L) = \partial\Sigma$, the interior of b misses λ , so both endpoints of b lie on one torus T_i and b is properly embedded in the annulus A_i obtained by cutting T_i along λ_i , whose boundary circles $\lambda_i^\pm$ are the two sides of λ_i . If both endpoints lie on the same circle, b together with an arc $\lambda_0 \subset \lambda_i$ bounds a disc $\Delta \subset A_i$ with $\text{Int } \Delta \cap \lambda = \emptyset$; then $D \cup \Delta$, pushed off $\partial E(L)$, is a disc meeting Σ exactly in its boundary $a \cup \lambda_0$.

If that curve is essential in Σ it is a compressing disc, contrary to incompressibility; if not, a cuts a disc off Σ together with an arc of $\partial\Sigma$. If the endpoints lie on the two different circles, perturb the loop $\partial D = a \cup b$ to a loop γ transverse to Σ by pushing $\text{Int } a$ off Σ into the side on which D lies, keeping b fixed; this is an isotopy in $E(L)$, so γ is still null-homologous there. Near a point of λ_i the pair $(E(L), \Sigma)$ is $(\{z \geq 0\}, \{x = 0, z \geq 0\})$, with $\lambda_i = \{x = z = 0\}$, so $\lambda_i^\pm$ are the traces of the two sides $\{\pm x > 0\}$ of Σ , and γ can meet Σ only near the two endpoints of b . At the endpoint on the side into which a was pushed, b and the pushed arc leave on the same side and there is no crossing; at the other endpoint they leave on opposite sides and there is exactly one, transverse. The algebraic intersection number of γ with Σ is therefore ± 1 . But Σ is Poincaré dual to the total linking homomorphism $\varphi: H_1(E(L)) \rightarrow \mathbb{Z}$, $\gamma \mapsto \text{lk}(\gamma, L)$, by a count of meridians, so that number is $\varphi(\gamma) = 0$, γ being null-homologous in $E(L)$. $\square$

For the unknot, IS is a single point: the exterior is a solid torus V and the preferred longitude a meridian of V , so an incompressible $S \subset V$ has $\pi_1(S) \rightarrow \pi_1(V) \cong \mathbb{Z}$ injective, forcing $g(S) = 0$; and two meridian discs may be isotoped to share their boundary, made disjoint by innermost-circle surgery, and then carried onto one another by the Alexander trick applied to the ball the resulting sphere bounds. For a non-trivial knot, $\text{IS}(K)$ is connected [10, Thm. A], that theorem being proved there for incompressible spanning surfaces and not only for those of minimal genus. It is non-empty, since Seifert's algorithm produces a Seifert surface [12, Thm. 2.2] and compression terminates by Lemma 4.4; a surface of least genus is incompressible, so $\text{IS}_\ell(K) \neq \emptyset$ for every $\ell \geq g(K)$.

Lemma 4.3. *Let $v_1, \dots, v_n$ be pairwise distinct vertices of $\text{IS}(K)$ and let A_i be a relative area minimiser in the class v_i , supplied by (E) one class at a time. If v_i and v_j are adjacent then $A_i \cap A_j = \emptyset$. Consequently:*

- (1) *if $v_1, \dots, v_n$ is a clique in the 1-skeleton, the A_i are simultaneously pairwise disjoint, so $\text{IS}(K)$ and every $\text{IS}_\ell(K)$ are flag;*
- (2) *if $\text{dist}(u, w) = 2$ and $x \in \mathcal{I}(u, w)$, then the minimisers $S \in x$, $S_+ \in u$, $S_- \in w$ satisfy $S \cap (S_+ \cup S_-) = \emptyset$.*

Proof. Adjacent classes contain disjoint representatives, and the A_i are relative area minimisers of incompressible surfaces, so Theorem 3.1(i) gives that A_i and A_j are disjoint or coincide. A Seifert surface for a knot is connected, so coincidence would mean $A_i = A_j$ as sets and hence $v_i = v_j$, which is excluded. The minimisers were fixed before any pair was examined and do not depend on (i, j) , which gives (1); and (2) is the case of the three classes x, u, w , the first being adjacent to each of the others. Fullness makes the truncations retain the simplices. $\square$

Only the two-surface clause of (U) is used: the choice made in advance converts pairwise information into simultaneous information. Connectedness of the surfaces is used once, and only there; for the links of Theorem D it is supplied by Lemma 6.1.

4.2. Compression and the robustness of neighbours. Throughout, A and B are transverse neatly embedded compact surfaces. Cutting both along $A \cap B$, the resulting components are *patches* and the intersection curves and arcs duplicated on their boundaries are *exchange seams*; a *disc patch* is a disc component of the result.

Lemma 4.4. *Let A, B be compact and transverse. Cut along $A \cap B$ and reglue by the orientation-compatible normal exchange, and let P_{tot} be the union of the output components. Then $\chi(P_{\text{tot}}) = \chi(A) + \chi(B)$, so that if $P_{\text{tot}} = P \sqcup C_1 \sqcup \dots \sqcup C_m$ then*

$$\chi(P) = \chi(A) + \chi(B) - \sum_j \chi(C_j). \quad (5)$$

Moreover, compressing a connected Seifert surface along an essential circle and keeping the component that carries the longitudinal boundary strictly lowers the genus, so iterated compression terminates at an incompressible Seifert surface.

Proof. Triangulate $A \cup B$ so that each intersection curve is a subcomplex. Cutting duplicates exactly the cells of the intersection trace and the normal regluing identifies the same cells again, so the Euler correction is the same before and after the exchange; additivity over components gives (5). Compressing along a non-separating circle removes a handle and keeps the unique boundary component, so the genus drops by one; compressing along a separating essential circle splits off a closed surface, orientable and of positive genus, since a closed side of genus 0 would exhibit the circle as the boundary of a disc in the surface, and the genus of the side carrying the longitudinal boundary drops by that amount. The genus is a non-negative integer, so the process is finite. $\square$

Lemma 4.5. *Let M be an irreducible orientable 3-manifold, let $P \subset M$ be connected, neatly embedded, with non-empty boundary, and let $Q \subset M$ be incompressible, neatly embedded and disjoint from P . Given any finite compression sequence for P , each step keeping the component with boundary, there is a compression sequence whose intermediate and final ambient isotopy classes agree with the given ones one by one and all of whose discs are disjoint from Q .*

Proof. It suffices to clean a single prescribed compressing disc D and then induct along the sequence, transporting the next prescribed disc to the current representative at each step. Put D transverse to Q ; as $\partial D \subset P$ and $P \cap Q = \emptyset$, the intersection $D \cap Q$ consists of circles. Each bounds a subdisc of D , hence is null-homotopic in M , hence bounds a disc in Q by incompressibility. Take an innermost such disc $E \subset Q$ and the subdisc $F \subset D$ it cobounds not containing ∂D ; the sphere $E \cup F$ bounds a ball B by irreducibility, and pushing F across B to a parallel copy of E is an isotopy of D rel ∂D which removes that circle and all intersection circles inside F . The intermediate discs are again compressing discs because B is disjoint from P , which is what makes the isotopy extendable to an ambient isotopy preserving P . Indeed P misses ∂B : E lies in Q , which is disjoint from P , and F lies in D , which meets P only in ∂D , while F is the subdisc *not* containing ∂D . Since P is connected it therefore lies either in $\text{Int } B$ or in the complement of B . The first is impossible: ∂P is non-empty and contained in ∂M , whereas $B \cap \partial M \subset \partial B$ — a point of ∂M interior to B would have a neighbourhood in M homeomorphic to $\mathbb{R}^3$, contradicting invariance of domain in a half-space chart — and $P \cap \partial B = \emptyset$. Repeating gives D' disjoint from Q and isotopic to D rel boundary, so the compressed surfaces are ambient isotopic; the component with boundary remains disjoint from Q and furnishes the input for the next step. $\square$

Corollary 4.6 (Robustness of neighbours). *Let P be connected, neatly embedded, with non-empty boundary and disjoint from an incompressible surface Q . Then every ambient isotopy class obtained by compressing P down to an incompressible descendant with boundary is, in $\text{IS}(K)$, equal or adjacent to $[Q]$.*

Proof. By Lemma 4.5 the final representative may be taken disjoint from Q ; if the classes do not coincide, those representatives span an edge. $\square$

4.3. The complexity.

Definition 4.7. For a vertex v of $\text{IS}(K)$ write $g(v)$ for the genus of any representative, determined by v through (4), and

$$A(v) := \inf\{\text{Area}(F) : F \in v, \partial F \subset J\}, \quad c(v) := (g(v), A(v)) \in \mathbb{N} \times \mathbb{R}_{>0},$$

the latter ordered lexicographically. Put $W := c(\text{IS}(K)^{(0)})$, the set of attained values, with the induced order.

The infimum is attained, by (E) applied to the normal isotopy class of a representative, which is the vertex; so $A(v)$ is a minimum, realised by a stable minimal surface, and both coordinates are invariants of v . It is W , not $\mathbb{N} \times \mathbb{R}_{>0}$, that Definition 2.1 requires to be well-ordered, and the distinction is not cosmetic, $\mathbb{R}_{>0}$ not being well-ordered.

Lemma 4.8. *Let K be a non-trivial knot, with the metric and foliation of §3. Then*

- (i) *for every $h \in \mathbb{N}$ and $a > 0$ the set $\{v \in \text{IS}(K)^{(0)} : g(v) = h, A(v) \leq a\}$ is finite;*
- (ii) *for each h the set $W_h := \{A(v) : g(v) = h\}$ is closed and discrete in $\mathbb{R}_{>0}$, of order type at most ω ;*
- (iii) *W is well-ordered of order type at most ω^2 ; in particular every non-empty subset of W has a least element;*
- (iv) *$\text{IS}(K)$ has countably many vertices.*

Proof. Fix h , let S be the surface of genus h with one boundary circle, and let $\mathcal{M}(S) \supseteq \mathcal{M}_a$ be as in (Fin), whose standing hypotheses hold here as recorded in §3. Let v be a vertex with $g(v) = h$ and $A(v) \leq a$, and Σ_v a minimiser supplied by (E); it is a stable minimal surface with boundary a leaf of J and area at most a , so $\Sigma_v \in \mathcal{M}_a$. By (Fin) the space $\mathcal{M}_a$ is the disjoint union of finitely many open and closed subsets, each consisting of mutually isotopic surfaces. Distinct vertices are distinct ambient isotopy classes, hence distinct proper isotopy classes by isotopy extension, so $v \mapsto$ the subset containing Σ_v is injective on the set in (i), which is therefore finite. For (ii), $W_h \cap (0, a]$ is the image of that finite set, and a subset of $\mathbb{R}_{>0}$ meeting every $(0, a]$ in a finite set is closed and discrete and is order-isomorphic to an initial segment of $\mathbb{N}$. For (iii), W is lexicographically the ordered sum $\sum_{h \in \mathbb{N}} W_h$ of well-orders of type at most ω indexed by $\mathbb{N}$, hence a well-order of type at most ω^2 . For (iv), each set in (i) is finite and $\text{IS}(K)^{(0)} = \bigcup_{h,n \in \mathbb{N}} \{v : g(v) = h, A(v) \leq n\}$. $\square$

4.4. The cyclic-cover frontier construction. The exchange lemma is handed a vertex x and two vertices u, w adjacent to it and at distance 2 from one another. What Lemma 4.3(2) supplies is one surface disjoint from two others which meet each other; the construction below turns that configuration into two new Seifert surfaces. It is stated for surfaces, not for classes, and no minimality is assumed or concluded.

Lemma 4.9 (Frontier construction). *Let S, S_+, S_- be Seifert surfaces in $E(K)$, in particular connected, with*

$$S \cap (S_+ \cup S_-) = \emptyset, \tag{6}$$

with S_+ meeting S_- , the pair S_+, S_- transverse, $\partial S_+ \cap \partial S_- = \emptyset$, and with no disc patch produced by cutting S_+ and S_- along $S_+ \cap S_-$. Then there are Seifert surfaces $P_\uparrow, P_\downarrow$ such that

- (1) *each is disjoint from S, S_+, S_- and from the other;*
- (2) *every compact surface Z disjoint from $S_+ \cup S_-$ is disjoint from some choice of representatives of $P_\uparrow, P_\downarrow$;*
- (3) *writing $T_\uparrow, T_\downarrow$ for the possibly disconnected frontier outputs containing $P_\uparrow, P_\downarrow$,*

$$\chi(T_\uparrow) + \chi(T_\downarrow) = \chi(S_+) + \chi(S_-), \tag{7}$$

and every component discarded from either output is a closed surface of Euler characteristic at most 0.

Proof. Let $p: \tilde{E} \rightarrow E(K)$ be the infinite cyclic cover determined by the linking number homomorphism and τ a generator of its positive deck group. A Seifert surface carries the primitive class dual to p , hence is non-separating, and cutting along S produces the fundamental domain C between a lift S_0 and $S_1 = \tau S_0$, with $p|_C: C \rightarrow E(K) \setminus S$ a homeomorphism. Hypothesis (6) is what puts S_+ and S_- in $E(K) \setminus S$, so the inverse of that homeomorphism lifts both to $S_0^+, S_0^- \subset C$; this is the only use of (6), and it is a use for both surfaces at once. Each $S_0^\pm$ separates $\tilde{E}$ into two parts $M_\uparrow^\pm, M_\downarrow^\pm$, the upper one containing S_1 . Put

$$\tilde{B}_\downarrow = \text{fr}(M_\downarrow^- \cap M_\downarrow^+), \quad \tilde{B}_\uparrow = \text{fr}(M_\uparrow^- \cap M_\uparrow^+). \quad (8)$$

These are assembled from complementary subsurfaces of $S_0^+ \cup S_0^-$ and are piecewise smooth with corners exactly along $\Gamma := S_0^+ \cap S_0^-$, which by transversality and $\partial S_+ \cap \partial S_- = \emptyset$ is a finite family of disjoint smooth closed curves in the interior. Round the corners by Lemma 5.4 and push each output into its own open region at scale σ by Lemma 5.5. The two regions are opposite, so the results are disjoint from the lifted inputs and from each other; they lie in C , on which p is injective, and project to embedded surfaces $T_\uparrow, T_\downarrow$ disjoint from S, S_+, S_- . By Lemma 5.5(3) all small σ give isotopic outputs, so σ labels a representative and not a class. Each frontier is the frontier of a region in an orientable cover, hence two-sided and consistently oriented across the resolved seams, and the algebraic intersection number of a lifted meridian with its boundary is $+1$.

Counting boundary curves in $\mathcal{A} := p^{-1}(\partial E(K))$. The linking-number homomorphism sends the meridian to 1 and kills the longitude, so its restriction to $\pi_1(\partial E(K))$ is onto with kernel generated by the longitude; hence $\mathcal{A}$ is connected and $p|_{\mathcal{A}}$ is the infinite cyclic cover of the torus whose core is a longitude, that is, an open annulus. The two curves ∂S_0^+ and ∂S_0^- are lifts of longitudes, hence core circles of $\mathcal{A}$, and they are disjoint because S_0^+ and S_0^- are. They cut $\mathcal{A}$ into three pieces, and $M_\downarrow^\pm \cap \mathcal{A}$ is the piece of $\mathcal{A} \setminus \partial S_0^\pm$ on the side away from ∂S_1 , so $(M_\downarrow^- \cap M_\downarrow^+) \cap \mathcal{A}$ is the piece lying beyond both circles on that side, and $(M_\uparrow^- \cap M_\uparrow^+) \cap \mathcal{A}$ the piece beyond both on the other. Their frontiers in $\mathcal{A}$ are, respectively, the nearer and the farther of the two circles. Since Γ is closed and contained in the interior, cutting along it creates no boundary; so $\partial \tilde{B}_\uparrow$ and $\partial \tilde{B}_\downarrow$ are the two circles ∂S_0^+ and ∂S_0^- , one each, in the order their positions in $\mathcal{A}$ dictate. Each output therefore has exactly one component with boundary, a connected oriented Seifert surface whose boundary is a longitude; these are $P_\uparrow, P_\downarrow$, and the others are closed. This is (1). For (2), a compact Z disjoint from the compact set $S_+ \cup S_-$ is at positive distance from it, so the push-off may be taken small enough to miss Z , and two sufficiently small push-offs on the same side are isotopic through the collar between them. For (3), the two frontiers exhaust the cut-open $S_0^+ \cup S_0^-$, so Lemma 4.4 gives (7). A discarded component is closed and, being cut from the orientable $S_+ \cup S_-$, orientable, so $\chi \leq 0$ unless it is a sphere. A sphere carrying no exchange seam would be a whole component of one of the inputs with the intersection curves deleted, and no such component is closed, the inputs being connected with non-empty boundary; and on a sphere carrying a seam an innermost seam bounds a seam-free disc, a disc patch, which the hypothesis excludes. $\square$

Lemma 4.9 says nothing about the genus of either output beyond what (7) forces, and nothing more is used below.

5. THE EXCHANGE LEMMA

This section proves axiom (N2) for $\text{IS}(K)$. Given two vertices at distance 2 and a common neighbour, it produces two new vertices, each an apex of the two-interval, whose complexities cannot both fail to drop. The mechanism is Lemma 4.9 applied to relative area minimisers, and

the difficulty is quantitative: rounding the corners of a frontier gains area, pushing the result off the inputs costs area, and the gain must be pinned down before the cost. Throughout, K is a non-trivial knot, and by Theorem A.6 and Lemma A.7 every minimiser supplied by (E) may be taken smooth up to $\partial E(K)$ and is then neatly embedded.

Theorem 5.1 (Incompressible exchange lemma). *Let K be a non-trivial knot and let x, u, w be vertices of $\text{IS}(K)$ with*

$$\text{dist}(x, u) = \text{dist}(x, w) = 1, \quad \text{dist}(u, w) = 2.$$

Then there are vertices $w_\uparrow, w_\downarrow$, fixed once x, u, w are given and in particular before the vertex z of clause (b) is presented, such that

- (a) *each of $w_\uparrow, w_\downarrow$ is equal or adjacent to each of x, u, w ;*
- (b) *every vertex z equal or adjacent to both u and w is equal or adjacent to both $w_\uparrow$ and $w_\downarrow$;*
- (c) *$g(w_\uparrow) + g(w_\downarrow) \leq g(u) + g(w)$;*
- (d) *if (c) is an equality then $A(w_\uparrow) + A(w_\downarrow) < A(u) + A(w)$.*

Clause (a) asserts adjacency pair by pair and not that the five vertices carry simultaneously disjoint representatives; they cannot, since $\text{dist}(u, w) = 2$ forces the two minimisers to meet. In clause (b) the two vertices are those of the existential quantifier above, fixed before z is presented. No canonical assignment $(x, u, w) \mapsto (w_\uparrow, w_\downarrow)$ is claimed: the proof fixes minimisers, a perturbation scheme and a compression sequence, and (E) supplies minimisers without asserting their uniqueness. What is claimed, and what clause (b) needs, is that those choices are made before z enters.

Proposition 5.2 (Apices of a two-interval). *Let K be a non-trivial knot and let u, w be vertices of $\text{IS}(K)$ with $\text{dist}(u, w) = 2$. Then*

- (1) *$\mathcal{I}(u, w)$ has an apex; indeed both outputs of Theorem 5.1, applied to u, w and any common neighbour, are apices;*
- (2) *some apex x of $\mathcal{I}(u, w)$ satisfies $c(x) < \max\{c(u), c(w)\}$;*
- (3) *if $g(u), g(w) \leq \ell$ then the apex in (2) has $g(x) \leq \ell$.*

Consequently $(\text{IS}(K), c)$ is an exchange complex in the sense of Definition 2.1.

Proof. (1) Let $x_0 \in \mathcal{I}(u, w)$, which is non-empty since $\text{dist}(u, w) = 2$, and let y be either output of Theorem 5.1 applied to $(x_0; u, w)$. By (a), y is equal or adjacent to u and to w ; it cannot equal u , since it would then be adjacent to w , contradicting $\text{dist}(u, w) = 2$, and likewise it cannot equal w ; so $y \in \mathcal{I}(u, w)$. By (b), every z equal or adjacent to both u and w — in particular every member of $\mathcal{I}(u, w)$ — is equal or adjacent to y . So y is an apex.

Local descent. Suppose $x \in \mathcal{I}(u, w)$ satisfies $c(x) \geq c(u)$ and $c(x) \geq c(w)$, and let $y_\uparrow, y_\downarrow$ be the outputs of Theorem 5.1 applied to $(x; u, w)$. Lexicographic comparison gives $g(x) \geq g(u)$ and $g(x) \geq g(w)$, so by (c) $g(y_\uparrow) + g(y_\downarrow) \leq g(u) + g(w) \leq 2g(x)$ and the minimum of the two genera is at most $g(x)$. If it is strictly smaller, the corresponding output y has $c(y) < c(x)$ and $g(y) \leq g(x)$. If it equals $g(x)$, then both genera equal $g(x)$, the sum is $2g(x)$, and (c) is an equality with $g(u) = g(w) = g(x)$; then $A(x) \geq A(u)$ and $A(x) \geq A(w)$, so (d) gives $A(y_\uparrow) + A(y_\downarrow) < A(u) + A(w) \leq 2A(x)$ and the output of smaller area has $g(y) = g(x)$, $A(y) < A(x)$, hence again $c(y) < c(x)$ and $g(y) \leq g(x)$.

(2) The set of complexities of apices of $\mathcal{I}(u, w)$ is a non-empty subset of W , so by Lemma 4.8(iii) it has a least element; let x_* be an apex attaining it. Suppose $c(x_*) \geq \max\{c(u), c(w)\}$. An apex is a common neighbour, so local descent applies to $(x_*; u, w)$ and produces an output y with $c(y) < c(x_*)$. By the argument in (1), y is itself an apex of $\mathcal{I}(u, w)$, contradicting the minimality of $c(x_*)$. Hence $c(x_*) < \max\{c(u), c(w)\}$, which is (2).

(3) From $c(x_*) < \max\{c(u), c(w)\}$ and the lexicographic order, $g(x_*) \leq \max\{g(u), g(w)\} \leq \ell$.

For the last sentence: $\text{IS}(K)$ is non-empty and connected by §4.1 and flag by Lemma 4.3(1); W is well-ordered by Lemma 4.8(iii); (N1) is clause (1) and (N2) is clause (2). $\square$

Remark 5.3. Theorem 5.1 produces its two outputs from minimisers chosen one class at a time, and different common neighbours may produce different apices; no map $(u, w) \mapsto x$ is defined anywhere above, and none is needed, because Definition 2.1 asks only for existence. This is the point at which the obstruction identified in [16] is bypassed rather than solved.

The rest of the section proves Theorem 5.1.

5.1. Rounding and pushing off. The frontier of Lemma 4.9 is piecewise smooth with corners, whereas $A(\cdot)$ is an infimum over smooth neatly embedded surfaces. Two steps separate the two, and their order is the pivot of the quantifier structure below. Lemma 5.4 is stated pointwise — varying angle, varying width, gain an integral — because in configuration (T2) the angle degenerates near a tangency point and a single global width would collapse with it. Its proof, and that of Lemma 5.5, are in Appendix B.

Lemma 5.4 (Corner rounding, pointwise form). *Let Γ be a finite disjoint union of compact C^2 curves without boundary in a compact Riemannian 3-manifold, let $\tilde{B}$ be an embedded surface with a corner along Γ , smooth off Γ and with its two sheets smooth up to Γ — every constant below depends on them only through their C^2 bounds — and let $\Gamma \cap \partial\tilde{B} = \emptyset$. Assume that in the normal slice at each $x \in \Gamma$ the trace of $\tilde{B}$ consists of the two edges of a wedge with vertex x and opening angle $\alpha(x) \in (0, \pi)$ — so that α , and with it the slope $m := \cot(\alpha/2)$, is C^1 along Γ — and put*

$$\theta(x) := \min\{\alpha(x), \pi - \alpha(x)\} > 0. \quad (9)$$

Then there are continuous $\eta_{\max}, \kappa: \Gamma \rightarrow (0, \infty)$, a continuous non-decreasing $g_0: (0, \pi/2] \rightarrow (0, \infty)$, and non-decreasing $\underline{\eta}, \underline{\kappa}: (0, \pi/2] \times (0, \infty)^2 \rightarrow (0, \infty)$ determined by the ambient metric, by the C^2 bounds of the two sheets on the normal slice at x , by the local reach

$$\begin{aligned} \varrho(x) &:= \sup\{\varrho > 0 : \exp^\perp \text{ is injective on} \\ &\quad \{(z, \xi) : z \in \Gamma, |\xi| < \varrho, d_\Gamma(z, x) < \varrho\}, \\ &\quad \text{and } B(x, \varrho) \cap \Gamma \text{ is connected}\} \end{aligned} \quad (10)$$

and by $\varrho_\partial(x) := d(x, \partial\tilde{B})$, such that

$$\eta_{\max}(x) \geq \underline{\eta}(\theta(x), \varrho(x), \varrho_\partial(x)), \quad \kappa(x) \geq \underline{\kappa}(\theta(x), \varrho(x), \varrho_\partial(x)), \quad (11)$$

and such that for every C^1 width function η subject to

$$0 < \eta(x) \leq \eta_{\max}(x) \quad \text{and} \quad |\eta'(x)| \leq \kappa(x) \quad (12)$$

there is a compact C^∞ neatly embedded surface $\hat{B} = \hat{B}(\eta)$ with:

- (1) $\hat{B}$ agrees pointwise with $\tilde{B}$ outside the tubular neighbourhood $\{d(\cdot, x) < \eta(x)\}$ of Γ ; in particular $\partial\hat{B} = \partial\tilde{B}$;
- (2) inside that neighbourhood $\hat{B}$ lies on the closed side of $\tilde{B}$ into which the wedge of angle α opens; consequently, if $\tilde{B}$ is the frontier of an open $U \subset E(K)$ and the wedge opens into U at every point of Γ , then $\hat{B} \subset \bar{U}$ and there is a one-sided collar $\hat{B} \times [0, \epsilon_1) \rightarrow E(K)$ into U ;
- (3) $\hat{B}$ is carried to $\tilde{B}$ by an ambient isotopy of the ambient manifold supported in that neighbourhood;

(4) the area gain is non-negative slice by slice, and

$$\text{Area}(\widehat{B}) \leq \text{Area}(\widetilde{B}) - \int_{\Gamma} g_0(\theta(s)) \eta(s) ds; \quad (13)$$

(5) (The bounded region is unchanged.) Write $T := \{d(\cdot, x) < \eta(x)\}$ for the tubular neighbourhood of (1). If $\widetilde{B}$ is a closed surface separating the ambient manifold, then so is $\widehat{B}$, and the complementary regions correspond under the ambient isotopy of (3): if R is a complementary region of $\widetilde{B}$ and $\widehat{R}$ the corresponding one of $\widehat{B}$, then $\widehat{R} \setminus T = R \setminus T$.

In particular the integral in (4) may be restricted to any measurable subset of Γ and the rest discarded.

Lemma 5.5 (Pushing off). *Let $\widehat{B}$ be a compact C^∞ neatly embedded surface satisfying clause (2) of Lemma 5.4 for an open region U , with $\partial\widehat{B}$ a leaf of the longitudinal foliation. Then there are $\sigma_0 > 0$ and $C < \infty$, depending only on $\widehat{B}$ and on the ambient metric near it, such that for every $\sigma \in (0, \sigma_0]$ there is a compact C^∞ neatly embedded $T(\sigma)$ with $T(\sigma) \subset U$; $\partial T(\sigma)$ a leaf; $T(\sigma)$ isotopic to $\widehat{B}$ and any two of them isotopic to each other; and $|\text{Area}(T(\sigma)) - \text{Area}(\widehat{B})| \leq C\sigma$.*

The order of the two steps is forced: a normal push-off of a cornered surface is not embedded, while the frontier of an inner parallel set is again cornered. Hence C is determined by $\widehat{B}$, that is, by the cornered frontier and the once-chosen width η : the rounding gain is fixed before σ is.

The next lemma excludes a disc in the symmetric difference. It carries an error term because in configurations (T1) and (T2) one of the two surfaces is a perturbation of a minimiser and not a minimiser.

Lemma 5.6 (No disc patch). *Let A_0, B be relative area minimisers in two distinct vertices of $\text{IS}(K)$, let A be neatly embedded, incompressible, normally isotopic to A_0 with ∂A a leaf of J , and put $e := \text{Area}(A) - \text{Area}(A_0) \geq 0$. Suppose A and B are transverse with $\partial A \cap \partial B = \emptyset$, and suppose that cutting A and B along $\Gamma := A \cap B$ produces a disc patch. For a closed curve $\gamma \subset \Gamma$ and a width function admissible along it, let $\mathcal{G}(\gamma)$ denote the rounding gain (13) along γ ; by (9) it is one and the same number for either of the two pairs of opposite sectors at γ . Then there are closed curves $a, b \subset \Gamma$, not necessarily distinct, with*

$$\mathcal{G}(a) + \mathcal{G}(b) \leq e. \quad (14)$$

The lemma is applied with a width function chosen for a *different* surface: in configurations (T1) and (T2) the gain along γ is computed for the cornered frontiers of Lemma 4.9, whereas the proof below rounds the cornered surfaces produced by a disc exchange. A single η serves both. By Lemma 5.4 the bounds $\eta_{\max}$ and κ at $x \in \gamma$ depend only on: the opening angle $\theta(x)$, which by (9) is the same for all four sectors at x and hence for both surfaces; the local reach (10) of γ , which depends on γ alone; the C^2 bounds of the two sheets meeting along γ , which in both cases are those of A and of B , the exchange surface differing from one of them by a push-off whose scale we let tend to 0 and in which all the estimates are continuous; and $\varrho_{\partial}(x)$. For the last, every surface occurring here — the two frontiers, A , B , and the exchanged surfaces — has its boundary on $\partial E(K)$, so $\varrho_{\partial}(x) \geq d(x, \partial E(K))$ in every case, and replacing ϱ_{∂} by $d(\cdot, \partial E(K))$ only decreases $\eta_{\max}$; we do so throughout. With that reading a width admissible along γ for one of these surfaces is admissible for all of them, and $\mathcal{G}(\gamma)$ is unambiguous.

Proof. No patch has a boundary arc on its frontier, since the endpoints of such an arc would lie in $\partial A \cap \partial B = \emptyset$; so Γ is a finite disjoint union of closed curves in $\text{Int } E(K)$ and the frontier of a disc patch is a single one of them.

Step 0: the discs $D_A(\gamma)$. A simple closed curve γ in the interior of A bounds at most one disc in A : the two sides of a separating γ cannot both be discs, or A would be a sphere, and the disc side is the one missing ∂A . Write $D_A(\gamma)$ for that disc when it exists and A_γ for the closure of $A \setminus D_A(\gamma)$; since A is connected, a component of A_γ missing γ would be a component of A , so A_γ and $A_\gamma \setminus \gamma$ are connected. Use the same notation for B . Let $\mathcal{D} \subset \Gamma$ be the set of curves bounding a disc in A or in B . For $\gamma \in \mathcal{D}$ both discs exist: γ bounds a disc, hence is null-homotopic in $E(K)$; a two-sided incompressible surface is π_1 -injective, so γ is null-homotopic in the other surface as well; and a null-homotopic simple closed curve on a surface bounds a disc there. A disc patch is $D_A(\gamma)$ or $D_B(\gamma)$ for γ its frontier, by the uniqueness just proved, so $\mathcal{D} \neq \emptyset$. Since $D_A(\gamma) \subset A$ and $A \cap B = \Gamma$, the sets

$$\mathcal{A} := \{\gamma \in \mathcal{D} : \text{Int } D_A(\gamma) \cap B = \emptyset\}, \quad \mathcal{B} := \{\gamma \in \mathcal{D} : \text{Int } D_B(\gamma) \cap A = \emptyset\}$$

are the sets of curves whose disc on the respective side is a disc patch.

Step 1: $\mathcal{A}$ and $\mathcal{B}$ are non-empty. Γ is finite, hence so is $\mathcal{D}$. Let $\gamma \in \mathcal{D}$ have $D_A(\gamma)$ minimal under inclusion among $\{D_A(\gamma') : \gamma' \in \mathcal{D}\}$. If a curve γ' of Γ met $\text{Int } D_A(\gamma)$ it would lie in it, and would cut off a subdisc of $D_A(\gamma)$, which is $D_A(\gamma')$ by Step 0; then $D_A(\gamma') \subsetneq D_A(\gamma)$ contradicts minimality. So $\gamma \in \mathcal{A}$, and symmetrically $\mathcal{B} \neq \emptyset$.

Step 2: one-sided exchange. Let $\gamma \in \mathcal{A}$ and write $D := D_A(\gamma)$, $D' := D_B(\gamma)$. Then $\text{Int } D \cap B = \emptyset$ forces $D \cap B = \gamma$, so $D \cup D'$ is an embedded sphere, and it lies in $\text{Int } E(K)$ because $A \cap \partial E(K) = \partial A$ is not met by D , and likewise for D' . It has a corner along γ ; round it by the qualitative part of Lemma 5.4, at a width small enough that the tube T about γ misses $\partial E(K)$. By irreducibility the resulting smooth sphere bounds a ball Q , and by clause (5) of that lemma Q agrees off T with the region bounded by $D \cup D'$; moreover $Q \cap \partial E(K) = \emptyset$, since a point of $\partial E(K)$ in Q would have to lie on $\partial Q \subset \text{Int } E(K)$. Now

$$B \cap \text{Int } Q = \emptyset. \tag{15}$$

Indeed, $B_\gamma \setminus \gamma$ is connected by Step 0 and is disjoint from $D \cup D'$ — from D' because $B_\gamma \cap D' = \gamma$, and from D because $D \cap B = \gamma$. It is therefore disjoint from ∂Q : outside T the two coincide, and inside T the sphere ∂Q lies, by clause (2) of Lemma 5.4, in the closed sector of the normal slice bounded by D and D' , meeting the boundary of that sector only in $D \cup D'$, whereas the two sheets of B at γ bound the two *other* sectors and so meet that closed sector only along γ , which is not on ∂Q . Hence $B_\gamma \setminus \gamma$ lies in a single complementary region of ∂Q , and that region is not Q because it contains ∂B , which lies on $\partial E(K)$ while Q does not meet $\partial E(K)$. As for D' : outside T it lies on ∂Q , and inside T it lies on the boundary of the sector while ∂Q lies in the sector's interior, so it is disjoint from $\text{Int } Q$ as well. Since $B = D' \cup B_\gamma$, this is (15).

By (15), pushing D' across Q to a parallel copy of D is an ambient isotopy of $E(K)$, supported in a neighbourhood of Q ; this is the only point at which the position of the second surface matters. The isotopy carries B to a neatly embedded B' with $\partial B' = \partial B$ and with $\text{Area}(B') \rightarrow \text{Area}(B) - \text{Area}(D') + \text{Area}(D)$ as the push-off tends to 0. Along the corner circle B' has a genuine corner, since $A \pitchfork B$; rounding by Lemma 5.4 gives a smooth neatly embedded B'' , isotopic to B , with boundary still the same leaf and $\text{Area}(B'') \leq \text{Area}(B) - \text{Area}(D') + \text{Area}(D) - \mathcal{G}(\gamma)$. As B minimises in its class, $\text{Area}(B'') \geq \text{Area}(B)$, whence

$$\text{Area}(D_B(\gamma)) + \mathcal{G}(\gamma) \leq \text{Area}(D_A(\gamma)), \quad \gamma \in \mathcal{A}. \tag{16}$$

The argument used only that the disc in the *first* surface is a patch, so exchanging the roles of A and B it applies to $\gamma \in \mathcal{B}$ and produces a smooth neatly embedded A'' , isotopic to A and hence to A_0 , with the same boundary leaf and $\text{Area}(A'') \leq \text{Area}(A) - \text{Area}(D_A(\gamma)) + \text{Area}(D_B(\gamma)) - \mathcal{G}(\gamma)$. Here $\text{Area}(A'') \geq \text{Area}(A_0) = \text{Area}(A) - e$, whence

$$\text{Area}(D_A(\gamma)) + \mathcal{G}(\gamma) \leq \text{Area}(D_B(\gamma)) + e, \quad \gamma \in \mathcal{B}. \tag{17}$$

Step 3: the two curves. Put $\varphi(\gamma) := \text{Area}(D_A(\gamma)) - \text{Area}(D_B(\gamma))$ on $\mathcal{D}$, so that (16) reads $\varphi \geq \mathcal{G}$ on $\mathcal{A}$ and (17) reads $\varphi \leq e - \mathcal{G}$ on $\mathcal{B}$. Choose $a \in \mathcal{A}$ with $\text{Area}(D_A(a))$ least and $b \in \mathcal{B}$ with $\text{Area}(D_B(b))$ least. Then $\text{Area}(D_A(a)) \leq \text{Area}(D_A(b))$: if $b \in \mathcal{A}$ this is the choice of a , and otherwise $\text{Int } D_A(b)$ meets Γ , and choosing among the curves of Γ inside it one, c , whose disc $D_A(c)$ — a subdisc of $D_A(b)$, by Step 0 — is minimal under inclusion, the argument of Step 1 gives $c \in \mathcal{A}$ and hence $\text{Area}(D_A(a)) \leq \text{Area}(D_A(c)) < \text{Area}(D_A(b))$. Symmetrically $\text{Area}(D_B(b)) \leq \text{Area}(D_B(a))$. Adding the two, $\varphi(a) \leq \varphi(b)$, so $\mathcal{G}(a) \leq \varphi(a) \leq \varphi(b) \leq e - \mathcal{G}(b)$, which is (14). $\square$

The exchange is performed one surface at a time because (15) is available only on that side: a simultaneous replacement of both discs would require $D_A(\gamma)$ and $D_B(\gamma)$ to be patches for one and the same γ , which no innermost choice supplies. What replaces it is the pair (16), (17) at two possibly distinct curves, glued by the monotonicity of φ . The proof uses only that A_0 and B separately minimise, not that A does.

5.2. Proof in the transverse configuration.

Proof of Theorem 5.1 when the minimisers are transverse. By (E) fix, one class at a time, relative area minimisers $S \in x$, $S_+ \in u$, $S_- \in w$; by Lemma 4.3(2) the single surface S is disjoint from both others, and $S_+ \cap S_- \neq \emptyset$ since $\text{dist}(u, w) = 2$. Assume in this proof that S_+ and S_- are transverse with $\partial S_+ \cap \partial S_- = \emptyset$.

Write $\Gamma := S_+ \cap S_-$, which by neatness and $\partial S_+ \cap \partial S_- = \emptyset$ is a finite family of disjoint smooth closed curves in the interior, of total length $\ell_0 > 0$. In the normal slice at $y \in \Gamma$ the two tangent planes cut the plane into four sectors of angles $\alpha, \pi - \alpha, \alpha, \pi - \alpha$ with $\alpha(y) \in (0, \pi)$, and transversality together with the compactness of Γ makes

$$\theta_0 := \min_{\Gamma} \min\{\alpha, \pi - \alpha\} > 0. \quad (18)$$

Choose a constant width $\eta \leq \min_{\Gamma} \eta_{\max}$ once and for all: the minimum is positive because Γ is compact and $\theta \geq \theta_0 > 0$ on it, and a constant width satisfies the constraint on η' in (12) vacuously. With this width the gain (13) along a closed curve $\gamma \subset \Gamma$ is at least $g_0(\theta_0)\eta \text{length}(\gamma) > 0$, uniformly in γ . Now S_- is itself a relative area minimiser, so Lemma 5.6 applies with $e = 0$: were there a disc patch it would supply closed curves a, b with $\mathcal{G}(a) + \mathcal{G}(b) \leq 0$, whereas both gains have just been bounded below by a positive number. So cutting S_+ and S_- along Γ produces no disc patch, Lemma 4.9 applies, and it yields $P_{\uparrow}, P_{\downarrow}$.

By (7) and $\chi \leq 0$ for the discarded closed components, $\chi(P_{\uparrow}) + \chi(P_{\downarrow}) \geq \chi(S_+) + \chi(S_-)$; substituting (4) gives

$$g(P_{\uparrow}) + g(P_{\downarrow}) \leq g(u) + g(w). \quad (19)$$

Compress each P along essential circles, always keeping the component with the longitudinal boundary, until it is incompressible; by Lemma 4.4 the genus does not increase and the process terminates. Call the resulting vertices $w_{\uparrow}, w_{\downarrow}$; with (19) this is (c). Each P is disjoint from each of S, S_+, S_- , so Corollary 4.6, applied separately to the three, gives (a). For (b): if $z \notin \{u, w\}$ is equal or adjacent to both, take a minimiser $Z \in z$; by Theorem 3.1(i) it is disjoint from S_+ and from S_- , so by clause (2) of Lemma 4.9 the coarse outputs have representatives disjoint from Z , and Corollary 4.6 applies again. That the push-off scale depends on Z is harmless: all small push-offs represent the same coarse classes and the chosen compression sequence is transported along the isotopy between them, so the output *vertices* do not move. If $z \in \{u, w\}$, (a) already gives the conclusion.

Suppose now (c) is an equality. Then (19) and every compression inequality is an equality, so no compression occurred and each P is already incompressible and represents the corresponding

w . The two frontiers (8) pick out, in the normal slice at $y \in \Gamma$, a pair of opposite sectors, so both have the same opening angle $\alpha(y)$ and θ_0 of (18) bounds the angle for both. The two cornered frontiers partition $S_+ \cup S_-$ up to the measure-zero set Γ , so

$$\text{Area}(\tilde{B}_\uparrow) + \text{Area}(\tilde{B}_\downarrow) = \text{Area}(S_+) + \text{Area}(S_-). \quad (20)$$

Round both frontiers with the width η already fixed; by (13) the total gain is at least $2\varepsilon_0$ with $\varepsilon_0 := g_0(\theta_0) \eta \ell_0 > 0$, which depends on $S_\pm$ and on this one choice and on nothing later. Apply Lemma 5.5 to the two rounded surfaces separately, take σ_0 to be the smaller of the two thresholds and C the sum of the two constants, so that the total cost of the two push-offs is at most $C\sigma$; both are fixed before σ . Choose $\sigma \in (0, \sigma_0]$ with $C\sigma < \varepsilon_0$. Combining, and discarding the closed components, which have positive area,

$$\begin{aligned} \text{Area}(P_\uparrow) + \text{Area}(P_\downarrow) &\leq \text{Area}(S_+) + \text{Area}(S_-) - 2\varepsilon_0 + C\sigma \\ &< A(u) + A(w) - \varepsilon_0. \end{aligned}$$

These are smooth neatly embedded competitors with boundary a leaf of J , so $A(w_\uparrow) + A(w_\downarrow)$ is at most the left-hand side, which is (d). $\square$

5.3. The non-transverse configurations. The proof just given assumed two things, and both may fail. Both ∂S_+ and ∂S_- are leaves of the foliation J , and two leaves of a foliation are equal or disjoint. So either $\partial S_+ \cap \partial S_- = \emptyset$ or $\partial S_+ = \partial S_-$, and in each case the two surfaces may or may not be transverse in the interior. Three configurations therefore remain to be treated beyond the one just settled:

- (T0) $\partial S_+ \cap \partial S_- = \emptyset$ and $S_+ \pitchfork S_-$; this is the case proved above;
- (T1) $\partial S_+ \cap \partial S_- = \emptyset$, with S_+ and S_- tangent at one or more interior points;
- (T2) $\partial S_+ = \partial S_-$, so that the two surfaces meet along a whole curve of common boundary and the argument above cannot even start.

The surfaces of (T2) need not be tangent along the shared leaf, and in general are not: by (25) below they are transverse at all but finitely many of its points. What fails in (T2) is not transversality but the disjointness of the boundaries, on which the cutting and pasting of §5.2 depends throughout. In (T1) it is transversality that fails: two distinct minimal surfaces with disjoint boundaries may touch at an interior point. Both configurations are disposed of by perturbing S_- to a transverse competitor and running the transverse argument on the perturbation; what has to be controlled in each case is that the area lost to the perturbation stays below the area gained by rounding, and it is Lemma 5.6, with its error term e , that makes the two comparable. The structure of the tangency set is the following.

Throughout, write $A := S_-$ and $B := S_+$, keeping S fixed and disjoint from both.

The interior picture. Let p be a point of $\text{Int } A \cap \text{Int } B$ at which $T_p A = T_p B$. In normal coordinates (x_1, x_2, y) at p with the common tangent plane $\{y = 0\}$, both surfaces are graphs over a disc D about the origin, say $B = \{y = \beta\}$ and $A = \{y = \hat{f}\}$ with $\beta(0) = \hat{f}(0) = 0$ and $D\beta(0) = D\hat{f}(0) = 0$; both are smooth by Theorem A.6. Both graphs are minimal, so subtracting the two minimal graph equations and applying the mean value theorem along the segment joining the two gradients gives, classically on D ,

$$L_p f_p := \partial_i (a^{ij} \partial_j f_p) + b^i \partial_i f_p + c f_p = 0, \quad f_p := \hat{f} - \beta, \quad (21)$$

with a^{ij} smooth and uniformly elliptic and b^i, c smooth. The tangency points of A and B in D are exactly the points of $\{f_p = 0, df_p = 0\}$.

The collar picture, in configuration (T2) only. Suppose the two boundary leaves coincide along a longitude L . Take a vector field on $\partial E(K)$ transverse to J whose flow carries leaves

to leaves, extend it into the collar, and use its direction as a coordinate y , the inward distance $r \geq 0$, and arclength s along L . The resulting chart (s, r, y) is bi-Lipschitz onto its image with a constant $C'_{\text{met}} < \infty$ depending only on the ambient metric near L and on the chosen vector field. By Lemma A.7 both surfaces are neat, so the extension may be taken transverse along L to both tangent planes, and both are graphs over one half-cylinder:

$$\partial E(K) = \{r = 0\}, \quad B = \{y = 0\}, \quad A = \{y = f(s, r)\}, \quad f(s, 0) = 0. \quad (22)$$

By Theorem A.6 the function f is smooth up to $\{r = 0\}$. Both graphs are minimal on $\{r > 0\}$, so subtracting the two minimal graph equations and applying the mean value theorem along the segment joining the two gradients gives, classically on the closed half-disc,

$$Lf := \partial_i(a^{ij}\partial_j f) + b^i\partial_i f + cf = 0, \quad (23)$$

with a^{ij} smooth and uniformly elliptic and b^i, c smooth, all up to $\{r = 0\}$.

Proposition 5.7 (Structure of the tangency set). *Let A and B be as above: distinct relative area minimisers in two vertices at distance 2, meeting each other. Then:*

- (1) (Interior; no hypothesis on the boundaries.) *Write $\mathcal{T} \subset \text{Int } A \cap \text{Int } B$ for the set of interior tangency points. Every $p \in \mathcal{T}$ is isolated in $\text{Int } A \cap \text{Int } B$ and has an integer order $m_p \geq 1$ and constants $c_1, c_2 > 0$ with $|df_p| \geq c_1\rho^{m_p} - c_2\rho^{m_p+1} > 0$ at distance ρ from p on a punctured neighbourhood, f_p being the local difference of (21); so p is the only critical point of f_p in a disc D_p . Moreover*

$$\text{for each } p \in \mathcal{T}, \quad f_p \text{ takes both positive and negative values in every neighbourhood of } p. \quad (24)$$

- (2) (Collar; in configuration (T2).) *If $\partial A = \partial B = L$, then with f as in (22) there is $r_0 > 0$ with $\mathcal{T} \subset \{r \geq r_0\}$ and*

$$\begin{aligned} &\text{in the open collar } \{0 < r < r_0\} \text{ one has } df \neq 0 \text{ everywhere,} \\ &\text{while the critical points of } f \text{ on } \{r = 0\} \text{ are finite in} \\ &\text{number and all lie on } \{f = 0\}. \end{aligned} \quad (25)$$

- (3) *In both configurations $\mathcal{T}$ is finite; put $N := \#\mathcal{T}$.*

We refer to (25) and (24) as the collar conditions, though strictly (25) concerns the collar and (24) the interior tangency set. In configuration (T2) they are used together; in configuration (T1) only (24) is available, and only it is needed, the whole intersection being then at a positive distance from $\partial E(K)$.

The bounded case of the exchange-and-roundoff argument is developed in [3, §7] under an existence statement which the authors state they have not proved, and in any case for boundaries held fixed; neither covers a common boundary leaf, nor a competitor that is not itself minimal. Proposition 5.7 and the two configurations (T1), (T2) are therefore proved here, in Appendix B; the three configurations (T0), (T1), (T2) exhaust the possibilities, so this completes the proof of Theorem 5.1.

6. PROOFS OF THEOREMS A, C AND D

6.1. Theorem A.

Proof of Theorem A. Let K be a non-trivial knot and let $c(v) = (g(v), A(v))$ be the complexity of Definition 4.7, with values in the set W of attained values ordered lexicographically. We check Definition 2.1 clause by clause. The complex $\text{IS}(K)$ is non-empty and connected by §4.1,

the connectedness being [10, Thm. A] in its incompressible case, and its edge-path metric is Kakimizu's distance [10, Prop. 3.1(1)]. It is flag by Lemma 4.3(1), which uses (U)(i). The set W is well-ordered by Lemma 4.8(iii), which uses (E) and (Fin). Axiom (N1) is Proposition 5.2(1) and axiom (N2) is Proposition 5.2(2); both rest on Theorem 5.1, whose proof uses (U), (E), (R1) — that is, Theorem A.6 — and, in the non-transverse configurations, Proposition 5.7. So $(\text{IS}(K), c)$ is an exchange complex, and Theorem B gives the result. $\square$

6.2. Theorem C.

Proof of Theorem C. Fix $\ell \geq g(K)$ and put $D_\ell = \{v : g(v) \leq \ell\}$. The order is lexicographic with the genus in the first coordinate, so $c(u) \leq c(v)$ and $g(v) \leq \ell$ force $g(u) \leq \ell$; thus D_ℓ is downward closed, and it is non-empty because $\ell \geq g(K)$. By Theorem A and Lemma 2.2 we may take the complexity injective; D_ℓ remains downward closed for the refined complexity, which orders each fibre of c internally and so satisfies $c'(u) < c'(v) \Rightarrow c(u) \leq c(v)$, while D_ℓ is a union of fibres of c . Then Corollary 2.3 shows that the full subcomplex $\text{IS}_\ell(K)$ on D_ℓ is isometrically embedded in $\text{IS}(K)$ and connected, while Lemma 2.4(1), applied to the downward closed set D_ℓ , shows that it is an exchange complex. Theorem B gives contractibility.

For the second family fix in addition $a_0 > 0$ and put $D_{\ell, a_0} = \{v : g(v) < \ell\} \cup \{v : g(v) = \ell, A(v) \leq a_0\}$, again a union of fibres of c and downward closed, and non-empty as soon as $\ell > g(K)$, or $\ell = g(K)$ and a_0 is at least the least relative area in genus $g(K)$. The same two citations apply verbatim. $\square$

Taking $\ell = g(K)$ recovers the case $E = E(K)$ of [16, Thm. 1.1], which is stated there for the complex of spanning surfaces of minimal Thurston norm; that case is not claimed here.

6.3. Theorem D. Kakimizu's complexes are defined for links, and the obstacle to extending the above is not the combinatorics but the definition of a vertex. Throughout, $L = L_1 \cup \dots \cup L_n$ is a link in S^3 , $E(L)$ its exterior, and a *spanning surface* is a properly embedded orientable surface in $E(L)$ whose boundary is the union of the n longitudes.

One point of vocabulary has to be settled first, because for $n \geq 2$ the boundary slope of a spanning surface is not the preferred longitude. The intersection $\lambda_i := \Sigma \cap \partial N(L_i)$ of a spanning surface with the i -th boundary torus is the *surface-framed longitude*: since $\Sigma \subset E(L)$ misses L_i , its boundary bounds in the complement of L_i , so $\text{lk}(\partial \Sigma, L_i) = 0$ and therefore

$$\text{lk}(\lambda_i, L_i) = - \sum_{j \neq i} \text{lk}(L_j, L_i), \quad (26)$$

a slope depending only on L ; the foliation J_i of the i -th torus used in §3 is taken by curves of this slope. The total linking number defines a homomorphism $\varphi : H_1(E(L)) \rightarrow \mathbb{Z}$ sending every meridian to 1, and, $[c] = \sum_j \text{lk}(c, L_j) [m_j]$ for a curve c in $E(L)$, equation (26) gives $\varphi(\lambda_i) = \text{lk}(\lambda_i, L_i) + \sum_{j \neq i} \text{lk}(L_i, L_j) = 0$. So each λ_i lifts to a closed curve in the infinite cyclic cover determined by φ , and the lifting argument of Lemma 4.9 applies verbatim.

Lemma 6.1. *Let L be linking-indecomposable. Then every spanning surface for L in $E(L)$ is connected and has exactly n boundary curves; in particular so are the surfaces produced by the frontier construction of Lemma 4.9.*

Proof. Suppose F is a spanning surface with a decomposition $F = F' \sqcup F''$ into non-empty unions of components. A spanning surface has no closed component, so every component carries at least one longitude; let I be the set of indices whose longitude lies in $\partial F'$ and I^c the set for F'' . These are non-empty and, the components being disjoint, partition $\{1, \dots, n\}$.

Write $L_I = \bigcup_{i \in I} L_i$. The chain F' lies in $E(L)$, hence misses L , and $\partial F'$ is the union of the λ_i , $i \in I$, each isotopic to L_i inside $N(L_i)$ by an isotopy disjoint from every other component. Therefore $[L_I] = [\partial F'] = 0$ in $H_1(S^3 \setminus L_{I^c})$. For $j \in I^c$ the linking number $\text{lk}(L_I, L_j)$ is the image of $[L_I]$ under $H_1(S^3 \setminus L_j) \cong \mathbb{Z}$, and the inclusion $S^3 \setminus L_{I^c} \hookrightarrow S^3 \setminus L_j$ carries the vanishing class to it; so $\text{lk}(L_j, L_I) = 0$ for every $j \in I^c$. The same argument applied to F'' gives $\text{lk}(L_m, L_{I^c}) = 0$ for every $m \in I$. Since I and I^c are the two parts of a partition into non-empty sets, this says $\text{lk}(L_m, L_J) = 0$ for every index m , with J the part not containing m , contradicting linking-indecomposability. Hence F is connected, and its boundary consists of the n longitudes. The frontier construction outputs spanning surfaces for L , so the same applies to them. $\square$

Lemma 6.2. *Let L be linking-indecomposable and let F be a connected spanning surface for L . Compressing F along an essential circle that separates F produces two disjoint surfaces of which exactly one carries boundary, and that one carries all n boundary curves; its genus strictly decreases. Compression along a non-separating circle keeps the surface connected and again lowers the genus. So “keep the component with boundary” is unambiguous, and iterated compression terminates.*

Proof. If both sides carried boundary curves, their union would be a disconnected spanning surface for L — properly embedded, orientable, with boundary the union of the n longitudes, compression having altered F only in the interior — contradicting Lemma 6.1. So exactly one side F_1 carries boundary, and it carries all n curves; F_2 is closed and is discarded. The compressing circle being essential, F_2 is not a sphere, so $\chi(F_2) \leq 0$ and $\chi(F_1) = \chi(F) + 2 - \chi(F_2) \geq \chi(F) + 2$; with n fixed and $\chi = 2 - 2g - n$ the genus drops by at least one. The non-separating case is that of Lemma 4.4. $\square$

Kakimizu’s vertices are isotopy classes of spanning surfaces allowed to be disconnected and required only to be incompressible componentwise, whereas ours are connected; on a linking-indecomposable link Lemma 6.1 makes the two notions coincide, and with them the two edge conventions, the convention of [10, 1.3] that disjointness is equivalent to distance 1 being false for links only because of disconnected surfaces.

Proof of Theorem D. Let L be non-split, linking-indecomposable and not the unknot. By Lemma 6.1 the vertices of $\text{IS}(L)$ are the isotopy classes of connected incompressible spanning surfaces. Connectedness of $\text{IS}(L)$ is [10, Thm. A] for a non-split link, which is where non-splitness is used, through the irreducibility of $E(L)$; flagness is Lemma 4.3, whose proof uses only (U), (E) and the connectedness of the surfaces, now supplied by Lemma 6.1. The complexity is again (g, A) , with g the genus of the connected spanning surface.

The geometric inputs are available in n -component form: (U) and (E) are stated in [17] with the constraint $\partial F \subset J$ and with ∂M allowed to be a union of tori, the convexity hypothesis being imposed on each boundary component separately [17, Thm. 4 = Thm. A.5 and Thm. 2 = Cor. A.4]. Both rest on an appendix which assumes the surface connected and meeting each boundary torus in a single curve, and both hypotheses hold here, the first by Lemma 6.1 and the second because a connected spanning surface for an n -component link meets each of the n tori in exactly one curve. (R1) transfers in the n -component form recorded after Theorem A.6. The compression step of Lemma 4.4 is replaced by Lemma 6.2, and the connectedness hypothesis of Lemma 4.9 by Lemma 6.1, which also gives that each frontier output has exactly one component with boundary and that this component carries all n longitudes. Two further modifications are needed.

The exchange lemma on n boundary tori. Everything in §5.3 and in Appendix B is local to a single boundary torus apart from finitely many global constants. On each torus T_i the curves

$\partial_i A$ and $\partial_i B$ are leaves of J_i , hence equal or disjoint; let $E \subseteq \{1, \dots, n\}$ be the set of indices at which they are equal. The trichotomy of §5.3 becomes: $E = \emptyset$ and $A \pitchfork B$, which is (T0); $E = \emptyset$ with interior tangencies, which is (T1) and needs no change, its proof never mentioning the boundary except through (45); and $E \neq \emptyset$, which is (T2). In the last case Proposition 5.7(1) is unchanged, being local at an interior tangency point; clause (2) is applied on each T_i with $i \in E$ separately, in the collar chart (s, r_i, y) of that torus, the collars taken disjoint, and yields $r_0^{(i)} > 0$ with $\mathcal{T} \subseteq \bigcap_{i \in E} \{r_i \geq r_0^{(i)}\}$; clause (3) then gives $\mathcal{T}$ finite. Lemma B.1 gives on each such torus a sign ς_i , an arc $I_i \subset L_i$ and constants $a_0^{(i)}, r_1^{(i)}$, and the perturbation (39) becomes

$$A_t = \left\{ y = f + t \sum_{i \in E} \varsigma_i \chi_i(r_i) - t \sum_{p \in \mathcal{T}} \psi_p \right\},$$

with χ_i supported in the collar of T_i ; on a torus with $i \notin E$ nothing is done, $\partial_i A_t = \partial_i A$ being already disjoint from $\partial_i B$. The sign of the interior bumps is immaterial, (24) being a statement about both signs at once, so the boundary signs may be chosen independently torus by torus. Lemma B.2 and Lemma B.3(i),(ii) are proved one collar at a time. In Lemma B.3(iii) a closed component γ of Γ_t may visit several collars, so the three cases are read per torus: if γ meets no set $\{r_i < r_1^{(i)}\}$, Case A gives $\text{length}(\gamma \cap G) \geq 3\rho_0/8$; otherwise fix an i with $\gamma \cap \{r_i < r_1^{(i)}\} \neq \emptyset$, and, the collars being disjoint, either γ stays inside the collar of T_i , which is Case C for that torus, or it leaves it and then $\max_\gamma r_i > r_0^{(i)}/2$, which is Case B. Both use only the behaviour of γ inside that one collar, so $c_\star$ is the minimum of $3\rho_0/8$, of the numbers $r_0^{(i)}/2$ and of the arclengths $\ell(I_i)$, $i \in E$, divided by the largest of the n bi-Lipschitz constants of the collar charts. Corollary B.4 is unchanged, its constants being minima over finitely many tori, and the ramp (44) acquires one collar term for each $i \in E$, the entry $d(x, \{r \leq r_1/2\})$ being replaced by $\min_{i \in E} d(x, \{r_i \leq r_1^{(i)}/2\})$. With these readings (42) and the proof of Theorem 5.1 go through word for word. Lemma 5.6 needs no change: of A and B its proof uses only that each is connected, that each has non-empty boundary on $\partial E(L)$, and that $E(L)$ is irreducible. In Lemma 4.9 the boundary count is made torus by torus: over T_i the lifted boundary is an annulus containing the two disjoint circles $\partial_i S_0^+$ and $\partial_i S_0^-$, one above the other, and the argument given there sends them to opposite frontiers; which frontier receives $\partial_i S_0^+$ may depend on i , and either way each frontier meets each T_i in exactly one curve.

Primitivity. Through Lemma 4.9 the knot argument invokes the assertion that a Seifert surface represents a primitive generator of $H_2(E(K), \partial E(K))$. For a link $H_2(E(L), \partial E(L)) \cong \mathbb{Z}^n$ and the assertion is false as stated; it is replaced by what the argument uses, namely that the total linking number defines a homomorphism $\pi_1(E(L)) \rightarrow \mathbb{Z}$ and that a spanning surface for L is Poincaré dual to the image of the generator under the induced map on H^1 , a primitive class because the homomorphism is onto, each meridian mapping to 1.

Axioms (N1) and (N2) are then obtained exactly as in the proof of Theorem A, and Theorem B gives the result. $\square$

APPENDIX A. BOUNDARY REGULARITY OF RELATIVE AREA MINIMISERS

This appendix proves the two regularity statements used in §5: (R1), that a relative area minimiser is smooth up to $\partial E(K)$ and meets it exactly in its own boundary (Theorem A.6); and (R2), that it meets $\partial E(K)$ transversally (Lemma A.7). They are proved from (E)(i) together with (Fin), and from no other result of [6] or [17]. Throughout, M is a compact orientable P^2 -irreducible Riemannian 3-manifold carrying the collar data (2), and $\lambda \subset \partial M$ is a smooth closed 1-manifold.

A.1 Piecewise smooth surfaces.

Definition A.1. A compact surface F properly embedded in a Riemannian 3-manifold X is *piecewise smooth* if there is a finite graph $\Gamma_F \subset F$, embedded as a finite union of smooth arcs and circles and *properly* — so that $\Gamma_F \cap \partial F$ is the finite set of endpoints of its arcs, and no arc runs along ∂F — such that $F \setminus \Gamma_F$ is a smooth submanifold of X and the closure of each of its components, its *pieces*, is a compact surface with corners, smooth in its interior and embedded of class C^1 up to its own boundary, the two edges at each corner having distinct tangent rays. A smooth surface is piecewise smooth, with $\Gamma_F = \emptyset$. Such an F admits a smooth triangulation compatible with Γ_F , and *isotopy* of piecewise smooth surfaces is understood in the piecewise smooth category, in which Alexander’s theorem and the Alexander trick are available.

The term is Hass and Scott’s: the class $\mathcal{F}$ of (E)(i) is, in their words, “the set of piecewise smooth surfaces in M which are isotopic to F rel ∂F ” [6, Thm. 6.12]; no definition of the term is printed there. Definition A.1 is the reading their construction supports: in the proof of [6, Thm. 5.1], from which Theorem 6.12 is obtained, the members of a minimising sequence are made transverse to the frontiers of finitely many balls and the outermost discs so cut off are replaced by least area discs with the same boundary, and what is manipulated there is a surface smooth off a finite union of circles. We use (E)(i) with $\mathcal{F}$ read as Definition A.1. Of that reading the arguments below use one clause,

$$\text{the surface } F' \text{ of (E)(i) satisfies Definition A.1,} \quad (27)$$

which is the membership assertion of the cited theorem: Hass and Scott conclude that there is a surface F' in $\mathcal{F}$ of area I , and $\mathcal{F}$ is by their own description a class of piecewise smooth surfaces. In particular Lemma A.2 uses only that $\mathcal{F}$ is closed under diffeomorphisms of M fixing ∂M pointwise and that F' minimises over it, and both hold whatever the term means. What would replace (27) is not a regularity theorem, which is what this appendix supplies, but the existence statement that the infimum of area over the surfaces of Definition A.1 properly isotopic to F rel ∂F is attained within that class; the convergence in [6, Lem. 6.9] is smooth away from the face of the half-ball carrying the varying data, and the surfaces of [6, §§4–5] are closed, so the membership clause of Theorem 6.12 is not reached by the argument printed for it.

Throughout the rest of this appendix $F, \mathcal{F}, I$ are as in (E)(i) and $F' \in \mathcal{F}$ has $\text{Area}(F') = I$; we write $\Gamma := \Gamma_{F'}$ for a graph as in Definition A.1 and call the closures of the components of $F' \setminus \Gamma$ the pieces. Each is a compact smooth surface with corners, smoothly embedded up to its own boundary, and part of that boundary may lie in λ .

A.2 The first variation, and two local lemmas.

Lemma A.2 (Stationarity). *Let X be a smooth vector field on M with compact support in $\text{Int } M$. Then*

$$\sum_P \int_P \text{div}_P X \, dA = 0, \quad (28)$$

the sum being over the pieces P of F' . Consequently

- (i) *every piece is a minimal surface; and*
- (ii) *along the interior of every arc of Γ that lies in $\text{Int } F'$, the two sheets of F' meeting there have opposite outward conormals, $\eta_1 + \eta_2 = 0$.*

Proof. Let $\{\varphi_t\}$ be the flow of X . Each φ_t is a diffeomorphism of M equal to the identity near ∂M , so $\varphi_t(F')$ is properly embedded, is piecewise smooth — Definition A.1 being invariant under diffeomorphisms of M , with $\Gamma_{\varphi_t(F')} = \varphi_t(\Gamma)$ — and is isotopic to F' rel $\partial F'$. Hence $\varphi_t(F') \in \mathcal{F}$ and $\text{Area}(\varphi_t(F')) \geq I = \text{Area}(F')$, with equality at $t = 0$; the derivative at $t = 0$ therefore

vanishes, and it equals $\sum_P \int_P \operatorname{div}_P X$, the integrand being continuous on each piece because the piece is C^1 up to its boundary.

For (i), let $q \in \operatorname{Int} P$. Since F' is compact and embedded there is a ball $B \subset \operatorname{Int} M$ about q with $F' \cap B \subset \operatorname{Int} P$. For X supported in B only one term survives in (28), and the divergence theorem on $\operatorname{Int} P$ turns it into $-\int_P \langle H_P, X \rangle = 0$; as X is arbitrary, $H_P = 0$ near q .

For (ii), let q be an interior point of an arc $e \subset \Gamma \cap \operatorname{Int} F'$. The surface F' is a disc near q and e is an embedded arc in it, so e separates that disc into exactly two sheets, lying in pieces P_1, P_2 ; and there is a ball $B \subset \operatorname{Int} M$ about q with $F' \cap B$ contained in their union. Fix X supported in B and exhaust each P_i by compact subsurfaces $P_i^\delta \subset \operatorname{Int} P_i$ with $\partial P_i^\delta \rightarrow \partial P_i$ in C^1 . Since $H_{P_i} = 0$ by (i), the divergence theorem gives $\int_{P_i^\delta} \operatorname{div}_{P_i} X = \int_{\partial P_i^\delta} \langle \eta_i, X \rangle$; both sides converge as $\delta \rightarrow 0$, the left because the integrand is continuous on P_i and the right because η_i extends continuously to ∂P_i . So (28) becomes $\int_{e \cap B} \langle \eta_1 + \eta_2, X \rangle = 0$ for every such X , whence $\eta_1 + \eta_2 = 0$ on $e \cap B$. $\square$

Lemma A.3 (Sheets with opposite conormals glue smoothly). *Let $P_1, P_2 \subset M$ be smooth minimal surfaces with boundary, meeting along a common boundary arc e and otherwise disjoint near an interior point q of e , and suppose their outward conormals satisfy $\eta_1 + \eta_2 = 0$ along e . Then $P_1 \cup P_2$ is a smooth minimal surface near q .*

Proof. At each point of e the vectors η_1, η_2 are unit, orthogonal to the tangent of e , and tangent to P_1, P_2 respectively; $\eta_2 = -\eta_1$ therefore gives $TP_1 = \operatorname{span}(\dot{e}, \eta_1) = \operatorname{span}(\dot{e}, \eta_2) = TP_2$ along e , so P_1 and P_2 have a common tangent plane T at each point of e near q .

Choose coordinates near q with $T = \{y = 0\}$. For a small enough neighbourhood, P_1 and P_2 are graphs $y = u_1, y = u_2$ over the two closed sides of the projection of e in $\{y = 0\}$, with $u_1 = u_2$ and $Du_1 = Du_2$ on that projection; so the function u equal to u_i on the i th side is C^1 on a full disc D . Write the minimal surface equation for a graph in the ambient metric in divergence form, $\partial_i(a^i(x, u, Du)) + b(x, u, Du) = 0$, with a and b smooth and the equation uniformly elliptic on bounded gradients. Each u_i solves it classically on its own side; for a test function φ supported in D , integrating by parts on each side produces two interface integrals of $a^i(x, u, Du)\nu_i\varphi$ along the projection of e , and these cancel because Du is continuous across it and $\nu_1 = -\nu_2$. Hence u is a Lipschitz weak solution on all of D .

Difference quotients of u solve linear divergence-form equations with bounded measurable uniformly elliptic coefficients, so De Giorgi–Nash–Moser gives $Du \in C^{0,\alpha}$; the coefficients $a^i(x, u, Du)$ are then $C^{0,\alpha}$ and Schauder bootstrapping gives $u \in C^\infty$ [4, Chs. 6 and 8]. So $P_1 \cup P_2$ is a smooth surface near q , minimal because u solves the equation. $\square$

Lemma A.4 (Hopf at a boundary point). *Let $p \in \partial M$, let $T \subset T_p M$ be a plane, and let $U \subset T$ be a closed region with $0 \in \partial U$ satisfying the interior sphere condition at 0: some open disc $B \subset \operatorname{Int} U$ has $0 \in \partial B$. Let W be the graph over $U \cap B_\delta(0)$ of a C^1 map into the orthogonal complement of T vanishing to first order at 0, so that $p \in W$ and $T_p W = T$. Assume that $W \subset M$, that W is a minimal surface over the interior of U , and that W meets ∂M only over ∂U . Then*

$$dr_p|_T \neq 0, \quad \text{equivalently} \quad T \neq T_p \partial M.$$

Proof. Put $u := r \circ \Phi$ on $U \cap B_\delta(0)$, where Φ is the graph map. Since $W \subset M = \{r \geq 0\}$ we have $u \geq 0$; u vanishes at 0 and, by hypothesis, nowhere on $\operatorname{Int} U$, so $u > 0$ there; and u is smooth over $\operatorname{Int} U$ and C^1 up to 0. For a minimal surface W ,

$$\Delta_W u = \operatorname{tr}_W \operatorname{Hess} r + \langle \operatorname{grad} r, H \rangle = \operatorname{tr}_W \operatorname{Hess} r,$$

H vanishing. In the collar $\text{Hess } r = \varphi\varphi'g_{\partial M} \leq 0$ on vectors tangent to the level sets of r , by (3), while $\text{Hess } r(\partial_r, \cdot) = 0$ because r is a geodesic coordinate; so $\text{Hess } r \leq 0$ as a quadratic form and $\Delta_W u \leq 0$. Read in the coordinates of U , the operator Δ_W is uniformly elliptic with continuous coefficients and no zeroth-order term, so $-u$ is a subsolution.

Take the disc B of the interior sphere condition. On B we have $-u < 0 = -u(0)$, and the outward normal derivative at 0 exists because u is C^1 up to 0. Hopf's boundary point lemma [4, Lem. 3.4] gives $\partial_\nu(-u)(0) > 0$, that is $\partial_\nu u(0) < 0$ for the outward ν . The differential of Φ at 0 is the inclusion $T \hookrightarrow T_p M$, so $\partial_\nu u(0) = dr_p(\nu)$, whence $dr_p|_T \neq 0$. $\square$

Lemma A.4 uses only $\text{Hess } r \leq 0$, that is $\varphi' \leq 0$ in (2). Its hypothesis on U is stated as the interior sphere condition rather than as an opening angle because the regions it is applied to are bounded by curved arcs. Two families satisfy it, and they are the two applications: a closed circular sector of opening angle at least π , which is the case of Lemma A.7 with the sector a half-disc; and, more generally, a region bounded near 0 by two C^2 arcs through 0 whose tangent rays bound a closed sector S of opening angle at least π with $U \subset S$ near 0, which is the case arising inside Proposition A.5. For the second, let $B_\rho \subset S$ be the disc of radius ρ internally tangent to S at 0 and centred on its bisector. The half-angle of S being at least $\pi/2$, a point of B_ρ at distance t from 0 is at distance at least ct^2/ρ from ∂S , with $c > 0$ depending only on the angle, while the two bounding arcs, being C^2 with those tangent rays, stay within Ct^2 of ∂S . So $B_\rho \subset \text{Int } U$ near 0 once $\rho < c/C$.

A.3 Minimisers with fixed boundary.

Proposition A.5. *Let $F, \mathcal{F}, I, F'$ be as above. Then F' is a smooth embedded minimal surface, smooth up to ∂M , with $F' \cap \partial M = \partial F' = \lambda$; and F' is stable.*

Proof. By Lemma A.2(i) every piece is minimal, and by Lemma A.2(ii) with Lemma A.3 the surface F' is smooth and minimal near every interior point of every arc of Γ lying in $\text{Int } F'$. Three things remain: the vertices of Γ in $\text{Int } F'$, the points of λ , and stability.

Vertices in the interior. Let q be a vertex of Γ in $\text{Int } F'$. Finitely many arcs of Γ emanate from q and cut a disc neighbourhood N of q in F' into closed sectors $S_1, \dots, S_m$ in cyclic order, each contained in a piece, hence minimal and smoothly embedded up to q . Along the interior of each shared arc the two adjacent sectors have the same tangent plane, and the tangent plane of a sector extends continuously to q ; so all the S_i have one and the same tangent plane T at q .

Take coordinates with $q = 0$ and $T = \{y = 0\}$, let pr be the orthogonal projection onto T , and put $\Sigma_i := \text{pr}(S_i)$. Shrinking N , each S_i is the graph of a C^1 function over Σ_i , and the Σ_i are consecutive. These regions are not circular sectors: Σ_i is bounded near 0 by the projections of the two arcs of Γ bounding S_i , which are smooth curves through 0 with distinct tangent rays there, by the corner clause of Definition A.1. We call the angle $\theta_i \in (0, 2\pi)$ spanned by those two tangent rays the *opening angle* of Σ_i , and write $\widehat{\Sigma}_i$ for the sector they bound, the tangent cone of Σ_i at 0; no bound $\theta_i < \pi$ is available, and none is used. Then pr carries $N \setminus \{q\}$ onto a punctured neighbourhood of 0 in T as a covering map of some degree k , with $2\pi k = \sum_i \theta_i$, the total angle swept by the tangent cones $\widehat{\Sigma}_i$. Fix a disc D_ε inside that neighbourhood. Over each point of $D_\varepsilon \setminus \{0\}$ the k preimages are distinct points of the embedded surface F' , hence have pairwise distinct heights; the heights depend continuously on the point and never coincide, so they order the sheets consistently and the covering is trivial. But $N \setminus \{q\}$ is connected, so $k = 1$.

Hence, after shrinking N to the part lying over D_ε , it is the graph of a function u over D_ε . It is C^1 , the sectors having matching tangent planes along the arcs; it is smooth off the arcs; and it is a weak solution of the minimal surface equation on $D_\varepsilon \setminus \{0\}$, the interface terms cancelling as in Lemma A.3. It is then a weak solution on all of D_ε : cutting a test function off at radius δ

about 0 changes the weak formulation by at most $C\delta^{-1} \cdot \pi\delta^2 \rightarrow 0$, u being Lipschitz. By the regularity quoted in Lemma A.3, u is smooth; so F' is a smooth minimal surface on all of $\text{Int } F'$.

Points of λ . Since Γ is properly embedded, a point $p \in \lambda$ either lies off Γ or is one of the finitely many endpoints of arcs of Γ on λ . In the first case some relative neighbourhood of p misses Γ , so lies in $F' \setminus \Gamma$, a smooth submanifold of M by Definition A.1 — smooth up to ∂M and including its boundary points — and F' is smooth up to ∂M at p . So only the finitely many endpoints remain; fix such a p .

Those arcs cut a half-disc neighbourhood N of p in F' into closed sectors $S_0, \dots, S_m$, where S_0 and S_m carry the two arcs ℓ_0, ℓ_m into which λ is divided at p . As in the interior case all the S_i share a tangent plane T at p ; take coordinates with $p = 0$, $T = \{y = 0\}$ and, as there, put $\Sigma_i := \text{pr}(S_i) \subset T$, a region over which S_i is a C^1 graph, bounded near 0 by two smooth arcs whose tangent rays span the opening angle $\theta_i \in (0, 2\pi)$, and write $\widehat{\Sigma}_i$ for the tangent sector. Put $\Theta := \sum_i \theta_i$. The extreme edges of $\widehat{\Sigma}_0$ and $\widehat{\Sigma}_m$ are the two tangent rays of λ at p , which are opposite; hence

$$\Theta \equiv \pi \pmod{2\pi}, \quad \text{and in particular } \Theta \geq \pi. \quad (29)$$

What is wanted is a region of opening angle in $[\pi, 2\pi)$ over which F' is a graph. Let j be least with $\Theta_j := \theta_0 + \dots + \theta_j \geq \pi$; such a j exists by (29). If $\Theta_j < 2\pi$, put $U := \Sigma_0 \cup \dots \cup \Sigma_j$ and $W := S_0 \cup \dots \cup S_j$. If instead $\Theta_j \geq 2\pi$ then $j \geq 1$, since $\theta_0 < 2\pi$, and $\Theta_{j-1} < \pi$ by the minimality of j , so that θ_j alone exceeds π ; put $U := \Sigma_j$ and $W := S_j$.

Either way W is a C^1 graph over U , and U is bounded near 0 by two smooth arcs whose tangent rays span an angle $\Theta' \in [\pi, 2\pi)$, the intermediate arcs being interior to U ; so U satisfies the interior sphere condition at 0, by the second family recorded after Lemma A.4. Moreover W lies in M ; it is minimal over $\text{Int } U$, by the previous step; and W meets ∂M only over ∂U , because $F' \cap \partial M = \lambda$ by properness while $\lambda \cap N = \ell_0 \cup \ell_m$, and ℓ_0, ℓ_m are the extreme edges, carried by S_0 and S_m . Lemma A.4 applies and gives $dr_p|_T \neq 0$.

Now let v be a unit vector tangent at p to one of the sectors. That sector is C^1 at p and contained in $M = \{r \geq 0\}$, so its points at parameter t in the direction v have $r = t dr_p(v) + o(t) \geq 0$, forcing $dr_p(v) \geq 0$. The sectors are in cyclic order about p and consecutive ones share an edge, so the set of such v is the image in the unit circle of T of an interval of length Θ , swept monotonically from the tangent ray of ℓ_0 ; and it lies in the closed half-circle $\{dr_p \geq 0\}$, whose length is π because $dr_p|_T \neq 0$. Were $\Theta \geq 2\pi$ that sweep would cover the whole circle, which no half-circle contains; so $\Theta < 2\pi$, the sweep is injective, its image is an arc of angular length Θ , and $\Theta \leq \pi$. With (29), $\Theta = \pi$; in particular each $\theta_i < \pi$, which is a conclusion here and not a hypothesis.

So the Σ_i tile a region $H \subset T$ with $0 \in \partial H$, whose tangent cone at 0 is a half-plane, and N is the graph of a function u over H . It is C^1 on $\overline{H}$ and, by the previous step and the cancellation of the interface terms, a weak solution of the minimal surface equation on $\text{Int } H$. Near 0 the boundary of H is the projection of λ , a smooth curve, the arcs interior to H having been absorbed; so there H has smooth boundary, and there u takes the boundary values given by λ , which are smooth. A $C^{0,1}$ weak solution of a quasilinear divergence-form elliptic equation with smooth data on a smooth boundary portion is $C^{1,\alpha}$ up to that portion by the boundary form of De Giorgi–Nash–Moser, and then C^∞ up to it by boundary Schauder estimates [4, Chs. 6 and 8]. Hence F' is a smoothly embedded surface with boundary up to ∂M near p .

Globally and stability. The two steps give a smooth structure near every point of the compact set F' , so F' is a smooth embedded minimal surface, smooth up to ∂M ; and $F' \cap \partial M = \partial F' = \lambda$ is part of the hypothesis that F' is properly embedded, which is the membership condition of $\mathcal{F}$. Finally F' has least area among the piecewise smooth surfaces isotopic to it rel $\partial F'$, so all nearby normal deformations fixing the boundary increase area: F' is stable. $\square$

A.4 Minimisers with sliding boundary.

Theorem A.6 ((R1)). *Let $F \subset E(K)$ be properly embedded, incompressible and piecewise smooth with ∂F a leaf of J , and let*

$$A := \inf \{ \text{Area}(G) : G \text{ piecewise smooth and properly embedded,} \\ \text{properly isotopic to } F, \partial G \subset J \}.$$

Then there is a surface Σ properly isotopic to F , with $\partial \Sigma$ a leaf of J and $\text{Area}(\Sigma) = A$, which is the image of a smooth proper embedding. In particular Σ is a stable minimal surface, for some $\rho > 0$ the set $\Sigma \cap \{0 \leq r \leq \rho\}$ is an embedded smooth surface with boundary up to $\{r = 0\}$, and $\Sigma \cap \partial E(K) = \partial \Sigma$.

Proof. Let G_i be a minimising sequence for A . Each G_i is properly embedded, piecewise smooth and incompressible, incompressibility being invariant under isotopy, with ∂G_i a leaf of J . Apply (E)(i) to G_i to obtain F'_i of least area $I(G_i)$ in the class $\mathcal{F}(G_i)$ of piecewise smooth surfaces isotopic to G_i rel ∂G_i , and apply Proposition A.5 to it. Then F'_i is a smooth embedded stable minimal surface, smooth up to $\partial E(K)$, with $F'_i \cap \partial E(K) = \partial F'_i = \partial G_i$ a leaf of J ; and F'_i is properly isotopic to F , so that

$$A \leq \text{Area}(F'_i) = I(G_i) \leq \text{Area}(G_i) \longrightarrow A.$$

In particular each F'_i is a smooth proper embedding of the compact connected surface S underlying F , whose preimage of $\partial E(K)$ is the whole of ∂S , whose image is incompressible, and whose boundary is a curve of J ; that is, $F'_i \in \mathcal{M}(S)$ in the notation of §3, and $\text{Area}(F'_i) \leq a := A + 1$ for large i puts it in $\mathcal{M}_a$. Membership has two requirements. The first is that F'_i be a smooth proper embedding, which is Proposition A.5. The second is stability under variations fixing the boundary: it is the fixed-boundary least area surface of [6, Thm. 6.12] that is called a stable minimal surface in [17, App.], and the proof of [17, Prop. A.1] uses stability only through Schoen's interior estimates at finitely many interior points. Having least area rel $\partial F'_i$, F'_i is stable in that sense.

By (Fin) the space $\mathcal{M}_a$ is the disjoint union of finitely many open and closed subsets, each consisting of mutually isotopic surfaces, so after passing to a subsequence all the F'_i lie in one such subset $\mathcal{N}$; and $\mathcal{M}_a$ is compact, so after a further subsequence $F'_i \rightarrow \Sigma$ in the C^1 topology with $\Sigma \in \mathcal{M}_a$. As $\mathcal{N}$ is closed, $\Sigma \in \mathcal{N}$, so Σ is isotopic to the F'_i and hence properly isotopic to F . Area is continuous in the C^1 topology, so $\text{Area}(\Sigma) = \lim \text{Area}(F'_i) = A$. Membership in $\mathcal{M}(S)$ means that Σ is a stable minimal surface which is the image of a smooth proper embedding $f: (S, \partial S) \rightarrow (E(K), \partial E(K))$ with $f^{-1}(\partial E(K)) = \partial S$ and $f|_{\partial S}$ a parametrised curve of J : this is smoothness up to $\partial E(K)$ in a collar, together with $\Sigma \cap \partial E(K) = \partial \Sigma$. $\square$

Nothing in the proof used that $\partial E(K)$ is connected, or that S has a single boundary circle: the standing hypothesis of [17, App.] is that $f^{-1}(\partial_i M)$ be a single component of ∂S for each boundary component $\partial_i M$ of M , which is what a spanning surface for an n -component link satisfies. Theorem A.6 therefore holds verbatim in the n -component setting used in §6, and we refer to that case as the n -component form of (R1).

Lemma A.7 ((R2)). *Let Σ be as in Theorem A.6. Then $T_p \Sigma \neq T_p \partial E(K)$ for every $p \in \partial \Sigma$; that is, Σ meets $\partial E(K)$ transversally along $\partial \Sigma$, and is neatly embedded.*

Proof. By Theorem A.6, Σ is a smooth surface with boundary, smooth up to $\partial E(K)$, with $\Sigma \cap \partial E(K) = \partial \Sigma$. Let $p \in \partial \Sigma$ and $T := T_p \Sigma$. In a boundary chart at p the surface Σ is the graph over a half-disc $U \subset T$, of opening angle π and with p on its straight edge, of a smooth map vanishing to first order at p ; a circular sector of opening angle π satisfies the interior sphere

condition at its vertex. The graph lies in $E(K)$, is minimal, and meets $\partial E(K)$ exactly in the image of that straight edge. Lemma A.4 gives $dr_p|_T \neq 0$, that is $T_p\Sigma \neq T_p\partial E(K)$. $\square$

Remark A.8. One could instead minimise over the homotopy class $\text{rel } \partial F$; make the resulting map an embedding by [3]; and return to the isotopy class by the rigidity theorem of [19], whose hypothesis of ∂ -incompressibility is Lemma 4.2. We have not taken that route. Since it is the natural alternative to the one taken here, we record where each of its steps stands.

Its first step is available with a complete printed proof, once one supplies two things that the usual citations do not. The statement [6, Thm. 6.13] usually cited for it is given without proof, and its content at ∂F is the boundary regularity established above; but [3, Thm. 7.2] asserts, for a proper incompressible map $f: (F, \partial F) \rightarrow (M, \partial M)$ into a P^2 -irreducible Riemannian 3-manifold whose boundary has non-negative mean curvature and with $f(\partial F)$ a collection of disjoint simple closed curves, a least area map homotopic to f by a homotopy fixing the image of the boundary and parametrisable as a smooth immersion. That theorem is attributed there to [11, Thm. (1.7)], which is proved; but two clauses of it are not clauses of Lemaire's theorem. Lemaire's ambient manifold is homogeneously regular and without boundary, which a compact manifold with boundary is not, and his conclusion [11, Lem. (5.1)] is a minimal branched *immersion*, branch points not being excluded. Appendix C supplies both under the strict convexity of §3: the ambient boundary is removed by Lemma C.1 and the immersion clause is proved in Proposition C.12. The existence statement of [3, §7] that its authors say they have not proved is Existence Statement 7.3, and that one is the *free* boundary case: the class of maps properly homotopic to g , with the boundary free to move in ∂M . It is not needed for the fixed boundary case at issue here.

It is the second step that is not available. Theorem 5.1 of [3], and with it all of [3, §§1–6], is proved for closed surfaces. Section 7 asserts that the bounded case follows and lists the alterations needed section by section; but at [3, Lems. 1.3–1.6], where the local structure of a tangency between two sheets of a least area map is established from Bers's theorem at an *interior* point, the alteration offered is the single clause that one needs to note that a least area immersion can fail to be transverse on the boundary of the surface. No argument is printed for that case. Appendix C closes that clause, in the strengthened form that under strict convexity the failure does not occur (Proposition C.12), and closes with it the alteration listed for [3, §4] (Corollary C.15); both conditionally on the one clause of [11, Thm. (1.7)] which that theorem asserts without printing an argument for it (Remark C.4). What remains unproved is the global part of the announced alterations, listed in Remark C.16; so the bounded form of [3, Thm. 5.1] is, as things stand, in the same position as (27): asserted, and not reached by the argument printed for it.

APPENDIX B. ROUNDING ESTIMATES AND THE NON-TRANSVERSE CONFIGURATIONS

This appendix proves Lemmas 5.4 and 5.5, Proposition 5.7, and Theorem 5.1 in the configurations (T1) and (T2). The notation is that of §5; in particular $A = S_-$ and $B = S_+$ are the two minimisers that meet, S is the minimiser disjoint from both, and f, f_p are the graph differences of (22) and (21).

B.1 Corner rounding.

Proof of Lemma 5.4. Take an odd $\chi \in C^\infty([-1, 1], [-1, 1])$ with $0 \leq \chi \leq 1$ on $[0, 1]$, $\chi \equiv 1$ near 1 and $\chi(0) = 0$, and put $\Theta(t) = c_0 + \int_0^t \chi$ with $c_0 := 1 - \int_0^1 \chi \geq 0$. Then Θ is even, $\Theta(\pm 1) = 1$, $\Theta'(\pm 1) = \pm 1$ and $\Theta^{(k)}(\pm 1) = 0$ for $k \geq 2$, so Θ meets $|\cdot|$ to infinite order at $|t| = 1$; and

$t \mapsto \Theta(t) - t$ is non-increasing on $[0, 1]$ with value 0 at 1, so $\Theta \geq |\cdot|$; finally $\Theta'(0) = 0$ and $|\Theta'| \leq 1$. For $m > 0$ put

$$g(m) := \int_{-1}^1 \left(\sqrt{1+m^2} - \sqrt{1+m^2\Theta'(\tau)^2} \right) d\tau,$$

positive because $|\Theta'| \leq 1$ and $\Theta'(0) = 0$, and continuous in m . It is non-decreasing: differentiating under the integral sign,

$$\partial_m g(m) = \int_{-1}^1 m \left[(1+m^2)^{-1/2} - \Theta'(\tau)^2 (1+m^2\Theta'(\tau)^2)^{-1/2} \right] d\tau \geq 0,$$

since $u \mapsto u(1+m^2u)^{-1/2}$ has derivative $(1+m^2u)^{-3/2}(1+\frac{1}{2}m^2u) > 0$ and $\Theta'(\tau)^2 \leq 1$. Set $g_0(\theta) := \frac{1}{2} \min\{g(m) : m \in [\tan(\theta/2), \cot(\theta/2)]\} > 0$, the interval being non-empty because the two-sided bound $\alpha(x) \in [\theta(x), \pi - \theta(x)]$ presupposes $\theta \leq \pi/2$; since $\cot$ is decreasing, that bound puts the slope $m(x) = \cot(\alpha(x)/2)$ in the interval. The minimum is at the left endpoint, so $g_0(\theta) = \frac{1}{2}g(\tan(\theta/2))$ is non-decreasing, and $g_0(\theta) \rightarrow 0$ as $\theta \downarrow 0$.

Straighten first. Each sheet is smooth up to Γ and the two meet there at the angle $\alpha \in (0, \pi)$, hence transversally. Extend both across Γ to smooth surfaces $\hat{A}_\pm$ meeting transversally in Γ . Straighten $\hat{A}_+$ to $\{y=0\}$ with Γ the z -axis; transversality makes $\hat{A}_-$ a graph $\{x=G(y, z)\}$ with $G(0, z)=0$, and $(x, y, z) \mapsto (x-G(y, z), y, z)$ carries it to $\{x=0\}$ while fixing $\{y=0\}$. So there is a smooth chart, fixing Γ pointwise, in which the two sheets are half planes meeting along the Γ -axis.

The metric in that chart is arbitrary. Both sheets are planes *containing* the Γ -axis, so both are preserved by linear maps of the normal plane and by shears along Γ , $(z, x, y) \mapsto (z+ax+by, x, y)$; the shears are what remove the cross terms $h(\partial_z, \partial_x)$, $h(\partial_z, \partial_y)$, which the straightening leaves arbitrary. Apply Gram–Schmidt in the order forced by the sheets: normalise ∂_z ; project ∂_x and ∂_y h -orthogonally to it, a shear, so each sheet stays planar; then map the two resulting rays to the rays at angles $\pm\alpha(s)/2$ from $+\xi_1$, a linear map of the now h -orthogonal normal plane. Done smoothly in s , this leaves s arclength on Γ , the slices $\{s=\text{const}\}$ h -orthogonal to Γ , and $h=\delta$ at every point of Γ , cross terms included, while the wedge is the standard one of opening angle $\alpha(s)$ measured in h . Writing (s, ξ_1, ξ_2) for this chart, with the bisector along $+\xi_1$,

$$\tilde{B} = \{\xi_1 = m(s)|\xi_2|\} \text{ near } \Gamma, \quad h = \delta + O(|\xi|). \quad (30)$$

The first is an equality, not an approximation: a replacement profile matched to $m|\xi_2|$ while the true sheets were $m|\xi_2| + O(|\xi|^2)$ would miss them by $O(\eta^2)$ at the seam.

On $|\xi_2| \leq \eta(s)$ replace the two sheets by $\xi_1 = m(s)\eta(s)\Theta(\xi_2/\eta(s))$. At $|\xi_2| = \eta(s)$ this equals $m(s)|\xi_2|$; and the difference of the two profiles is $m(s)\eta(s)\Psi(\xi_2/\eta(s))$ with $\Psi := \Theta - |\cdot|$, every derivative of which carries a factor $\Psi^{(k)}(t)$, all vanishing at $|t|=1$. The glued surface is therefore C^∞ across the seam and coincides with $\tilde{B}$ outside the tube. Since $\Theta \geq |\cdot|$ the new sheet lies in the closed wedge side $\{\xi_1 \geq m(s)|\xi_2|\}$, and in its interior where $|\xi_2| < \eta$, which is the local half of (2), the global half following from it under the additional hypothesis stated there. Taking $\eta_{\max}$ less than half the normal injectivity radius keeps the tubes disjoint and embedded, and less than half ϱ_∂ keeps them clear of $\partial\tilde{B}$. This is (1)–(2), the collar in (2) coming from compactness.

For (3): inside the tube the old and the new sheets are, in each normal slice, two graphs over $|\xi_2| \leq \eta(s)$ agreeing to infinite order at the two ends, so the linear interpolation between them is a family of embedded surfaces, constant outside the tube, and pushing across the region they cobound gives an ambient isotopy supported in the tube. Clause (5) follows, an ambient isotopy carrying complementary regions to complementary regions and being the identity outside T .

A varying width does not spoil the matching. With $t := \xi_2/\eta(s)$ the new sheet is the graph $\xi_1 = F(s, \xi_2)$, $F = m\eta\Theta(t)$, so

$$F_{\xi_2} = m(s)\Theta'(t), \quad F_s = \eta(s)m'(s)\Theta(t) + m(s)\eta'(s)(\Theta(t) - t\Theta'(t)), \quad (31)$$

and at $|t| = 1$ we have $\Theta = 1$, $\Theta' = \pm 1$, so $\Theta - t\Theta'$ vanishes and $F_s = \eta m'$, which by (30) is the corresponding derivative of the old sheet $m(s)|\xi_2|$ at $|\xi_2| = \eta$. The η' term switches itself off at the seam, and the two pieces still meet to infinite order along the now non-parallel curves $|\xi_2| = \eta(s)$. That term is $O(\eta')$ and not $O(\eta)$, so no shrinking of $\eta_{\max}$ controls it; this is why (12) constrains η' . Since $c_0 \leq \Theta \leq 1$ and $|\Theta'| \leq 1$ we have $|\Theta - t\Theta'| \leq 2$, whence

$$F_s^2 \leq 2\eta^2 m'^2 + 8m^2 \eta'^2. \quad (32)$$

One slice. Fix s . The two quantities compared below are area densities with respect to $ds d\xi_2$, not lengths of slice curves: for a graph $\xi_1 = u(s, \xi_2)$ the Euclidean area element is $\sqrt{1 + u_s^2 + u_{\xi_2}^2} ds d\xi_2$, and the u_s term must be carried on both sides. Over $|\xi_2| \leq \eta(s)$ the old surface is $\{\xi_1 = m(s)|\xi_2|\}$ by (30), with $u_s = m'\xi_2$ and $u_{\xi_2} = \pm m$, so its density is

$$\int_{-\eta}^{\eta} \sqrt{1 + m^2 + m'^2 \xi_2^2} d\xi_2 = 2\eta \sqrt{1 + m^2} + E_1, \quad 0 \leq E_1 \leq m'^2 \eta^3,$$

the upper bound by $\sqrt{a+b} \leq \sqrt{a} + b/(2\sqrt{a})$ with $a = 1 + m^2 \geq 1$. Since $E_1 \geq 0$, discarding it is conservative; we nevertheless retain an allowance $C_1 \eta^2$ below, with C_1 bounding the C^2 norms of the two sheets, so that the estimate survives a straightening chart in which the sheets are only C^2 -close to the wedge (30) instead of equal to it. The new profile has density

$$\eta \int_{-1}^1 \sqrt{1 + m^2 \Theta'^2 + F_s^2} dt \leq \eta \int_{-1}^1 \sqrt{1 + m^2 \Theta'^2} dt + \eta \max_t F_s^2,$$

by the same inequality with $a = 1 + m^2 \Theta'^2 \geq 1$ over an interval of length 2. So the Euclidean decrease in the slice is at least $\eta[g(m) - 2\eta^2 m'^2 - 8m^2 \eta'^2 - C_1 \eta]$, using (32) and the definition of g .

The true metric. By (30) $h = \delta + O(|\xi|)$, and inside the tube $|\xi| \leq \eta\sqrt{1 + m^2}$. Let a graph $\xi_1 = u$ have Euclidean first fundamental form Q and true one $Q + \mathcal{E}$. Every tangent vector v satisfies $|\mathcal{E}(v, v)| \leq \|h - \delta\| Q(v, v)$, so the eigenvalues of $Q^{-1}\mathcal{E}$ are bounded by $\|h - \delta\|$ irrespective of $|\nabla u|$, which is what lets the estimate survive $m \rightarrow \infty$, and

$$\left| \sqrt{\det(Q + \mathcal{E})} - \sqrt{\det Q} \right| \leq C_{\text{met}} \eta \sqrt{1 + m^2} \sqrt{\det Q}. \quad (33)$$

Both profiles have Euclidean length $O(\eta\sqrt{1 + m^2})$, so (33) costs at most $6C_{\text{met}}\eta^2(1 + m^2)$, and

$$\text{Area}(\tilde{B}) - \text{Area}(\hat{B}) \geq \int_{\Gamma} \eta \left[g(m) - 2\eta^2 m'^2 - 8m^2 \eta'^2 - C_1 \eta - 6C_{\text{met}} \eta (1 + m^2) \right] ds. \quad (34)$$

Choosing $\eta_{\max}$ and κ . Since $g(m(s)) \geq 2g_0(\theta(s))$, put

$$\eta_{\max} := \min \left\{ \frac{\varrho}{2\sqrt{1 + m^2}}, \sqrt{\frac{g_0(\theta)}{8m'^2}}, \frac{g_0(\theta)}{4C_1}, \frac{g_0(\theta)}{24C_{\text{met}}(1 + m^2)}, \frac{\varrho_{\partial}}{2} \right\}, \quad \kappa := \frac{\sqrt{g_0(\theta)}}{6m}, \quad (35)$$

all five entries read pointwise at $x \in \Gamma$, with ϱ the local reach (10), $\varrho_{\partial}(x) = d(x, \partial \tilde{B}) > 0$ by the hypothesis $\Gamma \cap \partial \tilde{B} = \emptyset$, and the second entry read as $+\infty$ where $m' = 0$. The second, third and fourth entries, together with κ , make the four error terms in (34) at most $\frac{1}{4}g_0(\theta)$ each, the last because $8m^2 \eta'^2 \leq 8m^2 \kappa^2 = \frac{2}{9}g_0(\theta) < \frac{1}{4}g_0(\theta)$. The first and fifth do not enter (34): the first secures the Fermi chart and the disjointness and embeddedness of the tubes, the fifth their avoidance of $\partial \tilde{B}$. So the bracket is at least $2g_0(\theta) - \frac{35}{36}g_0(\theta) > g_0(\theta) \geq 0$ for each s separately. That is (13) with the slicewise non-negativity asserted in (4), and it is what allows the integral

to be restricted to a measurable subset. Both ϱ and ϱ_∂ are positive and lower semicontinuous on Γ , and are stated pointwise because in §§B.4–B.5 the global normal injectivity radius and the global distance $d(\Gamma, \partial\tilde{B})$ degenerate while their values on the subset over which the gain is collected do not.

Finally (11). Where $\theta(x) \geq \theta_\bullet$, $\varrho(x) \geq \varrho_\bullet$ and $\varrho_\partial(x) \geq \varrho_{\partial,\bullet}$, the slope m lies in $[\tan(\theta_\bullet/2), \cot(\theta_\bullet/2)]$ and $g_0(\theta) \geq g_0(\theta_\bullet)$, so (35) is bounded below by the quantity obtained from it by replacing m with $\cot(\theta_\bullet/2)$, m' with $M' := \sup_\Gamma |m'| < \infty$, $g_0(\theta)$ with $g_0(\theta_\bullet)$, ϱ with $\varrho_\bullet$ and ϱ_∂ with $\varrho_{\partial,\bullet}$; call these $\eta(\theta_\bullet, \varrho_\bullet, \varrho_{\partial,\bullet})$ and $\kappa(\theta_\bullet, \varrho_\bullet, \varrho_{\partial,\bullet})$, both non-decreasing in each argument. The dependence on θ cannot be removed: $\eta_{\max} \lesssim \varrho/m \rightarrow 0$ as $\alpha \rightarrow 0$, and $g(m)/(1+m^2) \rightarrow 0$ at both ends $\alpha \rightarrow 0$, $\alpha \rightarrow \pi$. So Corollary B.4 must bound the angle before it can bound the width. $\square$

B.2 Pushing off.

Proof of Lemma 5.5. Let ν be the unit normal of $\hat{B}$ pointing into U and n the outward normal of $\partial E(K)$, extended through the boundary collar; put $X := \nu - \psi(r)\langle \nu, n \rangle n$ with $\psi \equiv 1$ near $r = 0$ and supported in the collar. Then X is tangent to $\partial E(K)$, and neatness with compactness gives $\langle \nu, n \rangle^2 \leq 1 - c$, so $\langle X, \nu \rangle \geq c > 0$. On $\partial E(K)$ replace X by a field transverse to J whose flow carries leaves to leaves, oriented towards U , and interpolate in r , shrinking the collar so that $\langle X, \nu \rangle \geq c/2$ persists. Its flow Ψ_s exists for $s \in [0, \sigma_0]$, preserves $\partial E(K)$ and carries leaves to leaves; put $T(\sigma) := \Psi_\sigma(\hat{B})$. Transversality makes $(x, s) \mapsto \Psi_s(x)$ an embedding into the one-sided collar for σ_0 small, which gives the first three clauses, the last because Ψ is an ambient isotopy and $\Psi_\sigma \Psi_\sigma^{-1}$ joins two push-offs. With $K_1 := \sup |\nabla X|$ near $\hat{B}$, the Jacobian of $\Psi_s|_{\hat{B}}$ obeys $|\partial_s \log J| = |\operatorname{div}_{T(s)} X| \leq 2K_1$, the divergence being the tangential one, $\sum_{i=1,2} \langle \nabla_{e_i} X, e_i \rangle$ over an orthonormal frame of $T(s)$, whence the factor 2; so $|\operatorname{Area}(T(\sigma)) - \operatorname{Area}(\hat{B})| \leq (e^{2K_1\sigma} - 1) \operatorname{Area}(\hat{B}) \leq C\sigma$ with $C := 2K_1 e^{2K_1\sigma_0} \operatorname{Area}(\hat{B})$, independent of σ , using $e^x - 1 \leq xe^x$ for $x \geq 0$. $\square$

B.3 The structure of the tangency set.

Proof of Proposition 5.7. Both clauses rest on two devices applied on different domains: the zeroth-order term is removed by dividing by a positive solution, and the resulting divergence-form equation is factorised in the sense of Vekua. We set the first up once, for a domain Δ which is either a disc $D_R(p)$ about an interior tangency point, carrying the operator L_p of (21), or a half-disc D_R^+ about a point of L , carrying the operator L of (23). In both cases we write L for the operator and f for its solution.

Removing the zeroth-order term. Solve $Lh = 0$ on Δ with $h = 1$ on $\partial\Delta$. No sign condition is assumed on the zeroth-order coefficient, so the existence theorems of [4, Ch. 8], which all carry hypothesis (8.8), are unavailable as stated; smallness of R replaces the sign condition. For $u \in W_0^{1,2}(\Delta)$ the Poincaré inequality gives $\|u\|_{L^2} \leq CR \|Du\|_{L^2}$, whence the associated bilinear form satisfies $B[u, u] \geq (\lambda - C\|b\|_\infty R - C^2\|c\|_\infty R^2) \|Du\|_{L^2}^2$, positive definite for R small; B is bounded on $W_0^{1,2}(\Delta)$, so the Lax–Milgram theorem [4, Thm. 5.8] yields a unique weak solution. Writing $h = 1 + \varpi$, the function ϖ solves $L\varpi = -c$ with zero boundary data. Rescaling $x \mapsto x/R$ to the unit disc or half-disc makes the drift $O(R)$ and the zeroth-order coefficient $O(R^2)$; testing the equation against ϖ gives $\|\varpi\|_{L^2} = O(R^2)$. The global maximum principle [4, Thm. 8.15], applied to ϖ and to $-\varpi$, both of which lie in $W_0^{1,2}$ and so are ≤ 0 on the boundary in the

generalised sense, then gives $\|\varpi\|_\infty \leq C(\|\varpi\|_{L^2} + \nu^{-1}\|c\|_{L^{q/2}}) = O(R^2)$, so $h \geq \frac{1}{2}$ for R small. Put $v := f/h$; expanding $L(hv) = 0$ with $Lh = 0$ gives, weakly,

$$\partial_i(h a^{ij} \partial_j v) + (a^{ij} \partial_j h + h b^i) \partial_i v = 0, \quad (36)$$

with Lipschitz leading coefficients, bounded drift and no zeroth-order term; in the half-disc case $v = 0$ on L , since f does. Multiplying by the reciprocal square root of the determinant of the leading matrix normalises that determinant to 1 and changes only the drift.

Proof of (1). Let $p \in \mathcal{T}$, run the previous paragraph on $\Delta = D_R(p)$ with L_p , and write $v_p := f_p/h_p$, so $v_p(p) = 0$. Choose a smooth Jordan subdomain $\Omega \Subset D_R(p)$ containing p . As in the paragraph “An isothermal chart up to L ” below — the argument is the same and shorter, no boundary arc being distinguished — Lichtenstein’s theorem [9, Thm. 4.1, case $m = 1$] supplies a $C^{1,\alpha}$ diffeomorphism of $\mathbb{D}$ onto $\bar{\Omega}$, conformal for the metric (37), and by the density law recorded there the equation for $\hat{v}_p$ becomes (38) on the disc, with no boundary condition and no reflection. The factorisation of the paragraph “Finite order” then applies verbatim: either $\mathfrak{w} := 2\partial_z \hat{v}_p$ has isolated zeros of finite integer order m with $c'_1|z - z_0|^m \leq |\mathfrak{w}| \leq c'_2|z - z_0|^m$ nearby, or $\mathfrak{w} \equiv 0$. In the second case $\hat{v}_p$ is constant, and the constant is 0; then $f_p \equiv 0$ near p , the set on which the two sheets agree is open by this argument applied at each of its points and closed by continuity, so the two surfaces coincide, contradicting $\text{dist}(u, w) = 2$. Transporting through the bi-Lipschitz chart gives $|d\hat{v}_p| \asymp \rho^m$ and $|\hat{v}_p| = O(\rho^{m+1})$ at distance ρ from p , and $f_p = h_p v_p$ with $h_p \geq \frac{1}{2}$ gives $|df_p| \geq c_1 \rho^m - c_2 \rho^{m+1} > 0$ on a punctured neighbourhood. So p is the only critical point of f_p in a disc D_p , and, the tangency points in D_p being critical points of f_p , it is isolated in $\text{Int } A \cap \text{Int } B$.

For (24), suppose $f_p \geq 0$ near p , the other sign being identical. Then $v_p \geq 0$ with $v_p(p) = 0$, so by the strong maximum principle at this regularity [4, Thm. 8.19], whose hypotheses are met since the operator is in divergence form with no zeroth-order term and $v_p \in W^{1,2}$, we get $v_p \equiv 0$, hence $f_p \equiv 0$ and the same contradiction.

Proof of (2). Assume $\partial A = \partial B = L$ and run the removal of the zeroth-order term on half-discs $\Delta = D_R^+$ against L , writing $v := f/h$.

An isothermal chart up to L . Choose a smooth Jordan subdomain Ω of the half-disc whose boundary contains a subarc of L around the point in question. Write $a = (a^{ij})$ for the leading matrix of (36) after the normalisation $\det a \equiv 1$, b for its drift, and

$$g := a^{-1}, \quad (37)$$

a Riemannian metric with $\det g = 1$; it is the *inverse* of the leading matrix that carries the relevant conformal structure. Being the inverse of a Lipschitz uniformly elliptic matrix, g is Lipschitz on $\bar{\Omega}$, hence $C^{0,\alpha}$, and $\partial\Omega$ is $C^{1,\alpha}$; so by the global form of Lichtenstein’s theorem [9, Thm. 4.1, case $m = 1$] there is a diffeomorphism $\tau: \mathbb{D} \rightarrow \bar{\Omega}$ of class $C^{1,\alpha}$, conformal from the flat metric to g . Let φ be a Möbius map of $\mathbb{D}$ onto the upper half-plane carrying the arc $\tau^{-1}(L \cap \partial\Omega)$ into $\mathbb{R}$ and τ^{-1} of the base point to 0, and put $F := \varphi \circ \tau^{-1}$. Then F is $C^{1,\alpha}$ up to L , sends L into $\mathbb{R}$ and the interior into the upper half-plane, and is bi-Lipschitz, DF being continuous and invertible on a compact set.

In the coordinate $z = F(\cdot)$ the operator becomes the flat Laplacian with a bounded drift. A divergence-form operator transforms under a $C^{1,\alpha}$ diffeomorphism by the density law: substituting $x = F^{-1}(z)$ in $\int (a \nabla v) \cdot \nabla \phi \, dx$ turns the leading matrix into $DF a DF^\top / |\det DF|$ and the drift into $DF b / |\det DF|$, no derivative of DF appearing. Hence the transformed drift is again in L^∞ ; the transformed leading matrix again has determinant 1, the factors $(\det DF)^2$ cancelling; and that matrix is a positive multiple of the identity exactly when $a = \lambda(DF^\top DF)^{-1}$ for a positive function λ , that is, exactly when $g = a^{-1}$ is a conformal multiple of the pullback $F^* \delta$ —

which is what the choice of τ arranged, φ being conformal. A positive multiple of the identity of determinant 1 is the identity, so (36) becomes

$$\Delta \hat{v} + \hat{b} \cdot \nabla \hat{v} = 0, \quad \hat{b} \in L^\infty, \quad \hat{v} = 0 \text{ on } \mathbb{R}. \quad (38)$$

Odd reflection. Extend $\hat{v}$ to the full disc by $\hat{v}(z) := -\hat{v}(\bar{z})$ for $\text{Im } z < 0$, and $\hat{b}$ by $\hat{b}_1(\bar{z}) = \hat{b}_1(z)$, $\hat{b}_2(\bar{z}) = -\hat{b}_2(z)$. For a test function ϕ supported in the full disc, split $\int \nabla \hat{v} \cdot \nabla \phi - \int (\hat{b} \cdot \nabla \hat{v}) \phi$ over the two half-discs and substitute $z \mapsto \bar{z}$ in the lower one: the interior terms cancel by (38), and the two conormal boundary terms along $\mathbb{R}$ cancel because the leading matrix is the identity, so its off-diagonal entry vanishes on $\mathbb{R}$ and the conormal derivative is $\partial_2 \hat{v}$, even under the reflection while the outward normals are opposite. Hence the extension solves (38) weakly on the full disc.

Finite order. Put $\mathfrak{w} := 2\partial_z \hat{v}$. Since $\partial_{\bar{z}} \partial_z = \frac{1}{4} \Delta$ and $\hat{b} \cdot \nabla \hat{v} = \frac{1}{2}[(\hat{b}_1 + i\hat{b}_2)\mathfrak{w} + (\hat{b}_1 - i\hat{b}_2)\bar{\mathfrak{w}}]$ for real $\hat{v}$, equation (38) reads

$$\partial_{\bar{z}} \mathfrak{w} = -\frac{1}{4}(\hat{b}_1 + i\hat{b}_2) \mathfrak{w} - \frac{1}{4}(\hat{b}_1 - i\hat{b}_2) \bar{\mathfrak{w}},$$

a generalized analytic function with coefficients in $L^\infty \subset L^p$ for every $p > 2$, which is Vekua's standing hypothesis [18, Ch. III, §1, (1.10)], the equation already being in the canonical form [18, Ch. III, (1.5)]. By the similarity principle [18, Ch. III, §4.1], $\mathfrak{w} = \Phi e^\omega$ with Φ holomorphic and ω bounded and Hölder; so unless $\mathfrak{w} \equiv 0$ its zeros are isolated of finite integer order m , with $c'_1|z - z_0|^m \leq |\mathfrak{w}| \leq c'_2|z - z_0|^m$ nearby [18, Ch. III, §4.2, Thm. 3.5]. The alternative $\mathfrak{w} \equiv 0$ forces $\hat{v} \equiv 0$, since $\hat{v} = 0$ on $\mathbb{R}$, hence $f \equiv 0$ on an open set, and the same contradiction as in (1) follows. Transporting through the bi-Lipschitz F gives $|d\hat{v}| \asymp \rho^m$ and $|\hat{v}| = O(\rho^{m+1})$ at distance ρ , and $f = h\hat{v}$ with $h \geq \frac{1}{2}$ gives $|df| \geq c_1\rho^m - c_2\rho^{m+1} > 0$ on a punctured half-neighbourhood.

Every critical point of f on L is therefore isolated in the closed half-neighbourhood, and at a non-critical boundary point continuity gives a neighbourhood free of critical points; L is compact, so finitely many remain and a uniform r_0 exists with no critical point in $\{0 < r < r_0\}$. They lie on $\{f = 0\}$ because f vanishes on L . A tangency point in the collar being a critical point of f , this also gives $\mathcal{T} \subset \{r \geq r_0\}$.

Proof of (3). In (T1) one has $A \cap B \cap \partial E(K) = \partial A \cap \partial B = \emptyset$, so $A \cap B$ is a compact subset of $\text{Int } E(K)$; in (T2), clause (2) confines $\mathcal{T}$ to the compact set $A \cap B \cap \{r \geq r_0\}$, which again misses $\partial E(K)$, a point of $A \cap B$ on $\partial E(K)$ lying on L , where $r = 0$. So $\mathcal{T}$ is a subset of a compact subset of $\text{Int } A \cap \text{Int } B$. Tangency is a closed condition, the tangent planes varying continuously, so $\mathcal{T}$ is closed there; being also discrete by clause (1), it is finite. $\square$

B.4 Configuration (T2): a common boundary leaf. The perturbation of A has a free sign, and fixing it is what keeps the intersection curves away from L .

Lemma B.1 (A sign, and an arc of L swept clear). *Put $a(s) := \partial_r f(s, 0)$ and $\Lambda := \sup |\partial_r^2 f| < \infty$. Then $a \not\equiv 0$, and there are $\varsigma \in \{\pm 1\}$, a closed arc $I \subset L$, and constants $a_0 > 0$, $r_1 > 0$, all depending only on A and B , such that $\varsigma f(s, r) \geq \frac{1}{2}a_0 r > 0$ for $s \in I$ and $0 < r \leq r_1$.*

Proof. Since $f(s, 0) \equiv 0$ we have $\partial_s f(s, 0) \equiv 0$, so $df(s, 0) = (0, a(s))$ and the critical points of f on $\{r = 0\}$ are the zeros of a . By (25) there are finitely many, so $a \not\equiv 0$. Pick s_0 with $a(s_0) \neq 0$, set $\varsigma := \text{sign } a(s_0)$ and $a_0 := \frac{1}{2}|a(s_0)|$, and choose by continuity a closed arc $I \ni s_0$ with $\varsigma a \geq a_0$ on I . Put $r_1 := \min\{a_0/(2\Lambda), r_0/4\}$. For $s \in I$ and $0 \leq \rho \leq r_1$ we have $\varsigma \partial_r f(s, \rho) \geq \varsigma a(s) - \Lambda \rho \geq \frac{1}{2}a_0$; integrate in r . Nothing here refers to the perturbation parameter. $\square$

Take a cut-off $\chi(r)$ equal to 1 near $r = 0$, supported in the collar $\{r < r_0\}$, with its transition band placed where $\{f = 0\}$ is regular, and pairwise disjoint discs $D_p \ni D'_p$ around the points of $\mathcal{T}$ with bumps ψ_p equal to 1 on D'_p ; with ς as above set

$$A_t := \left\{ y = f + \varsigma t \chi(r) - \varsigma t \sum_{p \in \mathcal{T}} \psi_p \right\}. \quad (39)$$

Near the boundary $A_t \cap B$ is the level set $\{f = -\varsigma t\}$, regular for small $t \neq 0$ by (25); on the transition band, if transversality failed along $t_n \rightarrow 0$ the bad points would converge to a point with $f = 0$ and $df = 0$, excluded by the placement of the band. On D'_p one has $A_t \cap B = \{f = \varsigma t\}$, and by Proposition 5.7(1) the only critical point of f in D_p is p , with critical value 0, so every $t \neq 0$ is a regular value of either sign; on $D_p \setminus D'_p$ one has $|df| \geq c > 0$, hence $|d(f - \varsigma t \psi_p)| \geq c/2$ for t small. Elsewhere $A_t = A$ and transversality is open. So there is $\delta^{(1)} > 0$ such that for $t \in (0, \delta^{(1)})$ the surface A_t is neat, incompressible, transverse to B , with ∂A_t the leaf $\{y = \varsigma t\}$ disjoint from ∂B , disjoint from S , and $A_t \rightarrow A$ in C^1 . The interior perturbation is of subtract-a-constant type: f_t and f differ by a constant on each D'_p , which the two proofs below use.

Lemma B.2 (No small closed intersection curves). *There are $\rho_0 > 0$ and $\delta_1 > 0$ such that for every $t \in (0, \delta_1]$ every closed component γ of $A_t \cap B$ has $\text{diam}(\gamma) \geq \rho_0$.*

Proof. Otherwise there are $t_n \rightarrow 0$ and closed components γ_n with $\text{diam}(\gamma_n) \rightarrow 0$; by C^0 convergence, points $x_n \in \gamma_n$ subconverge to some $x \in A \cap B$. If x is a transverse point outside the collar, a fixed ball at x meets $A \cap B$ in a single arc crossing it, and C^1 stability keeps that true for small t , so γ_n cannot lie in the ball. If $x = p \in \mathcal{T}$, then for large n the curve γ_n bounds a disc $D_n \subset D_p$ on which f_t vanishes on the boundary. If $f_t \equiv 0$ on D_n then $df_t \equiv 0$ there, impossible since p is the only critical point of f_t in D_p ; otherwise f_t attains a strict interior extremum at a critical point, hence at p — but on D'_p it differs from f by the constant $-\varsigma t$, and by (24) p is not a local extremum of f . If x lies in the collar, then D_n does not meet ∂B , so $D_n \subset \{r > 0\}$, where by (25) $df \neq 0$ and hence $|df_t| \geq c/2$ for small t ; but f_t vanishes on ∂D_n , so its maximum and its minimum on $\overline{D_n}$ are both 0, forcing $df_t \equiv 0$. Each case is a contradiction. $\square$

Choice of ρ_2 . Choose $\rho_2 > 0$ with the balls $B_{2\rho_2}(p)$, $p \in \mathcal{T}$, pairwise disjoint,

$$4N\rho_2 < \rho_0/8, \quad \rho_2 \leq r_0/8, \quad \rho_2 \leq r_1/(2C'_{\text{met}}), \quad (40)$$

and put $P := \bigcup_p B_{2\rho_2}(p)$, $G := E(K) \setminus P$ and $G' := E(K) \setminus \bigcup_p B_{\rho_2}(p)$, so $G \subset G'$. Only ρ_0 , r_0 , r_1 and $\mathcal{T}$ enter this choice. Lemma B.3 quotes Corollary B.4(1), stated after it; the proof of that corollary uses only Lemma B.2, Proposition 5.7 and Lemma 5.4, so there is no circularity.

Lemma B.3 (Every closed component has uniform length away from L). *Let ς, I, a_0, r_1 be as in Lemma B.1 and write $\mathcal{C} := L \times (0, r_0)$ for the open collar. Then*

- (i) *for every $t > 0$ the set $\Gamma_t := A_t \cap B$ is disjoint from $I \times (0, r_1]$;*
- (ii) *no closed component of Γ_t contained in $\mathcal{C}$ is null-homotopic in $\mathcal{C}$;*
- (iii) *there is $c_\star > 0$, depending only on A , B and the perturbation scheme, such that for every $t \in (0, \min\{\delta_1, \delta^{(1)}\})$ every closed component γ of Γ_t satisfies $\text{length}(\gamma \cap G \cap \{r \geq r_1\}) \geq c_\star$.*

Proof. (i) In the collar $\Gamma_t = \{f = -\varsigma t\}$, that is $\varsigma f = -t < 0$, whereas $\varsigma f \geq \frac{1}{2}a_0r > 0$ on $I \times (0, r_1]$ by Lemma B.1.

(ii) If a closed component $\gamma \subset \mathcal{C}$ bounded a closed disc $D \subset \mathcal{C}$, then $f \equiv -\varsigma t$ on ∂D ; a value of f on D above or below $-\varsigma t$ would give an interior extremum and a critical point, contradicting (25), so $f \equiv -\varsigma t$ on D and $df \equiv 0$ there, contradicting (25) again.

(iii) First, $\gamma \cap \partial E(K) = \emptyset$, since $A_t \cap \partial E(K) = \partial A_t$ is the leaf $\{y = \varsigma t\}$ while $B \cap \partial E(K) = L = \{y = 0\}$. Next, $\mathcal{T} \subset \{r \geq r_0\}$ by Proposition 5.7(2) and $\rho_2 \leq r_0/8$ by (40), so $P \subset \{r > 3r_0/4\}$ and hence

$$\{r_1 \leq r \leq r_0/2\} \subset G \cap \{r \geq r_1\} =: Z. \quad (41)$$

Three cases exhaust the possibilities.

Case A: $\min_\gamma r \geq r_1$. Then $\gamma \cap G \subset Z$ and Corollary B.4(1) gives $\text{length}(\gamma \cap Z) \geq 3\rho_0/8$.

Case B: $\min_\gamma r < r_1$ and $\max_\gamma r > r_0/2$. The function r is continuous on the circle γ and takes a value below r_1 and one above $r_0/2$, so γ contains two disjoint subarcs joining $\{r = r_1\}$ to $\{r = r_0/2\}$ inside $\{r_1 \leq r \leq r_0/2\}$. In the collar chart r is 1-Lipschitz, so each has length at least $r_0/4$ up to the bi-Lipschitz constant C'_{met} ; both lie in Z by (41), and together they give $\text{length}(\gamma \cap Z) \geq r_0/(2C'_{\text{met}})$.

Case C: $\min_\gamma r < r_1$ and $\max_\gamma r \leq r_0/2$. Then $\gamma \subset \mathcal{C}$, so by (ii) it is not null-homotopic in the annulus $\mathcal{C}$ and its projection to L is onto. So γ has points over every $s \in I$, and by (i) each has $r > r_1$, while by hypothesis each has $r \leq r_0/2$; so the part of γ over I lies in Z by (41). The projection is 1-Lipschitz in the collar chart, so that part has length at least the arclength $\ell(I)$ of I , up to C'_{met} .

Take $c_\star := \min\{3\rho_0/8, r_0/2, \ell(I)\}/C'_{\text{met}}$. □

Corollary B.4 (Bounds independent of t). *With ρ_2, P, G, G' as above, there are $\theta_1 \in (0, \pi/2]$, $\eta_1^{\text{geo}} > 0$, $\kappa_1 > 0$ and $\delta_2 > 0$ such that for every $t \in (0, \delta_2]$ and every closed component γ of $A_t \cap B$:*

- (1) $\text{length}(\gamma \cap G) \geq 3\rho_0/8$;
- (2) *the opening angle α along $\gamma \cap G' \cap \{r \geq r_1/2\}$ satisfies the two-sided bound $\min\{\alpha, \pi - \alpha\} \geq \theta_1$;*
- (3) *along $\gamma \cap G' \cap \{r \geq r_1/2\}$ the functions of Lemma 5.4, read pointwise, satisfy $\eta_{\max} \geq \eta_1^{\text{geo}}$ and $\kappa \geq \kappa_1$.*

Proof. (1) By Lemma B.2 there are $a, b \in \gamma$ with $d(a, b) \geq \rho_0/2$. Collapse each of the N components of P to a point and let d' be the resulting quotient pseudometric. Collapsing N sets of diameter at most $4\rho_2$ decreases distances by at most $4N\rho_2$, so $d(a, b) \leq d'(a, b) + 4N\rho_2$; and the image of an arc of γ from a to b is a path whose length is at most $\text{length}(\gamma \cap G) =: \Lambda$, the parts of the arc inside P being collapsed and contributing nothing. Hence $\rho_0/2 \leq \Lambda + 4N\rho_2 < \Lambda + \rho_0/8$, so $\Lambda > 3\rho_0/8$. The quotient is taken because a curve may enter and leave the same ball an unbounded number of times, whereas each component is collapsed once.

(2) Put $G'' := E(K) \setminus \bigcup_p B_{\rho_2/2}(p) \supset G'$ and $\mathcal{K} := A \cap B \cap G'' \cap \{r \geq r_1/4\}$, with r extended by $r \equiv r_0$ outside the collar. The set $\mathcal{K}$ is compact, and A meets B transversally at every point of it: inside the collar the two sheets are the graphs $\{y = f\}$ and $\{y = 0\}$ of (22), and transversality along $\{f = 0\}$ is the condition $df \neq 0$, which holds by (25), the tangencies on L being excluded by $r \geq r_1/4$; outside the collar Proposition 5.7 gives transversality away from $\mathcal{T}$, and $\mathcal{T}$ has been excised by G'' . Hence $\min\{\alpha, \pi - \alpha\}$, a continuous positive function on $\mathcal{K}$, has a positive infimum $2\theta_1$; the upper bound is included because the two frontiers occupy opposite sectors and Lemma 5.4 needs α bounded away from π as well as from 0. For small t the curve $A_t \cap B$ lies in the $\min\{\rho_2/4, r_1/4\}$ -neighbourhood of $A \cap B$ and on G'' the C^1 distance from A_t to A is $O(t)$, so the bound persists with θ_1 along $\gamma \cap G' \cap \{r \geq r_1/2\}$.

(3) By the last paragraph of the proof of Lemma 5.4 the lower bound $\underline{\eta}$ of (11) involves, besides the angle, the local reach ϱ of (10), the distance ϱ_∂ to $\partial\tilde{B}$, a constant C_1 bounding the C^2 norms of the two sheets, and $\sup|m'|$. For ϱ_∂ : on $\{r \geq r_1/2\}$ the distance to $\partial E(K)$ is at least $r_1/(2C'_{\text{met}})$, and $\tilde{B}$ may be taken to contain the $r_1/(4C'_{\text{met}})$ -neighbourhood of that set. For

C_1 and $\sup |m'|$: the graph functions of A_t over A have C^3 norms bounded independently of $t \leq \delta^{(1)}$ and B is fixed, which together with (2) bounds the second fundamental forms of the two sheets and the derivative of the angle along γ , uniformly in t . For ϱ : by the previous sentence the curvature of γ is bounded by some K_1 independent of t , so the normal exponential map of γ is injective on a tube of radius $\varrho' > 0$ around any arc of γ of length at most ϱ' , with ϱ' depending only on K_1 and the ambient injectivity radius; and two arcs far apart in γ cannot come close in $E(K)$. Otherwise there are $t_n \downarrow 0$ and points x_n, y_n on components γ_n of $A_{t_n} \cap B$, both in $G' \cap \{r \geq r_1/2\}$, with $d_{\gamma_n}(x_n, y_n) \geq \varrho'$ and $d(x_n, y_n) \rightarrow 0$; passing to a subsequence, $x_n, y_n \rightarrow x$, which lies in $A \cap B$ by C^1 convergence and hence in $\mathcal{K}$, so A meets B transversally at x and a ball $B(x, \zeta)$ meets $A \cap B$ in a single embedded arc; by C^1 convergence so does $A_{t_n} \cap B$ for large n , and then x_n, y_n lie on one arc of diameter at most 2ζ , of length $2\zeta(1 + o(1)) < \varrho'$ once ζ is small, a contradiction. Hence $\varrho \geq \varrho_1 > 0$, uniformly in t . Take $\eta_1^{\text{geo}} := \underline{\eta}(\theta_1, \varrho_1, r_1/(4C'_{\text{met}}))$ and $\kappa_1 := \underline{\kappa}$ of the same three arguments, both positive since $\theta_1 > 0$. $\square$

Proof of Theorem 5.1, configuration (T2). Put $\eta_1 := \min\{\eta_1^{\text{geo}}, \kappa_1 \rho_2/4\}$ and

$$\varepsilon_* := g_0(\theta_1) \cdot \eta_1 \cdot c_* > 0, \quad (42)$$

with c_* from Lemma B.3, depending on A, B and on the once-chosen perturbation scheme and on nothing later.

No disc patch, uniformly in t . Suppose cutting A_t and B along $A_t \cap B$ produces a disc patch. Every closed component γ of $A_t \cap B$ satisfies, by Corollary B.4, Lemma B.3(iii) and the slicewise non-negativity in Lemma 5.4,

$$\mathcal{G}(\gamma) \geq g_0(\theta_1) \eta_1 \text{length}(\gamma \cap G \cap \{r \geq r_1\}) \geq \varepsilon_*,$$

the gain being computed with the width (44) below, which equals η_1 on $G \cap \{r \geq r_1\}$, and the rest of the curve being discarded. So (14) gives $2\varepsilon_* \leq \mathcal{G}(a) + \mathcal{G}(b) \leq e(t) := \text{Area}(A_t) - \text{Area}(A)$, and $e(t) \rightarrow 0$ by C^1 convergence; so there is $\delta_3 > 0$ with $e(t) < \varepsilon_*$, hence no disc patch, *for every* $t \in (0, \delta_3]$. Here ε_* is pinned down before t : the patch exists only after t , and both the length of its frontier and the angle along it vary with t .

The output vertices do not depend on t . For t in a compact subinterval of $(0, \delta^{(1)})$ the surfaces A_t are transverse to B with disjoint boundaries and disjoint from S , so there is an ambient isotopy H_t with $H_{t_0} = \text{id}$, $H_t(A_{t_0}) = A_t$, $H_t(B) = B$ and $H_t|_S = \text{id}$, preserving $\partial E(K)$: take $X_t = \partial_t j_t$ for embeddings j_t with image A_t , correct it along the compact intersection track by a tangential term so that it becomes tangent to B there — possible because transversality makes the relevant map onto $T_p E(K)/T_p B$ surjective and the metric supplies a smooth right inverse — and extend by Hadamard's lemma and a partition of unity to a time-dependent field tangent to B and to $\partial E(K)$ and vanishing on S , whose flow is the required isotopy. Lift H_t to the infinite cyclic cover starting at the identity. It fixes S_0 and S_1 and the fundamental domain between them, commutes with the deck generator, and cannot interchange the two sides of a wall, the upper side being the one containing S_1 ; so it carries the frontiers (8) at t_0 to those at t and small push-offs to admissible push-offs. Hence the two coarse classes are independent of $t > 0$. Fix once and for all, at one value of t , a compression sequence from each coarse surface to an incompressible descendant with boundary, giving $w_\uparrow, w_\downarrow$; transporting the coarse surfaces and the discs along the isotopy gives the same two vertices at every $t > 0$. This is what makes (b) hold in the order stated: given z with minimiser Z disjoint from $A \cup B$, one may choose $t_z > 0$ after Z is known so that $Z \cap A_{t_z} = \emptyset$, and take the push-offs at t_z small enough to avoid Z ; the classes, and therefore the vertices, are the ones already fixed, so t_z changes only a representative.

The area estimate. Since $\partial A_t \cap \partial B = \emptyset$, the set $\Gamma_t := A_t \cap B$ is a finite union of closed curves, non-empty, for otherwise A_t and B would be disjoint representatives of w and u . Corollary B.4

and Lemma B.3 apply to every component and the slicewise gain is non-negative everywhere, so rounding the two cornered frontiers along Γ_t gains at least ε_* in total. The order of choices is

$$\begin{aligned} A, B &\Rightarrow (\text{perturbation scheme, including } \varsigma) \Rightarrow \rho_0 \Rightarrow r_1 \Rightarrow \rho_2 \Rightarrow (\theta_1, \varrho_1) \\ &\Rightarrow (\eta_1^{\text{geo}}, \kappa_1) \Rightarrow \eta_1 \Rightarrow c_* \Rightarrow \varepsilon_* \Rightarrow \delta \Rightarrow t \Rightarrow \eta_2 \\ &\Rightarrow (\widehat{B}_\uparrow, \widehat{B}_\downarrow) \Rightarrow C \Rightarrow \sigma, \end{aligned}$$

each step using only what precedes it. Choose

$$\begin{aligned} \delta &\leq \min\{\delta^{(1)}, \delta_1, \delta_2, \delta_3\} \quad \text{and small enough that} \\ \text{Area}(A_t) &< \text{Area}(A) + \varepsilon_*/2 \quad \text{for } t \in (0, \delta], \end{aligned} \tag{43}$$

and fix such a t . The width must equal η_1 on $G \cap \{r \geq r_1\}$, where the gain is collected, and fall to an admissible value inside the balls $B_{\rho_2}(p)$, where the angle degenerates and $\eta_{\max}$ with it, and inside the collar $\{r \leq r_1/2\}$, where the two sheets meet $\partial E(K)$. Put

$$\begin{aligned} w(x) &:= \min\{\rho_2, d(x, \bigcup_p B_{\rho_2}(p)), d(x, \{r \leq r_1/2\})\}, \\ \eta_2 &:= \min\{\eta_1, \min_{\Gamma_t} \eta_{\max}, \frac{1}{2} \min \varrho\} > 0, \end{aligned}$$

so that w is 1-Lipschitz on $E(K)$ and hence along Γ_t , and set

$$\eta(s) := \min\{\eta_1, \eta_2 + \frac{\kappa_1}{2} w(x(s))\}, \tag{44}$$

smoothed at a scale $\ll \rho_2$ so as to be C^1 with $|\eta'| \leq \kappa_1$. On $\Gamma_t \cap G \cap \{r \geq r_1\}$ we have $w = \rho_2$: the second entry is ρ_2 because G omits the balls $B_{2\rho_2}(p)$, and the third is at least $r_1/(2C'_{\text{met}}) \geq \rho_2$ because r is C'_{met} -Lipschitz in the collar chart. Since $\eta_2 + \kappa_1 \rho_2/2 \geq 2\eta_1$ by $\eta_1 \leq \kappa_1 \rho_2/4$, we get $\eta \equiv \eta_1$ there. Where $w = 0$ we have $\eta \equiv \eta_2 \leq \eta_{\max}$ pointwise. In between, $w > 0$ forces the point into G' and into $\{r > r_1/2\}$, so Corollary B.4(3) gives $\eta \leq \eta_1 \leq \eta_1^{\text{geo}} \leq \eta_{\max}$ and $|\eta'| \leq \kappa_1 \leq \kappa$. So (12) holds along all of Γ_t , and by slicewise non-negativity the gain over $\Gamma_t \cap G \cap \{r \geq r_1\}$ alone already accounts for ε_* . The bound on η' in (12) is absolute rather than relative because the descent from η_1 to η_2 must fit in the fixed transition length $\rho_2/2$, which needs $(\eta_1 - \eta_2)/\kappa_1 \leq \eta_1/\kappa_1$, independent of η_2 , whereas a bound $|\eta'| \leq \Lambda\eta$ would need $\Lambda^{-1} \log(\eta_1/\eta_2)$, unbounded as $\eta_2 \rightarrow 0$.

Now obtain C from Lemma 5.5 and choose $C\sigma < \varepsilon_*/4$. Running the transverse argument of §5.2 with (A_t, B) in place of (S_-, S_+) and ε_0 replaced by ε_* gives

$$\text{Area}(P_\uparrow) + \text{Area}(P_\downarrow) \leq \text{Area}(A_t) + \text{Area}(B) - \varepsilon_* + C\sigma < A(w) + A(u) - \varepsilon_*/4,$$

whence (d). The genus computation is topological and is transported by the isotopy, so (c) is unchanged, and (a) follows as before. $\square$

B.5 Configuration (T1): disjoint boundaries, interior tangencies. The device is that of §B.4, with the collar apparatus — the sign ς and the arc I , the length estimate inside the collar, the third scale of the width ramp — replaced by the single positive number (45).

Assume $\partial A \cap \partial B = \emptyset$ and $\mathcal{T} \neq \emptyset$. Since $A \cap \partial E(K) = \partial A$ and $B \cap \partial E(K) = \partial B$ we have $A \cap B \cap \partial E(K) = \emptyset$, and $A \cap B$ is compact, so

$$d_0 := \text{dist}(A \cap B, \partial E(K)) > 0. \tag{45}$$

By Proposition 5.7(1) each of the finitely many $p \in \mathcal{T}$ carries a disc D_p , inside the graph chart of (21), on which p is the only critical point of f_p and $f_p(p) = 0$. Shrinking, take the D_p pairwise disjoint and inside $\{d(\cdot, \partial E(K)) > d_0/2\}$, choose concentric $D'_p \Subset D_p$, and take bumps

ψ_p supported in D_p and equal to 1 on D'_p . Let A_t be obtained from A by replacing, over each D_p , the graph $\{y = f_p\}$ by

$$\{y = f_p - t\psi_p\}. \quad (46)$$

Then $\partial A_t = \partial A$, still a leaf of J disjoint from ∂B ; A_t is smooth, neatly embedded and normally isotopic to A , hence incompressible and a representative of w ; and $A_t = A$ outside a compact subset of $\text{Int } E(K)$, so $A_t \cap S = \emptyset$ for small t and $A_t \rightarrow A$ in C^∞ . As in (39) the perturbation is of subtract-a-constant type on D'_p , which Lemma B.5(1) uses.

Transversality holds for all small $t > 0$: on D'_p one has $A_t \cap B = \{f_p = t\}$, a regular value because the only critical point of f_p in D_p is p , with critical value 0; on $D_p \setminus D'_p$ one has $|df_p| \geq c > 0$, hence $|d(f_p - t\psi_p)| \geq c/2$ for t small; elsewhere $A_t = A$, which meets B transversally away from $\mathcal{T}$. Fix $\delta^{(1)} > 0$ accordingly. For $t \in (0, \delta^{(1)})$ the set $\Gamma_t := A_t \cap B$ is a finite union of disjoint smooth closed curves — closed, because an arc would have its endpoints in $\partial A_t \cap \partial B = \emptyset$ — lying in $\{d(\cdot, \partial E(K)) \geq d_0/2\}$, and non-empty, for otherwise A_t and B would be disjoint representatives of w and u , contradicting $\text{dist}(u, w) = 2$. Finally

$$e(t) := \text{Area}(A_t) - \text{Area}(A) \rightarrow 0 \quad (t \downarrow 0), \quad (47)$$

and $\text{Area}(A) = A(w)$, A being a minimiser.

Lemma B.5 (Bounds independent of t , tangency case). *There are $\rho_0 > 0$, $\rho_2 > 0$, $\theta_1 \in (0, \pi/2]$, $\eta_1^{\text{geo}} > 0$, $\kappa_1 > 0$ and $\delta_2 > 0$, depending only on A , B and the perturbation scheme, such that, writing $P := \bigcup_p B_{2\rho_2}(p)$, $G := E(K) \setminus P$ and $G' := E(K) \setminus \bigcup_p B_{\rho_2}(p)$, for every $t \in (0, \delta_2]$ and every closed component γ of Γ_t :*

- (1) $\text{diam}(\gamma) \geq \rho_0$ and $\text{length}(\gamma \cap G) \geq 3\rho_0/8$;
- (2) the opening angle satisfies the two-sided bound $\min\{\alpha, \pi - \alpha\} \geq \theta_1$ along $\gamma \cap G'$;
- (3) along $\gamma \cap G'$ the functions of Lemma 5.4, read pointwise, satisfy $\eta_{\max} \geq \eta_1^{\text{geo}}$ and $\kappa \geq \kappa_1$.

Proof. Each clause is the corresponding statement of §B.4 with the collar deleted.

Diameter. Run the proof of Lemma B.2: its third case is vacuous here, Γ_t missing the collar by (45), and its first two cases apply verbatim, (46) being again of subtract-a-constant type on D'_p . This gives $\rho_0 > 0$ for all small t . Choose $\rho_2 > 0$ with the balls $B_{2\rho_2}(p)$ pairwise disjoint, $4N\rho_2 < \rho_0/8$ and $2\rho_2 < d_0/2$; the collapsing argument of Corollary B.4(1), which uses only ρ_0 and this choice, gives $\text{length}(\gamma \cap G) \geq 3\rho_0/8$.

Angle. Put $G'' := E(K) \setminus \bigcup_p B_{\rho_2/2}(p)$ and $\mathcal{K} := A \cap B \cap G''$, compact and with A meeting B transversally at each of its points, $\mathcal{T}$ having been excised. The restriction $\{r \geq r_1/4\}$ of Corollary B.4(2) has no analogue here, $\partial A \cap \partial B = \emptyset$. So $\min\{\alpha, \pi - \alpha\}$ has a positive infimum $2\theta_1$ on $\mathcal{K}$, and for t small the two-sided bound persists with θ_1 along $\gamma \cap G'$.

Widths. As in Corollary B.4(3), $\underline{\eta}$ and $\underline{\kappa}$ are controlled by the angle, by bounds on the C^2 norms of the two sheets and on $\sup |m'|$, by the local reach ϱ and by ϱ_∂ . The first three are bounded exactly as there; for ϱ_∂ the collar estimate is replaced by (45), the boundary of the cornered frontier lying on $\partial E(K)$ and Γ_t being at distance at least $d_0/2$ from it; and for ϱ the compactness argument of Corollary B.4(3) applies verbatim with this $\mathcal{K}$. $\square$

Proof of Theorem 5.1, configuration (T1). Put $\eta_1 := \min\{\eta_1^{\text{geo}}, \kappa_1\rho_2/4\}$ and

$$\varepsilon_* := g_0(\theta_1) \cdot \eta_1 \cdot \frac{3}{8}\rho_0 > 0, \quad (48)$$

fixed before t , in the order $A, B \Rightarrow$ perturbation scheme $\Rightarrow \rho_0 \Rightarrow \rho_2 \Rightarrow \theta_1 \Rightarrow (\eta_1^{\text{geo}}, \kappa_1) \Rightarrow \eta_1 \Rightarrow \varepsilon_* \Rightarrow \delta \Rightarrow t$.

For the width take (44) with the third entry of w deleted, that is

$$w(x) := \min\{\rho_2, d(x, \bigcup_p B_{\rho_2}(p))\}, \quad \eta_2 := \min\{\eta_1, \min_{\Gamma_t} \eta_{\max}, \frac{1}{2} \min_{\Gamma_t} \varrho\} > 0,$$

the first again 1-Lipschitz and the second chosen after t . The verification of (12) given in (T2) goes through with Lemma B.5(3) in place of Corollary B.4(3) and the restriction $\{r \geq r_1/2\}$ deleted throughout, so by the slicewise non-negativity of Lemma 5.4(4) the gain along a closed component γ satisfies $\mathcal{G}(\gamma) \geq g_0(\theta_1) \eta_1 \text{length}(\gamma \cap G) \geq \varepsilon_*$, the rest of the curve being discarded.

If cutting A_t and B along Γ_t produced a disc patch, Lemma 5.6 — applicable because A is a minimiser, A_t is normally isotopic to it with ∂A_t a leaf, and $A_t \pitchfork B$ with $\partial A_t \cap \partial B = \emptyset$ — would give closed curves a, b of Γ_t with $2\varepsilon_* \leq \mathcal{G}(a) + \mathcal{G}(b) \leq e(t)$. By (47) there is $\delta_3 > 0$ with $e(t) < \varepsilon_*$, hence no disc patch, for every $t \in (0, \delta_3]$.

Choose $\delta \leq \min\{\delta^{(1)}, \delta_2, \delta_3\}$, small enough in addition that $\text{Area}(A_t) < \text{Area}(A) + \varepsilon_*/2$ for $t \in (0, \delta]$, and fix such a t . Lemma 4.9 applies to (A_t, B) and yields $P_\uparrow, P_\downarrow$; rounding the two cornered frontiers along Γ_t with the width η gains at least ε_* in total, and Lemma 5.5 then costs $C\sigma$, which we take below $\varepsilon_*/4$. Exactly as in configuration (T2), an ambient isotopy carrying A_{t_0} to A_t , fixing B and S and preserving $\partial E(K)$, shows that the two coarse classes, and hence the two output vertices $w_\uparrow, w_\downarrow$, do not depend on t ; its construction used only transversality, the disjointness of the boundaries and the disjointness from S , all of which hold here. Then

$$\text{Area}(P_\uparrow) + \text{Area}(P_\downarrow) \leq \text{Area}(A_t) + \text{Area}(B) - \varepsilon_* + C\sigma < A(w) + A(u) - \varepsilon_*/4,$$

which is (d); the genus computation is topological and transported by the isotopy, giving (c); (a) follows as in the transverse case; and (b) follows from the t -independence just established, exactly as in (T2). $\square$

APPENDIX C. THE FIXED BOUNDARY LEAST AREA MAP ALONG $\partial E(K)$

This appendix is not used elsewhere in the paper. It settles the first of the two steps of the alternative route recorded in Remark A.8, and it supplies the local input at ∂M which [3, §7] lists as the one alteration needed in [3, Lems. 1.3–1.6] and for which no argument is printed there: *that a least area immersion can fail to be transverse on the boundary of the surface*. Under the strict convexity of §3 that failure does not occur, and we prove so. What is *not* settled is the second step of Remark A.8, and one clause of the analytic input is asserted by the theorem we cite rather than proved in it (Remark C.4); §C.3 reduces that clause to a statement about a limit in moduli space, but does not prove it. The status after this appendix is recorded in Remark C.16.

Throughout, $M := E(K)$ carries the relative metric (2) with $\varphi(r) = 1 - r$ and $\varepsilon < 1$, the collar coordinate r is inward, $\lambda \subset \partial M$ is a leaf of the real-analytic foliation J , and S is a compact connected orientable surface with connected boundary. Only $\varphi(0) = 1$, $\varphi' < 0$, $\varphi > 0$ and the collar form of the metric are used.

C.1 An ambient manifold without boundary. The existence theorems for least area maps are stated for an ambient manifold without boundary; [3, Thm. 7.2] states one for an ambient manifold whose boundary has non-negative mean curvature, attributing it to [11], whose Theorem (1.7) is stated for a homogeneously regular manifold. Strict convexity lets us remove the ambient boundary once and for all, and so keep the theorem we cite the one that is printed.

Extend the collar coordinate to $r \in (-\infty, \varepsilon)$ and set

$$\widehat{E} := M \cup_{\partial M} ((-\infty, 0] \times \partial M), \quad \widehat{h} := dr^2 + (1 - r)^2 g_{\partial M} \text{ on } \{r \leq 0\}, \quad (49)$$

the metric agreeing with (2) on the overlap, so that $\widehat{h}$ is smooth, $M = \{r \geq 0\} \subset \widehat{E}$ and $\partial \widehat{E} = \emptyset$. Let $\rho: \widehat{E} \rightarrow M$ be the retraction which is the identity on M and sends $(r, x) \mapsto (0, x)$ for $r \leq 0$. Recall that a Riemannian manifold is *homogeneously regular* if there are $0 < \kappa \leq \Lambda$ and $\rho_0 > 0$

such that every point admits a coordinate chart on the Euclidean ball of radius ρ_0 in which $\kappa \delta \leq h \leq \Lambda \delta$.

Lemma C.1. (i) $(\hat{E}, \hat{h})$ is complete and homogeneously regular.

- (ii) The inclusion $M \hookrightarrow \hat{E}$ is a homotopy equivalence; in particular $\pi_1(M) \rightarrow \pi_1(\hat{E})$ is an isomorphism and $\pi_2(\hat{E}) = 0$.
- (iii) $\text{Hess } r \leq 0$ as a quadratic form on all of $\{r < \varepsilon\}$, with equality on a unit vector v exactly when $v = \pm \partial_r$.
- (iv) ρ is 1-Lipschitz, hence does not increase area; and for any continuous $\psi: (S, \partial S) \rightarrow (\hat{E}, \lambda)$ the maps ψ and $\rho \circ \psi$ are relatively homotopic in the sense of [11, Def. (1.6)].

Proof. (iii) On $\{r < \varepsilon\}$ the metric is the warped product $dr^2 + \varphi(r)^2 g_{\partial M}$ with $\varphi(r) = 1 - r > 0$, so the computation of (3) holds there verbatim: $\text{Hess } r = \varphi \varphi' g_{\partial M}$ on vectors tangent to the level sets of r and $\text{Hess } r(\partial_r, \cdot) = 0$, with $\varphi \varphi' = -(1 - r) < 0$. For a unit $v = a \partial_r + X$ this gives $\text{Hess } r(v, v) = -(1 - r)|X|_{g_{\partial M}}^2 \leq 0$, vanishing exactly when $X = 0$.

(i) Since $\hat{h} \geq dr^2$, the function r is 1-Lipschitz, so a Cauchy sequence stays in some $\{-A \leq r < \varepsilon/2\}$, which is compact; completeness follows. For homogeneous regularity, let $\iota > 0$ be such that $g_{\partial M}$ -normal coordinates $y = (y^1, y^2)$ at any $x_0 \in \partial M$ are defined for $|y| < \iota$ and satisfy $\frac{1}{2}\delta \leq g_{\partial M} \leq 2\delta$ there. Fix $p_0 = (r_0, x_0)$ with $r_0 \leq 0$ and put

$$\Psi(s, \eta) := \left(r_0 + s, \exp_{x_0}^{g_{\partial M}}(\eta/(1 - r_0)) \right), \quad |s| < \frac{1}{2}, \quad |\eta| < \iota.$$

This is a chart about p_0 , defined because $1 - r_0 \geq 1$, and in it

$$\hat{h} = ds^2 + \left(\frac{1 - r_0 - s}{1 - r_0} \right)^2 (g_{\partial M})_{ij}(\eta/(1 - r_0)) d\eta^i d\eta^j,$$

where the warping factor lies in $[\frac{1}{4}, \frac{9}{4}]$ because $|s| < \frac{1}{2}$ and $1 - r_0 \geq 1$, and the matrix lies between $\frac{1}{2}\delta$ and 2δ . So $\frac{1}{8}\delta \leq \hat{h} \leq \frac{9}{2}\delta$ on the ball of radius $\rho_0 := \min\{\frac{1}{2}, \iota\}$, with constants depending only on $g_{\partial M}$. These charts cover $\{r \leq 0\}$; the points of the compact M admit such charts with uniform constants by compactness.

(ii) $\hat{E}$ deformation retracts to M along the lines $r \mapsto (r, x)$. Hence $\pi_2(\hat{E}) \cong \pi_2(M)$, and $\pi_2(M) = 0$: were $\pi_2(M) \neq 0$, the sphere theorem [8, Ch. 4] would give an embedded 2-sphere in the orientable M that is not null-homotopic, whereas M is irreducible [10], so every embedded 2-sphere in it bounds a ball and is null-homotopic.

(iv) On $\{r \leq 0\}$ write $v = a \partial_r + X$ at (r, x) , so that $|v|^2 = a^2 + \varphi(r)^2 |X|_{g_{\partial M}}^2$. Then $d\rho(v)$ is the vector X at $(0, x)$, of squared length $\varphi(0)^2 |X|_{g_{\partial M}}^2 = |X|_{g_{\partial M}}^2$, which is at most $\varphi(r)^2 |X|_{g_{\partial M}}^2 \leq |v|^2$ because $\varphi(r) = 1 - r \geq 1$ there. A 1-Lipschitz map does not increase the area of a surface. For the last clause put $\rho_t(r, x) := ((1 - t)r, x)$ for $r \leq 0$ and $\rho_t := \text{id}$ on M , a continuous family with $\rho_0 = \text{id}_{\hat{E}}$ and $\rho_1 = \rho$; then $\psi_t := \rho_t \circ \psi$ is a homotopy, each ρ_t is the identity on $M \supset \lambda$, and so $\psi_t|_{\partial S} = \psi|_{\partial S}$ throughout, with image in λ and monotone. $\square$

C.2 The analytic input. We isolate what is used about the minimiser, so that everything below is unconditional given it.

Definition C.2. A map $f: (S, \partial S) \rightarrow (\hat{E}, \lambda)$ is a *Lemaire minimiser* for λ if

- (L1) $f(\partial S) \subset \lambda$, the map $f|_{\partial S}$ is onto λ with connected point preimages, and f minimises area among the maps relatively homotopic to it in the sense of [11, Def. (1.6)];
- (L2) for some metric g on S , smooth up to ∂S , the map f is harmonic on $\text{Int } S$ and weakly conformal there, that is $f^* \hat{h} = \mu g$ for a function $\mu \geq 0$;
- (L3) $f \in C^{2,\alpha}(S, \hat{E})$ up to ∂S , for some $\alpha \in (0, 1)$.

Harmonicity of a map from a surface depends only on the conformal class of g , since the tension field scales as $\tau_{cg}(f) = c^{-1}\tau_g(f)$ for a positive function c ; so (L2) is a condition on a conformal structure, and any smooth metric in the class may be used below. By (L3) both the harmonic map equation and the relation $f^*\hat{h} = \mu g$, asserted on $\text{Int } S$, hold on S by continuity of df ; μ is continuous on S .

Proposition C.3 (Existence; citation). *Let $F \subset M$ be properly embedded, incompressible and connected with $\partial F = \lambda$, and let S be its underlying surface. Then the class of maps relatively homotopic to the inclusion of F contains a Lemaire minimiser for λ , and its area is at most that of any piecewise smooth surface in M properly isotopic to F rel ∂F .*

Proof. Apply [11, Thm. (1.7)] with $N = \hat{E}$ and the single Jordan curve $\Gamma_1 = \lambda$. The hypotheses hold: $\hat{E}$ is homogeneously regular with $\pi_2 = 0$ by Lemma C.1(i)–(ii); and the inclusion of F is a map of finite energy carrying ∂S homeomorphically onto λ and inducing an injection on π_1 , since F is incompressible and $\pi_1(M) \rightarrow \pi_1(\hat{E})$ is injective. For S of genus zero with one or two boundary circles the existence is classical, respectively is obtained by the same argument with a flat torus in place of the closed surfaces used there [11, §5]. The theorem produces a minimal map ϕ , relatively homotopic to the inclusion, minimising area in that class and smooth on $\text{Int } S$; by [11, Lem. (5.1)] it is harmonic and weakly conformal for a conformal structure on S ; and (L1) holds by construction. For (L3): λ is a leaf of the real-analytic foliation J , hence a $C^{s+\alpha}$ curve for every s , so the last clause of [11, Thm. (1.7)] gives $\phi \in C^{s,\alpha}(S, \hat{E})$. The clause of (L2) asserting that the conformal structure is represented by a metric *smooth up to ∂S* is taken from the statement of [11, Thm. (1.7)] and not from the argument printed for it; see Remark C.4.

A piecewise smooth G properly isotopic to F rel ∂F admits, by Definition A.1, a smooth triangulation compatible with Γ_G , and therefore carries a bi-Lipschitz homeomorphism from S : finitely many smooth compact pieces, glued along the arcs and circles of Γ_G . That is a map of finite energy, of area $\text{Area}(G)$, relatively homotopic to the inclusion of F through the isotopy, which fixes the boundary pointwise. So $\text{Area}(\phi) \leq \text{Area}(G)$. Finally, although $\hat{E}$ is not compact, no minimising sequence can escape: by Lemma C.1(iv) composing with ρ neither leaves the relative homotopy class nor increases area, so the infimum over the class equals the infimum over the maps into the compact M , and a minimising sequence may be taken to lie in M . $\square$

Remark C.4. The smoothness of g up to ∂S is where (L2) exceeds what is printed in [11], and everything after §C.3 rests on it: it is what makes Δ_g an operator with smooth coefficients in the Fermi coordinates (51), and so what lets Lemma C.6 feed Propositions C.11 and C.12. In the proof of [11, Thm. (1.7)] the conformal structure is produced as a limit: the structures t_k of a minimising sequence are doubled to closed surfaces by [11, Lem. (5.3)], the doubled structures are shown to lie in a compact subset of the moduli space of the closed surface, and a convergent subsequence gives a structure on the double, “inducing one on M by restriction”. The conclusion of [11, Thm. (1.7)] does assert $\varphi \in C^{s,\alpha}(M, N)$ for the limit, a statement which presupposes (L2); so (L2) is asserted by the theorem we cite, and is not reached by the argument printed for it. Definition C.2 is stated as a hypothesis precisely so that this dependence is visible; Proposition C.3 is a citation, and inherits it. §C.3 analyses the dependence: the regularity asserted by (L2) turns out to be automatic, and what is really missing is the compatibility of the limit in moduli space with the subsurface M , isolated there as (L2') and (L2'').

Remark C.5. Four points of bookkeeping. First, [11, Lem. (5.1)] concludes that ϕ is a minimal branched *immersion*: branch points are not excluded there. The interior ones are excluded for area minimisers by the methods of [15] and [5], as invoked in [3, p. 612] for the closed case; those are interior results, and the boundary is what Proposition C.12 supplies. Accordingly the clause

of [3, Thm. 7.2] that the minimiser “can be parameterized as a smooth immersion” is not a consequence of [11, Thm. (1.7)] as printed; with Proposition C.12 it does hold here. Second, the ambient manifold of [3, Thm. 7.2] has boundary, while that of [11, Thm. (1.7)] is homogeneously regular, which a compact manifold with boundary is not at its boundary; Lemma C.1 is what removes that discrepancy, and it uses strict convexity, not mean convexity. Third, (L1) pins the boundary curve, not its parametrisation: the boundary of a competitor slides along λ , and $f|_{\partial S}$ is merely monotone. That it is nevertheless a homeomorphism is part of Proposition C.12, not an assumption. The word *monotone* is not defined in [11]; we read it in the classical sense, that point preimages are connected, which is what (L1) says and all that Proposition C.12(iv) uses. Fourth, the interior regularity just cited is a statement about area minimisers, and f is one by (L1).

The next lemma is the computation that drives everything, in the form in which Lemma A.4 used it for an embedded minimal surface.

Lemma C.6. *Let f be a Lemaire minimiser for λ , let g be as in (L2), and put $u := r \circ f$. On the open set where f takes values in $\{r < \varepsilon\}$,*

$$\Delta_g u = \operatorname{tr}_g(f^* \operatorname{Hess} r) = \sum_{i=1,2} \operatorname{Hess} r(f_* e_i, f_* e_i) \leq 0 \quad (50)$$

for a g -orthonormal frame e_1, e_2 , with equality at a point exactly when both $f_ e_i$ are radial there. Moreover $u \in C^{2,\alpha}$ up to ∂S , and in any smooth chart Δ_g is uniformly elliptic with smooth coefficients and no zeroth-order term.*

Proof. In local coordinates on S , writing Γ for the Christoffel symbols of g and $\hat{\Gamma}$ for those of $\hat{h}$,

$$\Delta_g(r \circ f) = g^{ij}(\partial_i \partial_j(r \circ f) - \Gamma_{ij}^k \partial_k(r \circ f)) = g^{ij} \partial_{\alpha\beta}^2 r \partial_i f^\alpha \partial_j f^\beta + \partial_\gamma r \Delta_g f^\gamma.$$

Adding and subtracting $g^{ij} \hat{\Gamma}_{\alpha\beta}^\gamma \partial_\gamma r \partial_i f^\alpha \partial_j f^\beta$ turns the first term into $\operatorname{tr}_g(f^* \operatorname{Hess} r)$ and the second into $\langle \operatorname{grad} r, \tau_g(f) \rangle$, which vanishes as f is harmonic. The inequality and its equality case are Lemma C.1(iii). The regularity is (L3) and the smoothness of r and of g . $\square$

C.3 The conformal structure at ∂S . Clause (L2) asks for a metric on S smooth up to ∂S . Two things must be separated here. The smoothness itself is automatic, and is proved below; what is not automatic, and what [11] does not print, is that the limiting structure on the closed double may be restricted to S at all, that is, that the diffeomorphisms implicit in convergence in moduli space may be taken to respect the subsurface S . We isolate that statement, prove that a weaker and more classical form of it already suffices, and say what a proof of it would require. We do not prove it.

Throughout, $\hat{S} := S \cup_{\partial S} S^-$ is the double of S along its boundary, a closed smooth surface, and $i: \hat{S} \rightarrow \hat{S}$ is the swap involution: a smooth involution with $\operatorname{Fix}(i) = \partial S$ interchanging the two components of $\hat{S} \setminus \partial S$.

Lemma C.7. *Let Σ be a smooth surface and let t be a conformal structure on Σ , that is, a smooth field $p \mapsto t_p$ of conformal classes of inner products on $T_p \Sigma$. Then t is the conformal class of a smooth Riemannian metric on Σ . Consequently, if Σ is closed and $S \subset \Sigma$ is a compact smooth subsurface with boundary, then $t|_S$ is represented by a metric on S smooth up to ∂S .*

Proof. Smoothness of the field means that every point has a neighbourhood U_α carrying a smooth Riemannian metric g_α with $[g_\alpha]_p = t_p$ for all $p \in U_\alpha$. On an overlap the two metrics agree pointwise up to a positive factor, and that factor $e^{2\varphi_{\alpha\beta}}$ is smooth, being the quotient of $g_\alpha(X, X)$ by $g_\beta(X, X)$ for any smooth local non-vanishing vector field X ; the quotient does not depend on

X because the two metrics are pointwise conformal. Let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to $\{U_\alpha\}$ and put $g := \sum_\alpha \rho_\alpha g_\alpha$, a smooth symmetric 2-tensor. Fix a point p and an index β with $\rho_\beta(p) > 0$. Every α contributing at p satisfies $g_\alpha(p) = e^{2\varphi_{\alpha\beta}(p)} g_\beta(p)$, whence

$$g(p) = \left(\sum_\alpha \rho_\alpha(p) e^{2\varphi_{\alpha\beta}(p)} \right) g_\beta(p),$$

a positive multiple of $g_\beta(p)$. So g is a Riemannian metric and $[g]_p = t_p$ everywhere. For the consequence, take such a g on Σ and restrict it to S : the restriction of a metric smooth on Σ to a compact smooth subsurface is smooth up to ∂S , and its conformal class is $t|_S$. $\square$

Lemma C.8 (Normal form at a fixed point). *Let X be a Riemann surface and let $\iota : X \rightarrow X$ be an anti-holomorphic involution with $\iota(p) = p$. Then there is a holomorphic chart u , defined on an ι -invariant neighbourhood of p and with $u(p) = 0$, in which ι reads $u \mapsto \bar{u}$. In particular $\text{Fix}(\iota)$ is a 1-dimensional real-analytic submanifold of X , and each of its points has a neighbourhood which it separates into two half-discs interchanged by ι .*

Proof. Choose a holomorphic chart ψ at p with $\psi(p) = 0$, defined on a neighbourhood U ; replacing U by $U \cap \iota(U)$ we may assume $\iota(U) = U$. Then $G := \psi \circ \iota \circ \psi^{-1}$ is an anti-holomorphic involution of $\Omega := \psi(U)$ fixing 0, so $G(w) = g(\bar{w})$ for a holomorphic g on $\{\bar{w} : w \in \Omega\}$ with $g(0) = 0$. Write $\bar{g}(z) := g(\bar{z})$, which is again holomorphic and satisfies $\bar{g}(\bar{w}) = g(w)$ and $\bar{g}'(0) = g'(0)$. Then

$$G(G(w)) = g(\overline{G(w)}) = g(\overline{g(\bar{w})}) = g(\bar{g}(w)),$$

so $G \circ G = \text{id}$ says $g \circ \bar{g} = \text{id}$ near 0; both sides being germs of biholomorphisms fixing 0, also $\bar{g} \circ g = \text{id}$. Differentiating $g \circ \bar{g} = \text{id}$ at 0 gives $g'(0)\bar{g}'(0) = 1$, so $|g'(0)| = 1$.

Replace ψ by $c^{-1}\psi$ for a unit constant $c = e^{i\beta}$. The new chart has G replaced by $w \mapsto g_c(\bar{w})$ with $g_c(z) = c^{-1}g(\bar{c}z)$, so $g'_c(0) = e^{-2i\beta}g'(0)$; taking β to be half the argument of $g'(0)$ we may therefore assume $g'(0) = 1$.

Set $h(w) := w + \bar{g}(w)$. It is holomorphic near 0, with $h(0) = 0$ and $h'(0) = 1 + \overline{g'(0)} = 2 \neq 0$, hence a biholomorphism of a neighbourhood of 0 onto a neighbourhood of 0. Using $\bar{g} \circ g = \text{id}$ and $\bar{g}(\bar{w}) = g(w)$,

$$h(G(w)) = G(w) + \bar{g}(G(w)) = g(\bar{w}) + \bar{g}(g(\bar{w})) = g(\bar{w}) + \bar{w}, \quad \overline{h(w)} = \bar{w} + \overline{g(w)} = \bar{w} + g(\bar{w}),$$

so $h \circ G = \bar{h}$. Thus $u := h \circ \psi$ is a holomorphic chart in which ι reads $u \mapsto \bar{u}$; shrinking its image to a disc D about 0 makes its domain ι -invariant. In this chart the fixed point set is $u^{-1}(D \cap \mathbb{R})$, a real-analytic arc, and its complement in $u^{-1}(D)$ is the pair of open half-discs $\{\pm \text{Im } u > 0\}$, interchanged by ι . $\square$

Proposition C.9. *Let t be the conformal structure on S of clause (L2), and suppose either of the following holds.*

- (L2') *There are a conformal structure $\hat{t}$ on the closed surface $\hat{S}$ and a diffeomorphism Φ of $\hat{S}$ with $\Phi(S) = S$ such that $t = (\Phi^*\hat{t})|_S$.*
- (L2'') *There are a conformal structure $\hat{t}$ on $\hat{S}$ and an anti-conformal involution ι of $(\hat{S}, \hat{t})$, topologically conjugate to i , such that t is the pullback of $\hat{t}$ under a diffeomorphism of S onto the closure of one component of $\hat{S} \setminus \text{Fix}(\iota)$.*

Then t is represented by a metric on S smooth up to ∂S ; that is, (L2) holds. Under (L2'') the boundary ∂S is moreover real-analytic in the charts of that structure.

Proof. Under (L2'): $\Phi^*\hat{t}$ is a conformal structure on the closed smooth surface $\hat{S}$ and $S \subset \hat{S}$ is a compact smooth subsurface, so Lemma C.7 applies as it stands.

Under (L2''): by Lemma C.8 the set $C := \text{Fix}(\iota)$ is a closed 1-dimensional real-analytic submanifold of $\widehat{S}$. As ι is topologically conjugate to i , the complement $\widehat{S} \setminus C$ has exactly two components and ι interchanges them; let S_ι be the closure of one. Being bounded by the smooth closed 1-submanifold C , the set S_ι is a compact smooth subsurface of $\widehat{S}$ with $\partial S_\iota = C$. Lemma C.7 supplies a metric on $\widehat{S}$ representing $\widehat{t}$ and smooth on all of $\widehat{S}$, hence smooth up to C on S_ι ; its pullback under the given diffeomorphism $S \rightarrow S_\iota$ is the required metric. The last assertion is the normal form of Lemma C.8, in which C is a real-analytic arc. $\square$

Remark C.10. Neither (L2') nor (L2'') is proved here, and neither is proved in [11]; but the debt they name is a precise one. In the proof of [11, Thm. (1.7)] the structures t_k live on the fixed surface M , their doubles $\widehat{t}_k$ on the fixed closed surface $\widehat{M}$, and the fixed swap involution is anti-conformal for every $\widehat{t}_k$. Convergence is asserted in the *moduli space* of $\widehat{M}$, and what that provides is diffeomorphisms ψ_k of $\widehat{M}$ for which $\psi_k^* \widehat{t}_k$ converges. Nothing in a compactness statement about moduli makes ψ_k preserve $M \subset \widehat{M}$; and if it does not, then neither the phrase “inducing one on M by restriction” nor the boundary condition $\phi_k(\partial M) \subset \Gamma$ survives the passage to the limit. That is exactly (L2').

A classical route to (L2'') is visible. The involutions $\iota_k := \psi_k^{-1} \circ i \circ \psi_k$ are anti-conformal for $\psi_k^* \widehat{t}_k$, and each is topologically conjugate to i ; one would uniformise, observe that an anti-conformal involution is an isometry of the hyperbolic metric of its own conformal structure, extract by Arzelà–Ascoli a limit ι of a subsequence, and identify ι as an anti-conformal involution of the limit structure of the same topological type. Three inputs are needed and none of them is supplied here: that convergence in moduli space may be read as smooth convergence of the uniformising metrics after the ψ_k ; that the limiting isometry is again an involution conjugate to i ; and, in genus 0 and 1, a substitute for the hyperbolic metric. Until they are supplied, (L2) stays in the position of (27): asserted by the theorem cited, not reached by the argument printed for it. What Lemmas C.7 and C.8 change is the shape of the debt. It is not a regularity statement about a conformal structure at a boundary — that much is automatic — but a compatibility statement about a limit taken in moduli space.

C.4 The minimiser lies in $E(K)$.

Proposition C.11 (Barrier). *Let f be a Lemaire minimiser for λ . Then $f(S) \subset M$, that is $u := r \circ f \geq 0$.*

Proof. Let $\Omega := \{u < 0\}$, an open subset of $\text{Int } S$ with compact closure on which f takes values in $\{r < 0\}$, and let Ω' be a component of Ω . By Lemma C.6, $-u$ is a subsolution of Δ_g on Ω' . Its maximum over $\overline{\Omega'}$ is attained; if at an interior point, then the set where it is attained is closed in Ω' and, by the strong maximum principle [4, Thm. 3.5] applied in a chart about each of its points, open in Ω' , hence all of Ω' , so $-u$ is constant there and equal by continuity to its value on $\partial\Omega'$. Either way $\max_{\overline{\Omega'}}(-u) = \max_{\partial\Omega'}(-u)$. Now $u = 0$ on $\partial\Omega'$: a boundary point of Ω' lies either in $\text{Int } S$, where $u = 0$ by continuity, or in ∂S , where $u = 0$ because $f(\partial S) \subset \lambda \subset \{r = 0\}$. So $u \geq 0$ on Ω' , contradicting $u < 0$ unless $\Omega' = \emptyset$. $\square$

C.5 The Hopf package at ∂S . Fix once and for all, for g as in (L2), the Fermi coordinates (x, y) of g along ∂S : x is g -arclength on ∂S , y is the g -distance to ∂S , and

$$g = dy^2 + G(x, y) dx^2, \quad G > 0 \text{ smooth}, \quad (51)$$

a smooth chart on a collar of ∂S in which $\{y \geq 0\}$ is that collar, ∂_y is the inward g -unit normal and ∂_x is g -orthogonal to it.

Proposition C.12. *Let f be a Lemaire minimiser for λ and $u := r \circ f$. Then*

- (i) $u > 0$ on $\text{Int } S$; equivalently $f^{-1}(\partial M) = \partial S$;
- (ii) $\partial_y u(p) > 0$ for every $p \in \partial S$, in the coordinates (51);
- (iii) for every $p \in \partial S$ the differential df_p has rank 2, its image is not contained in $T_{f(p)}\partial M$, and $d(f|_{\partial S})_p \neq 0$;
- (iv) $f|_{\partial S}: \partial S \rightarrow \lambda$ is a homeomorphism, and f is an immersion on a neighbourhood of ∂S in S .

Consequently f is an immersion on all of S , is transverse to ∂M along ∂S , and meets ∂M exactly in $f(\partial S) = \lambda$.

Proof. (i) By Proposition C.11, $u \geq 0$. Put $Z := \{p \in \text{Int } S : u(p) = 0\}$ and let $p \in Z$. Near p the map f takes values in $\{r < \varepsilon\}$, so $\Delta_g u \leq 0$ there by Lemma C.6, and $u \geq 0 = u(p)$ exhibits an interior minimum of u , that is an interior maximum of the subsolution $-u$; the strong maximum principle [4, Thm. 3.5], applied in a chart about p , gives $u \equiv 0$ near p . So Z is open in $\text{Int } S$; it is closed there by continuity; and $\text{Int } S$ is connected. If $Z = \text{Int } S$ then $u \equiv 0$ on S , so $f(S) \subset \partial M = \{r = 0\}$ and every $f_* e_i$ is tangent to $\{r = 0\}$. Then (50) with (3) reads

$$0 = \Delta_g u = \varphi \varphi' \sum_{i=1,2} |f_* e_i|_{g_{\partial M}}^2 = - \sum_{i=1,2} |f_* e_i|_{g_{\partial M}}^2,$$

since $\varphi(0) = 1$ and $\varphi'(0) = -1$; so $df \equiv 0$ and f is constant on the connected S , contradicting that $f|_{\partial S}$ is onto λ . Hence $Z = \emptyset$.

(ii) Fix $p \in \partial S$ and work in the coordinates (51), shrunk so that f takes values in $\{r < \varepsilon\}$; let D^+ be a coordinate half-disc about $p = (0, 0)$. On D^+ we have $u \geq 0$ with $\Delta_g u \leq 0$, and $u > 0$ on the open half-disc by (i), while $u(p) = 0$. In these coordinates Δ_g is uniformly elliptic with smooth coefficients and no zeroth-order term, a half-disc satisfies the interior sphere condition at an interior point of its straight edge, and u is C^1 up to that edge by (L3). Hopf's boundary point lemma [4, Lem. 3.4], applied to the subsolution $-u$, gives $\partial_\nu(-u)(p) < 0$ for the inward normal $\nu = \partial_y$, that is $\partial_y u(p) > 0$.

(iii) By (ii), $dr_{f(p)}(f_* \partial_y) = \partial_y u(p) > 0$, so $f_* \partial_y \neq 0$ and $f_* \partial_y \notin T_{f(p)}\partial M = \ker dr_{f(p)}$. Since $f(\partial S) \subset \lambda$ and ∂_x is tangent to ∂S , the vector $f_* \partial_x$ lies in $T_{f(p)}\lambda \subset T_{f(p)}\partial M$. Weak conformality $f^* \hat{h} = \mu g$, valid up to ∂S , gives with (51)

$$|f_* \partial_y|^2 = \mu(p), \quad |f_* \partial_x|^2 = \mu(p)G(p), \quad \langle f_* \partial_x, f_* \partial_y \rangle = 0.$$

As $f_* \partial_y \neq 0$ we get $\mu(p) > 0$, hence $f_* \partial_x \neq 0$ because $G(p) > 0$. So $f_* \partial_x$ and $f_* \partial_y$ are non-zero and orthogonal: df_p has rank 2, its image contains $f_* \partial_y \notin T\partial M$, and $d(f|_{\partial S})_p = f_* \partial_x \neq 0$.

(iv) By (iii) the map $f|_{\partial S}: \partial S \rightarrow \lambda$ is an immersion of one circle into another, hence a local homeomorphism. By (L1) it is onto with connected point preimages; a point preimage is then a connected discrete set, so a single point. A continuous injection of a compact space is a homeomorphism onto its image, which is λ . For the second clause, df has rank 2 at each point of the compact ∂S by (iii) and is continuous up to ∂S by (L3), so rank 2 persists on a neighbourhood of ∂S .

For the final sentence: f has no branch point near ∂S by (iv) and none in the interior by [15] and [5], as in [3, p. 612]; transversality along ∂S is (iii); and $f^{-1}(\partial M) = \partial S$ with $f(\partial S) = \lambda$ is (i) with (iv). $\square$

Remark C.13. Proposition C.12(iii) is the assertion whose failure [3, §7] flags as the one alteration required in [3, Lems. 1.3–1.6]. What does the work is $\text{Hess } r \leq 0$ together with weak conformality: the first gives a strict sign for the *normal* derivative of $r \circ f$ at a boundary point, the second converts it into a strict sign for the *tangential* derivative of f , which is what excludes a boundary

branch point. Neither step survives for a free boundary, where $f(\partial S)$ is not confined to a curve and $r \circ f$ need not vanish on ∂S .

C.6 The intersections stay away from ∂M .

Corollary C.14. *Let f be a Lemaire minimiser for λ . There is $\delta > 0$ such that*

$$p, q \in S, \quad f(p) = f(q), \quad r(f(p)) < \delta \quad \implies \quad p = q. \quad (52)$$

In particular the double point set of f is a compact subset of $\{r \circ f \geq \delta\} \subset \text{Int } S$.

Proof. Suppose not: there are $p_n \neq q_n$ with $f(p_n) = f(q_n)$ and $r(f(p_n)) \rightarrow 0$. Passing to subsequences, $p_n \rightarrow p$ and $q_n \rightarrow q$ in the compact S , with $u(p) = u(q) = 0$, so $p, q \in \partial S$ by Proposition C.12(i); and $f(p) = f(q)$, so $p = q$ by Proposition C.12(iv). But f is injective near p : in the coordinates (51) extend f to a C^1 map on a full disc, which is possible as $f \in C^{2,\alpha}$ up to ∂S ; its differential at p has rank 2 by Proposition C.12(iii), so composing with a linear projection of $T_{f(p)}\hat{E}$ onto $df_p(T_p S)$ gives a map with invertible differential at p , injective near p by the inverse function theorem. This contradicts $p_n \neq q_n$ with $p_n, q_n \rightarrow p$. For the last clause, the double point set is contained in $\{r \circ f \geq \delta\}$, which is closed in the compact S and misses ∂S ; and it is itself closed, for if $p_n \rightarrow p$ with $f(p_n) = f(q_n)$ and $q_n \neq p_n$, then a subsequence of (q_n) converges to some q with $f(q) = f(p)$, and $q \neq p$ because f is an immersion on all of S and so injective near p , by the same inverse function theorem argument. $\square$

The separation (52) is stated through r , a function on $\hat{E}$, and not through the distance to ∂S ; so it passes to covering spaces unchanged. That is the form in which the embedding argument of [3, §5] would consume it, since that argument compares translates of a lift in the universal cover.

Corollary C.15. *Let $\pi: \tilde{E} \rightarrow \hat{E}$ be the universal covering, $\sigma: \tilde{S} \rightarrow S$ the universal covering of S , and $\tilde{f}: \tilde{S} \rightarrow \tilde{E}$ a lift of $f \circ \sigma$. Let g be a deck transformation of π . If there are $p, q \in \tilde{S}$ with $\tilde{f}(p) = g\tilde{f}(q)$ and $r(\tilde{f}(p)) < \delta$, then $g\tilde{f}(\tilde{S}) = \tilde{f}(\tilde{S})$. Hence two distinct translates of $\tilde{f}(\tilde{S})$ meet only inside $\{r \geq \delta\}$, and so do their images in any intermediate covering, r being invariant.*

Proof. Projecting, $f(\sigma p) = f(\sigma q)$ with $r(f(\sigma p)) < \delta$, so $\sigma p = \sigma q$ by (52) and $q = h p$ for a deck transformation h of σ . The universal covering is regular, so the two lifts $\tilde{f} \circ h$ and $\tilde{f}$ of $f \circ \sigma$ differ by a deck transformation: $\tilde{f} \circ h = \theta(h) \circ \tilde{f}$, where θ is the homomorphism induced by f . Then $\tilde{f}(p) = g\tilde{f}(hp) = g\theta(h)\tilde{f}(p)$, so $g\theta(h)$ fixes a point of $\tilde{E}$ and is therefore the identity, a deck transformation of a covering having no fixed point unless it is the identity. Hence $g = \theta(h)^{-1}$ and $g\tilde{f} = \tilde{f} \circ h^{-1}$, which has the same image as $\tilde{f}$. $\square$

In particular the intersection of two distinct translates contains no point of $\pi^{-1}(\partial M) = \{r = 0\}$ — the manifold $\tilde{E}$ has no boundary, and it is the old boundary that is meant — so it carries no arc meeting $\{r = 0\}$. That is the configuration whose appearance [3, §7] records as the alteration needed in [3, §4], “that one needs to consider arcs in $F_1 \cap F_2$ as well as circles”; in the fixed boundary case, under strict convexity, it does not appear.

C.7 What is closed, and what is not.

Remark C.16. Of the two steps of Remark A.8, the first is now available in the form needed, subject to one caveat: the ambient boundary is removed by Lemma C.1, the minimiser exists by [11, Thm. (1.7)] (Proposition C.3), it lies in $E(K)$ (Proposition C.11), and the immersion clause of [3, Thm. 7.2] is proved rather than asserted (Proposition C.12). The caveat is clause (L2), which is asserted by [11, Thm. (1.7)] and not reached by the argument printed for it (Remark C.4); everything from §C.4 onwards is conditional on it. What §C.3 contributes is a reduction: the

regularity that (L2) asserts is automatic (Lemma C.7), and (L2) follows from either of the two compatibility statements (L2'), (L2'') about the limit in moduli space (Proposition C.9), the second of them by way of the normal form for anti-conformal involutions (Lemma C.8). Neither compatibility statement is proved here (Remark C.10). The single alteration which [3, §7] flags in [3, Lems. 1.3–1.6], and for which no argument is printed there, is supplied in the strengthened form that the phenomenon does not occur (Remark C.13), so that those four lemmas are needed only at interior points, which is where they are proved; and Corollaries C.14 and C.15 dispose likewise of the alteration flagged for [3, §4].

The second step is still not available, and the reason is now narrower. The reduction it would serve is short: if the minimiser f of Proposition C.3 is an embedding whose image is properly isotopic to F rel ∂F , then that image is a smooth — hence piecewise smooth — member of $\mathcal{F}$ of area at most I , by Propositions C.11 and C.12 and the area comparison of Proposition C.3, so its area is exactly I and (27) is closed. Everything therefore rests on the embedding step. What remains unverified is the global part of the alterations announced in [3, §7]: [3, §2] with relative homology in place of absolute — that is, the regular neighbourhood, the tower of double coverings and the $\mathbb{Z}_2$ intersection theory of [3, Lems. 2.2–2.5], all for a surface meeting ∂M ; [3, Thm. 3.2] in the bounded case, which the proof of [3, Thm. 5.1] consumes at its first step, to know that the lift of the minimiser to M_F is an embedding; and the alterations in [3, §5] itself, which [3, §7] calls trivial. None of these is proved here, and this appendix does not assert that they are routine. Therefore the bounded form of [3, Thm. 5.1] is still not reached by a printed argument, the route of Remark A.8 is still not available, and (27) is not closed.

A third route deserves recording, if only to be set aside. One may try to minimise area directly over *embedded* surfaces in the isotopy class, after [14, Thm. 1], where the limit is embedded by construction and no tower argument is needed; the price is that the limit is a sum $n_1 \Sigma^{(1)} + \dots + n_R \Sigma^{(R)}$ of disjoint minimal surfaces with multiplicities, which must then be ruled out by incompressibility and irreducibility. But [14, Thms. 1, 1', 1''] are stated for *closed* embedded surfaces Σ_k ; the ambient manifold is allowed a mean convex boundary, the surfaces are not allowed one. The fixed boundary case needed for (27) is not printed there either. It is the bounded case of [6, §6] in a different guise, and we do not pursue it. The appendix on which the body of the paper depends is Appendix A, which goes through neither [3] nor this one.

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